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Simulation of a Partial Melting Mantle

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Abstract

Simulation of a Partial Melting Mantle

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This dissertation develops a multiphysics framework for modeling two-phase mantle dynamics with partial melting, emphasizing the handling of the degeneracy arising in no-melt regions and the relative movement between solid mantle and liquid magma. The model couples a degenerate Darcy–Stokes system describing static mechanics, a nonlinear transport system for chemical species and thermal energy, and a eutectic phase relation.

New mixed finite element spaces are introduced for the Darcy-Stokes system, including an $H(\mathrm{div})$-conforming finite element space for the Darcy flow and a modified Bernardi–Raugel finite element for the Stokes flow. Stability and approximation properties are established. The resulting saddle point system is solved by a modified Uzawa method. For nonlinear transport, we propose a new multilevel weighted essentially non-oscillatory (ML-WENO) reconstruction scheme. Coupling with the eutectic phase module links composition and enthalpy to porosity evolution, which closes the feedback loop between mechanical, transport, and phase processes.

Numerical experiments validate the method. Simplified melting functions reproduce compaction-driven porosity waves, while fully coupled simulations capture phase transitions and porosity channel formation in upwelling columns, demonstrating both stability and accuracy of the approach.

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Chapter 1: Introduction

Mantle melting—driven by decompression, heating, or water release—is a crucial geophysical process in lithospheric evolution and plate tectonics that shapes our planet, both today and in the past. Volcanism, for instance, the surface expression of magma, arises from mantle melting and melt segregation, producing Earth's crust and shaping its chemical evolution. Comparable processes may also occur on other terrestrial planets and moons. The following three key tectonic settings highlight the fundamental role of mantle melting and magmatism.

Mid-ocean ridges: Upwelling of the convecting mantle beneath mid-ocean ridges leads to decompression melting and generates the oceanic crust.

Hot-spots: Thermal anomalies induce melting beneath hot-spots and generate ocean islands and oceanic plateaus as well as flood basalt provinces on the continents.

Subduction zones: The addition of fluids that lower the melting point induces flux-melting above the dehydrating subducting oceanic crust and generates the continental crust.

However, prevailing estimates of upper mantle temperature, together with the inferred state of stress in the lithosphere, indicate that the asthenosphere is often partially molten (Mckenzie (1984) and Schmeling (2000)). In the partial melting scenarios, the porosity remains small enough that the solid mantle continues to transmits stress and maintain a pressure distinct from that of the full liquid magma.

Although a thorough understanding of partial melting is fundamental to comprehending the dynamics of magmatism and mantle convection, comparatively less effort has been devoted to systemically understanding and quantitatively modeling the genesis, segregation and emplacement of magma, along with the chemical transport it mediates (Katz (2022)). The governing equations for partial melting are highly complex and generally require numerical treatment. However, solving partial melting numerically in the mantle within a two-phase framework poses significant challenges, as porosity vanishes in melt-free regions, rendering the governing two-phase flow equations highly degenerate. In the absence of melt, the liquid velocity and pressure become undefined, which compromises the mathematical well-posedness of the mechanical flow equations and introduces discontinuities in transport quantities.

In this dissertation, building on the mathematical theory of Arbogast et al. (2017) for handling degenerate two-phase flow, we design, analyze and implement advanced numerical schemes capable of robustly handling such degeneracies. The aim is to study the time-dependent dynamics of a partially molten mantle. The simulations resolve the coupled evolution of porosity and phase transitions, providing a consistent framework for analyzing the thermomechanical processes associated with partial melting and melt segregation.

1.1 Previous Work

Fundamental theoretical developments on two-phase flow in deformable porous media emerged during the 1970s and 1980s. Sleep (1974), Stevenson (1980), Fowler (1984) and Mckenzie (1984) each derived governing equations for partially molten materials within a two-phase flow framework, differing slightly in formulation and geological context. These early models emphasized the coupling between a deformable creeping solid matrix, described by the Stokes equations, and fluid transport through porous flow governed by Darcy's law. A key feature of this framework is the compaction of the solid matrix, representing a unique mode of deformation.

Building on this two-phase solid–fluid framework, a variety of specific models were developed. Aharonov et al. (1995) proposed a single-species model emphasizing the combined effects of dissolution and porous flow, with less attention to compaction and energy transport. Later a reactive-dissolution model was advanced to further research in instability during melt segregation (Hesse et al. (2011); Liang et al. (2010)). Others approaches incorporated the enthalpy method to magma simulations enabling melting and freezing rates to be computed from equilibrium phase diagrams. In these formulations, the liquid velocity was often not solved directly but inferred from the solid velocity through a relative Darcy's law, with phase diagrams commonly parametrized with polynomial functions (Katz (2008); Hewitt and Fowler (2008)).

Geophysical studies have traditionally emphasized modeling aspects, with comparatively less focus on numerical methods. Finite difference and finite volume methods remain widely used among geophysicists. For example, Spiegelman (1993b) analyzed numerical convergence of finite difference schemes for solitary-wave solutions. More advanced approaches, such as total variation diminishing (TVD) flux-limiting schemes, were later applied to magma dynamics in Šrámek et al. (2010) and Katz and Weatherley (2012). A high performance parallel implementation of the finite difference method using PETSc is presented in Katz et al. (2007). Finite element methods have also been employed in magma simulations. Convergence and preconditioners for iterative solution of finite element discretizations of the magma dynamics equations are presented in Rhebergen et al. (2014) and Rhebergen et al. (2015). Applications of finite elements to subduction-zone modeling were demonstrated in Wilson et al. (2014). However, most existing formulations do not account for the degeneracy that arises in melt-free regions.

The treatment of melt-free regions has been approached in some works using several strategies. One strategy introduces a cutoff for material properties or primary variables (e.g. Keller et al. (2013); Dannberg and Heister (2016)). A different approach was proposed by Dannberg et al. (2019), in which a threshold scale is defined and compared with the Darcy coefficient at quadrature points. At each time step, computational cells are classified as either a "melt" or a "no melt" cell based on this threshold: the full two-phase system is solved in "melt" cells, while the Stokes system is solved in "no melt" cells. Although effective in practice, this method is less mathematically rigorous in that it classifies regions with actual melt as melt-free due to the imposed threshold. Moreover, the coexistence of two distinct sets of governing equations in melt and melt-free regions may require different finite element spaces or cause a loss of optimal convergence rates.

1.2 Related Software

We review some existing software packages that address related aspects of the problem.

1.2.1 General Purpose Software

Efficient numerical simulation of coupled geophysical processes requires robust and scalable computational frameworks. In this work, we primarily rely on PETSc, a software toolkit designed to support the scalable solution of scientific applications modeled by partial differential equations (Balay et al. (2025a); Balay et al. (2025b)). PETSc provides core functionality for parallel data management, iterative linear solvers, and discretization utilities, offering the flexibility and comprehensive tools needed for our simulations.

Alternative open-source platforms, such as FEniCS and deal.II, provide high-level environments for the rapid development of finite element solvers. FEniCS (Baratta et al. (2023)) offers a broad range of finite element function spaces and automated assembly tools, while deal.II (Arndt et al. (2021)) is designed for modular and efficient implementation of large-scale finite element methods. However, both frameworks are less suited to our specific requirements. In particular, our approach involves the use of a custom direct serendipity finite element space coupled with a finite volume method, which requires finer control over element assembly than these platforms provide.

For this reason, we directly implement the finite element assembly and couple it with our finite volume solver within the PETSc framework. This approach combines the flexibility needed for novel discretization strategies with the scalability of PETSc's linear algebra and solver infrastructure. This ensures both methodological innovation and computational efficiency in large-scale geophysical simulations.

1.2.2 Software for Mantle Dynamics

We now review several software frameworks specifically developed for simulations of mantle dynamics, but differ in scope and methodology from our approach.

ASPECT: One of the most advanced solvers based on deal.II, featuring excellent mesh adaptivity and numerical robustness, is capable of simulating partial melting processes. Following the framework outlined in Schubert et al. (2001), it solves a Stokes system for solid mantle flow coupled with an advection–diffusion–reaction system governing the transport of both energy and chemical species. Phase transitions are incorporated through the reaction term on the right-hand side of the transport equations. However, this approach does not account for melt extraction or relative motion between melt and solid, which limits its applicability in problems where melt transport is essential (Bangerth et al. (2018)). By contrast, modeling melt transport is a central focus of our solver.

CitcomS: A classical code used in simulation of mantle convection following the disscussion in Zhong et al. (2000) and Tan et al. (2006). The code solves an incompressible Stokes problem coupled with energy transport, however, melt extraction and chemical species transport are not explicitly represented.

TerraFERMA: A high level wrapper for FEniCS, PETSc and SPuD (a science neural options handling system) developed by Wilson et al. (2017). The primary aim of this framework is to enable rapid and reproducible construction of multiphysics models arising in geophysics scenarios. These packages do not handle the complexities of mantle convection in any direct way.

MELTS: The MELTS software family, originated by Ghiorso and Sack (1995), is designed to investigate phase relations in magmatic systems during melting and crystallization. However, mechanical behaviors and transport are not included.

1.3 New Framework

In this dissertation, we extend the classical two-phase Darcy-Stokes framework by emphasizing compaction and by a rigorous mathematically treatment of the degeneracy in no melt regions. Our model includes three key components:

- A degenerate two-phase Darcy–Stokes system describing the coupled interaction between the solid matrix and the interstitial melt;
- A nonlinear system governing the transport of chemical species and thermal energy;
- A eutectic phase relation characterizing the melting processes and connecting the transport and static mechanical system through enthalpy method.

We develop a local mass conservative numerical framework to address this complicated multiphysics system using a logically rectangular computational mesh of quadrilaterals. For the static mechanics system, we introduce $H(\mathrm{div})$-and an H^1- conforming mixed finite element spaces for Darcy and Stokes subproblems, respectively. The mixed finite element spaces are constructed using a noval direct serendipity space tailored for a quadrilateral mesh. A modified Bernardi–Raugel element is developed and analyzed. We provide detailed proofs of its stability and approximation properties. A new modified Uzawa iterative method is implemented to solve the singular saddle point problem induced by this degenerate coupled system.

To address the nonlinear transport of chemical species and energy in the presence of porous media, we study and modify the weighted essentially non-oscillatory

(WENO) method, described in detail by Shu (1998). We propose two simple alternatives to accelerate the evaluation of the smoothness indicator. Additionally, we develop a new multilevel WENO weighting scheme that is more versatile and flexible, allowing the combination of arbitrary stencils of different sizes.

We advance the multiphysics system in time by incorporating a eutectic phase module that couples compositional and thermal information with porosity, thereby capturing the evolution of porosity and phase transitions.

The code is implemented in C/C++ along with PETSc, and is designed to be easily adapted for large-scale parallel execution on supercomputers. Our code enables an accurate simulation of the degeneracy of the no melt regions. That is, the equations are solved irrespective of the value of the porosity, including zero porosity. In this way, a proper handling of melt extraction and chemical species segration are modeled by our code, leading to a computational framework for invectigating complex geologies settings involving melting and magmatism.

1.4 Outline

The rest of the dissertation is organized as follows.

In Chapter 2, we present a detailed description of the mathematical model used to describe the partial melting process: mechanical flow, transport phenomena and phase transitions. Section 2.1 derives a coupled Darcy-Stokes system with compaction, starting from the fundamental conservation laws. In Section 2.2, we describe transport of chemical species and energy within partial molten mantle. Section 2.3 introduces a simplified linearized parameterized two-component eutectic phase diagram. Finally, Section 2.4 presents a nondimensionalization of the physical model with carefully selected characteristic physical scales and formally present the nondimensional governing equations.

In Chapter 3, we present a new locally mass-conserving numerical framework that couples mixed finite element (MFEM) and finite volume (FVM) methods. In Section 3.1, we first describe the mesh partition intended for both the MFEM and the FVM solver. Section 3.3 to 3.5 develop a new MFEM formulation employing direct serendipity space for application to the degenerate Darcy-Stokes equation. In Section 3.6, we present a modified Uzawa algorithm applied to the linear system arising from the MFEM discretization. Section 3.10 to 3.12 discuss the FVM for solving the transport equations. We use the finite volume weighted essentially non-oscillatory methods. Section 3.10 introduces a new nonlinear weighting scheme for WENO reconstruction, and Section 3.11 demonstrates the flexibility of the new ML-WENO reconstruction scheme to address the reconstruction on the boundary.

In Chapter 4, we present numerical experiments coupling the previously defined MFEM and FVM solvers. Section 4.1 details the coupling algorithm and the strategy employed to handle discontinuous porosity. Section 4.2 introduces two numerical experiments on a pseudo one-dimensional mantle column: the first couples the solvers using a prescribed melting function, while the second employs the proposed eutectic phase package. Section 4.3 shows a simple two-dimensional numerical experiment coupling phase calculations.

Finally, in Chapter 5, we summarize our results, review our conclusions and propose some suggestions for future work. Appendix A is included to describe the code implementation in detail.

Chapter 2: Governing Equations in Mantle Dynamics

2.1 Mechanics in Mantle Dynamics

Extensive studies have been conducted on fluid dynamics of a two-phase mixture. We briefly discuss the four governing equations of mechanics, drawing on a similar discussion presented by Drew and Segel (1971), Mckenzie (1984) and Katz (2022).

2.1.1 Phase Fraction

We begin with defining the phase fraction using an indicator function, under the assumption that multiple phases cannot coexist at the same spatial location. Accordingly, the phases are considered immiscible. The indicator function is defined as

$$\chi_i(\mathbf{x}) = \begin{cases} 1 & \text{phase } i, \\ 0 & \text{not phase } i. \end{cases} \tag{2.1}$$

We then define the macroscopic phase fraction function of phase i within a representative volume element V as

$$\phi_i = \frac{1}{V}\int_V \chi_i(\mathbf{x}) d\mathbf{v}, \tag{2.2}$$

where $\sum_i \phi_i = 1$. In the incompressible setting, the phase fraction may be equivalently regarded as the volume fraction occupied by each phase within a representative elementary volume, reflecting the proportion of space occupied by each phase at the macroscopic scale. Phase-averaged (or volume-averaged) quantities, also referred to as bulk quantities, are defined as

$$\overline{\gamma} = \sum_i \phi_i \gamma_i, \tag{2.3}$$

where γ_i denotes a quantity that may be scalar- or vector-valued.

2.1.2 Conservation Laws

We consider the conservation equation for a single-phase system. For a given volumetric quantity γ defined within a representative volume element V, we can write a corresponding conservation equation as:

$$\frac{d}{dt}\int_V \gamma d\mathbf{v} = -\int_{\partial V} \mathbf{f} \cdot d\sigma + \int_V S d\mathbf{v}, \tag{2.4}$$

where $\mathbf{f}$ represents flux travelling across the boundary of a representative volume element V. The minus sign indicates that the positive direction aligns with normal unit vector pointing outwards. The source/sink term S, represents the volumetric production rate of the volumetric quantity γ. We apply Gauss's theorem to obtain the differential form

$$\frac{\partial \gamma}{\partial t} + \nabla \cdot \mathbf{f} = S. \tag{2.5}$$

We model the partially molten mantle as a two-phase mixture consisting of liquid magma and solid mantle. The phase fraction of liquid magma is denoted by ϕ_l and phase fraction of solid mantle is labelled as ϕ_s, such that the relation $\phi_l + \phi_s = 1$ is always obtained.

We rewrite the single-phase conservation law (2.5) with the bulk density as

$$\frac{\partial \overline{\rho}}{\partial t} + \nabla \cdot \overline{\rho \mathbf{v}} = 0, \tag{2.6}$$

where $\overline{\rho} = \phi_l \rho_l + \phi_s \rho_s$. Under the Boussinesq approximation, we neglect the compressibility of each individual phase. Furthermore, we extend the Boussinesq approximation by disregarding the density difference between the two phases, except in the body force term. Accordingly, the continuity equation for the two-phase mixture can be written as

$$\nabla \cdot \overline{\mathbf{v}} = 0. \tag{2.7}$$

Similarly, the conservation equation for the phase-averaged momentum takes the form

$$\frac{\partial \overline{\rho \mathbf{v}}}{\partial t} + \nabla \cdot \overline{\rho \mathbf{v} \otimes \mathbf{v}} = \overline{\rho}\mathbf{g} + \nabla \cdot \overline{\boldsymbol{\tau}} - \nabla \overline{p}, \tag{2.8}$$

where $\boldsymbol{\tau}$ represents the deviatoric stress. By neglecting inertial effects, justified by the low Reynolds number of magma and mantle, the static force balance equation reduces to

$$\overline{\rho}\mathbf{g} + \nabla \cdot \overline{\boldsymbol{\sigma}} = 0, \tag{2.9}$$

where $\boldsymbol{\sigma} = \boldsymbol{\tau} - p\boldsymbol{I}$ represents the total stress, consisting of the deviatoric component $\boldsymbol{\tau}$ and the isotropic pressure $p\boldsymbol{I}$.

2.1.3 Darcy's Law

The equations governing bulk quantities (2.7) and (2.9) alone are insufficient to simultaneously determine the two distinct velocities and pressures. Therefore, an additional constitutive relationship is required to describe the behavior of the liquid phase within the porous medium. We examine more closely the interphase forces between the solid and liquid phases. A brief overview of the derivation of the individual governing equations for each phase is provided here, while more detailed and rigorous treatments can be found in the mixture theory literature, such as Bowen (1980). We write momentum balances for each individual phase as:

$$\begin{aligned} 0 &= \phi_l \rho_l \mathbf{g} + \nabla \cdot (\phi_l \boldsymbol{\tau}_l) - \nabla(\phi_l p_l) + \mathbf{F}_l, \\ 0 &= \phi_s \rho_s \mathbf{g} + \nabla \cdot (\phi_s \boldsymbol{\tau}_s) - \nabla(\phi_s p_s) - \mathbf{F}_s, \end{aligned} \tag{2.10}$$

where $\mathbf{F}_l = \mathbf{F}_s$ are the interphase forces between solid and liquid phases. The interphase forces between the solid and liquid phases are equal and opposite, in accordance with Newton's third law. Neglecting the deviatoric stress in the liquid phase and retaining only the isotropic pressure component, we balance the remaining stress with the viscous drag force and the equilibrium body force. Accordingly, we obtain the following relation for the relative velocity between phases, in alignment with Darcy's law

$$\phi_l(\mathbf{v}_l - \mathbf{v}_s) + \frac{k_{\phi_l}}{\mu_l}(\nabla p_l - \rho_f \mathbf{g}) = 0, \tag{2.11}$$

where k_{ϕ_l} is the matrix permeability, which usually takes the quadratic form as $k_{\phi_l} = k_0 \phi_l^2$.

2.1.4 Compaction

In the context of mantle dynamics, an additional yet unique and essential relationship regulates melt migration. We now revisit the continuous equation (2.6), and consider the two phases separately with the effects of melting incorporated, i.e. ,

$$\begin{aligned} \frac{\partial \phi_l}{\partial t} + \nabla \cdot (\phi_l \mathbf{v}_l) &= \Gamma, \\ \frac{\partial (1-\phi_l)}{\partial t} + \nabla \cdot ((1-\phi_l)\mathbf{v}_s) &= -\Gamma, \end{aligned} \tag{2.12}$$

where Γ is the unknown melting rate. We focus on the solid phase equation to get the evolving compaction relationship as

$$\frac{\partial \phi_l}{\partial t} + \mathbf{v}_s \cdot \nabla \phi_l = \Gamma + (1-\phi_l)\nabla \cdot \mathbf{v}_s, \tag{2.13}$$

where the divergence term $\nabla \cdot \mathbf{v}_s$ is often referred to as the compaction rate. We approximate the static compaction relation as

$$\xi_{\phi_l} \nabla \cdot \mathbf{v}_s = p_l - p_s, \tag{2.14}$$

where ξ_{ϕ_l} is often referred to as bulk viscosity or compaction viscosity. The form of the compaction viscosity ξ_{ϕ_l} has been the subject of extensive discussion. Mckenzie (1984) popularized the concept of compaction viscosity and relating the term to the stress balance in a partially molten system. Bercovici et al. (2001) extended this formulation by incorporating surface tension effects. Rabinowicz et al. (2002) further examined compaction behavior in the limit of infinitesimally small porosity. Following the approaches of Hewitt and Fowler (2008) and Katz (2008), we approximate the compaction viscosity ξ_{ϕ_l} using a simplified constitutive relationship,

$$\xi_{\phi_l} = \frac{\mu_s}{\phi_l}, \tag{2.15}$$

which highlights the fact that increased melting facilitates the compaction of the solid phase, while in the limit of no melting ($\phi_l = 0$), compaction is entirely eliminated.

2.1.5 Full Static System

By combining equations (2.7), (2.9), (2.11) and (2.14), We finally arive at the full static flow mechanical system,

$$
\begin{aligned}
&\phi_l(\mathbf{v}_l - \mathbf{v}_s) + \frac{k_{\phi_l}}{\mu_l}(\nabla p_l - \rho_f \mathbf{g}) = 0, && (2.16)\\
&\nabla \cdot (\phi_l \mathbf{v}_l + \phi_s \mathbf{v}_s) \qquad\qquad = 0, && (2.17)\\
&\nabla \bar{p} - \nabla \cdot \boldsymbol{\tau}(\mathbf{v}_s) + \bar{\rho}\mathbf{g} \qquad\quad = 0, && (2.18)\\
&\mu_s \nabla \cdot \mathbf{v}_s - \phi_l(p_l - p_s) \qquad\quad = 0, && (2.19)
\end{aligned}
$$

where the deviatoric stress of the solid is

$$\boldsymbol{\tau}(\mathbf{v}_s) = 2\mu_s\phi_s\big(\mathcal{D}\mathbf{v}_s - \tfrac{1}{3}\nabla \cdot \mathbf{v}_s \mathbf{I}\big), \tag{2.20}$$

and $\mathcal{D}\mathbf{v}_s = \frac{1}{2}(\nabla \mathbf{v}_s + \nabla \mathbf{v}_s^T)$ is the symmetric gradient. Moreover, we adjust the relative permeability $k_{\phi_l} = k_0 \phi_l^{2+2\Theta}$ with constant $\Theta \in [0, 1/2]$. Here μ_s and μ_l are dynamic viscosities for solid and liquid respectively. This complete static mechanical system relates the velocities and pressures of the solid and liquid phases to a given porosity distribution, thereby laying the foundation for subsequent computations.

2.1.6 Scaled Formulation

The Darcy equation mathematically degenerates in the regions where the porosity is zero. To handle this issue, we use the formulation developed previously by Arbogast et al. (2017). We first define pressure potentials as

$$q_l = p_l - \rho_l g z \quad \text{and} \quad q_s = p_s - \rho_l g z, \tag{2.21}$$

where $g = |\mathbf{g}|$ is the magnitude of gravity acceleration and z is the depth, such that $\mathbf{g} = g\nabla z$. Note that ρ_l is used to define both potentials, so that with $q = \bar{q} = \bar{p} - \rho_l g z$,

$$p_l - p_s = q_l - q_s = \frac{1}{1-\phi_l}(q_l - q).$$

In the mechanical system, the subscript on ϕ_l will be omitted. and we will simply write it as ϕ. Accordingly, the solid phase fraction will be expressed as $1-\phi$, emphasizing the constraint that the two phase fractions sum to one. Thus

$$\phi \mathbf{v}_r + \frac{k_0 \phi^{2+2\Theta}}{\mu_l} \nabla q_l = 0, \tag{2.22}$$

$$\mu_s \nabla \cdot (\phi \mathbf{v}_r) + \frac{\phi}{1-\phi}(q_l - q) = 0, \tag{2.23}$$

$$\nabla q - \nabla \cdot \tau(\mathbf{v}_s) = -(1-\phi)\rho_r \mathbf{g}, \tag{2.24}$$

$$\mu_s \nabla \cdot \mathbf{v}_s - \frac{\phi}{1-\phi}(q_l - q) = 0, \tag{2.25}$$

where the relative velocity $\mathbf{v}_r = \mathbf{v}_l - \mathbf{v}_s$ and the density difference $\rho_r = \rho_s - \rho_l$. Scaled variables are also defined,

$$\tilde{\mathbf{v}}_r = \phi^{-\Theta} \mathbf{v}_r \quad \text{and} \quad \tilde{q}_l = \phi^{1/2} q_l, \tag{2.26}$$

and the problem is reformulated as

$$\tilde{\mathbf{v}}_r + \frac{k_0 \phi^{1+\Theta}}{\mu_l} \nabla (\phi^{-1/2} \tilde{q}_l) = 0, \tag{2.27}$$

$$\mu_s \phi^{-1/2} \nabla \cdot (\phi^{1+\Theta} \tilde{\mathbf{v}}_r) + \frac{1}{1-\phi}(\tilde{q}_l - \phi^{1/2} q) = 0, \tag{2.28}$$

$$\nabla q - \nabla \cdot \tau(\mathbf{v}_s) = -(1-\phi)\rho_r \mathbf{g}, \tag{2.29}$$

$$\mu_s \nabla \cdot \mathbf{v}_s - \frac{\phi^{1/2}}{1-\phi}(\tilde{q}_l - \phi^{1/2} q) = 0, \tag{2.30}$$

where the new scaled relative velocity maintains numerical stability in the zero prosority regions.

2.2 Transport of Chemical Species and Energy

With the velocities of solid and liquid phases determined, we are able to evolve our system forward in time. In our system, chemical species and energy are transported by the motion of the solid and the liquid phases.

2.2.1 Transport of Chemical Species

The realistic chemical compositions of mantle and magma are highly complicated and very hard to trace at the same time. Therefore, we present a simple two-species two-phase model. In this framework, we also negelect detailed description of the chemical reaction process and exclude the effect of the change of Gibbs free energies. Similar discussion can be found in the literature like Spiegelman and Elliott (1993), Hewitt and Fowler (2008), Katz (2008) and Hesse et al. (2022). Both species can exist in either the solid or liquid phase. They are assumed to be immiscible in the solid state but fully miscible in the liquid phase. By again extending the Boussinesq approximation, we consider a homogenous density ρ for both the solid phase and the liquid phase. Figure 2.1 illustrates the coexistence of species and phases in representative volume element. Within the representative volume element V, $\gamma_{i,j}$ represents

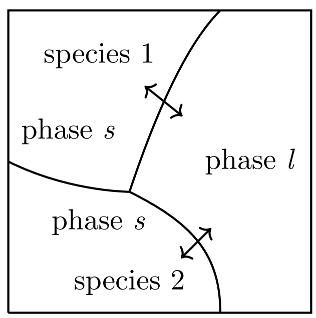


Figure 2.1: Phases and species in an representative volume element (RVE).

one quantity of the species i present in phase j, where $i \in 1, 2$ and $j \in \{s, l\}$. The mass fraction $c_{i,j}$ of a given species i in phase j is also regarded as a concentration. We can write transport equations for an individual species i, assuming a pseudo melting term Γ representing the transition between the solid and the liquid phase as

$$\frac{\partial}{\partial t}\big(\phi_l \rho_l c_{i,l}\big) + \nabla \cdot \big(\phi_l \rho_l c_{i,l} \mathbf{v}_l - \phi_l \rho_l \mathcal{D}_{i,l} \nabla c_{i,l}\big) = \Gamma, \tag{2.31}$$

$$\frac{\partial}{\partial t}\big(\phi_s \rho_s c_{i,s}\big) + \nabla \cdot \big(\phi_s \rho_l c_{i,s} \mathbf{v}_s - \phi_s \rho_s \mathcal{D}_{i,s} \nabla c_{i,s}\big) = -\Gamma, \tag{2.32}$$

where $\mathcal{D}_{i,l}$ and $\mathcal{D}_{i,s}$ are the diffusion coefficients of component i in the liquid and the solid phase respectively. The diffusion term will be dropped in the solid phase. By summing up equations (2.31) and (2.32), we obtain an advection-diffusion equation as

$$\frac{\partial \overline{\rho c_i}}{\partial t} + \nabla \cdot \left(\overline{\rho c_i \mathbf{v}} - \phi_l \rho_l \mathcal{D}_{i,l} \nabla c_{i,l} \right) = 0, \tag{2.33}$$

where the phase averaged quantities are $\overline{\rho c_i} = \sum_{j\in\{s,l\}} \phi_j \rho_j c_{i,j}$ and $\overline{\rho c_i \mathbf{v}} = \sum_{j\in\{s,l\}} \phi_j \rho_j c_{i,j} \mathbf{v}_j$. The subscript i will be omitted, as only one chemical species needs to be explicitly computed in this two-species system. By extending the Boussinesq approximation such that $\rho_l = \rho_s = \overline{\rho}$, the species transport equation can be simplified to the form

$$\frac{\partial \overline{c}}{\partial t} + \nabla \cdot \left(\overline{c\mathbf{v}} - \phi_l \mathcal{D}_l \nabla c_l \right) = 0. \tag{2.34}$$

2.2.2 Effective Velocity

The concept of effective velocity has long been employed in chemistry, particularly in the context of particle transport. In mantle dynamics, Spiegelman (1996) applied effective velocity for equilibrium transport of concentration within the liquid phases. We define a corresponding effective velocity for the transport of bulk concentration. To begin, we define the partition coefficient as $D_i = c_{i,s}/c_{i,l}$. The concentration of species i in the liquid and solid phases can be expressed in terms of the bulk concentration as

$$(\phi_l + \phi_s D_i) c_{i,l} = \overline{c}_i, \tag{2.35}$$

$$(\phi_l / D_i + \phi_s) c_{i,s} = \overline{c}_i. \tag{2.36}$$

By expressing $c_{i,l}$ and $c_{i,s}$ with bulk concentration given by equations (2.35) and (2.36), we can rewrite the bulk velocity as

$$\overline{c_i \mathbf{v}} = \phi_s \mathbf{v}_s c_{i,s} + \phi_l \mathbf{v}_l c_{i,l} = \left(\frac{\phi_l \mathbf{v}_l + D \phi_s \mathbf{v}_s}{\phi_l + \phi_s D} \right) \overline{c}_i. \tag{2.37}$$

We get the proper expression for the effective velocity, denoted as

$$\mathbf{v}_{\text{eff}} = \frac{\phi_l \mathbf{v}_l + D_i \phi_s \mathbf{v}_s}{\phi_l + \phi_s D_i}, \tag{2.38}$$

which is the appropriate velocity at which the bulk concentration should advect. In the absence of melting, where $\phi_l = 0$, we retrieve $\mathbf{v}_{\text{eff}} = \mathbf{v}_s$. Conversely, in the case of complete melting, where $\phi_l = 1$, we obtain $\mathbf{v}_{\text{eff}} = \mathbf{v}_l$. Therefore, the definition of effective velocity is well defined in those two limiting scenarios. We rewrite the transport equation (2.34) with effective velocity (2.38) as

$$\frac{\partial C}{\partial t} = \nabla \cdot (\mathbf{v}_{\text{eff}} C - \phi_l \mathcal{D}_l \nabla c_l) = 0, \tag{2.39}$$

where $C = \overline{c}$ for simplicity.

2.2.3 Transport of Energy

According to the first law of thermodynamics, we write the energy balance equation for the specific internal energy u_i in the phase-averaged form as

$$\frac{\partial \overline{\rho u}}{\partial t} + \nabla \cdot \overline{\rho u \mathbf{v}} = \overline{\rho \mathcal{Q}} + \nabla \cdot \overline{K} \nabla T + \nabla \cdot \overline{\boldsymbol{\sigma} \cdot \mathbf{v}} + \overline{\rho \mathbf{v}} \cdot \mathbf{g}, \tag{2.40}$$

where $\overline{K} = \phi_l K_l + \phi_s K_s$ is the phase averaged thermal conductivity. On the left-hand side, the equation represents the time-dependent rate of change of internal energy of a two-phase mixture as it moves with the flow. On the right-hand side, it includes the heat source term $\mathcal{Q}$, conductive heat transfer following Fourier's law, mechanical work done by mixture stress, and work against gravity, respectively.

In a thermochemical equilibrium scenario, we replace internal energy u_i with enthalpy h_i, and use hydrostatic P as a uniform pressure, i.e., $P = \rho_l g(z - z_0)$. The specific enthalpy is defined as

$$h_i = u_i + p_i \tilde{V}_i = u_i + P/\rho_i, \tag{2.41}$$

where $\tilde{V}_i = 1/\rho_i$ is the specific volume. We formulate the enthalpy equation by neglecting the external heating/cooling term $\mathcal{Q}$, mechanical work done by mixture stress and work against gravity, following the discussion presented by Katz (2008), so

$$\frac{\partial \overline{\rho h}}{\partial t} + \nabla \cdot \overline{\rho h \mathbf{v}} = \frac{\partial P}{\partial t} + \overline{\mathbf{v}} \cdot \nabla P + \nabla \cdot \overline{K} \nabla T, \tag{2.42}$$

using (2.17).

2.2.4 Temperature Formulation

We now connect enthalpy and temperature with thermodynamics relationships. To begin with, we write differential form of internal energy as

$$du_i = T ds_i - p_i d\tilde{V}_i. \tag{2.43}$$

We use (2.43) in the differential form of enthalpy as

$$dh_i = du_i + p_i d\tilde{V}_i + \tilde{V}_i dp_i = T ds_i - p_i d\tilde{V}_i + p_i d\tilde{V}_i + \tilde{V}_i dp_i = T ds_i + \frac{dp_i}{\rho_i}, \tag{2.44}$$

and entropy as

$$ds_i = c_p \frac{dT}{T} + \left(\frac{\partial s_i}{\partial p_i}\right)_T dp_i, \tag{2.45}$$

where c_p is the uniform specific heat capacity used for both phases. We replace $\left(\frac{\partial s}{\partial p}\right)_T$ with $-\left(\frac{\partial \tilde{V}_i}{\partial T}\right)_p$ according to Maxwell's fourth relation Campbell (1958). We assume a simple linear relationship between temperature and volume as

$$\frac{\partial \tilde{V}_i}{\partial T} = \alpha_\rho / \rho_i, \tag{2.46}$$

where α_ρ is the thermal expansion coefficient used uniformly for both the liquid and the solid phases. We substitute the entropy equation (2.45) into the enthalpy equation (2.44), so

$$dh_i = c_p dT + \frac{1 - \alpha_\rho T}{\rho_i} dp_i. \tag{2.47}$$

We can expand the divergence term in the equation (2.42) and write for phase i, $i \in \{s, l\}$ as

$$\nabla \cdot (\phi_i \rho_i h_i \mathbf{v}_i) = h_i \nabla \cdot (\phi_i \rho_i \mathbf{v}_i) + \phi_i \rho_i \mathbf{v}_i \nabla h_i, \tag{2.48}$$

where $\nabla h_i = c_p \nabla T + \frac{1-\alpha_\rho T}{\rho_i} \nabla p_i$. We now replace the term ∇h_i with the relation (2.46) as

$$\nabla \cdot \overline{\rho h \mathbf{v}} = (h_l - h_s) \nabla \cdot (\phi_l \rho_l \mathbf{v}_l) + h_s \nabla \cdot \overline{\rho \mathbf{v}} + c_p \overline{\rho \mathbf{v}} \cdot \nabla T + (1 - \alpha_\rho T) \overline{\mathbf{v}} \cdot \nabla p_i, \tag{2.49}$$

We substitute the new formulation back to the equation (2.42) and again use lithostatic pressure for approximation as

$$\frac{\partial \overline{\rho h}}{\partial t} + h_s \nabla \cdot \overline{\rho \mathbf{v}} + c_p \overline{\rho \mathbf{v}} \cdot \nabla T = -(h_l - h_s)\nabla \cdot (\phi_l \rho_l \mathbf{v}_l) + \frac{\partial P}{\partial t} + \nabla \cdot \overline{K} \nabla T + \alpha_\rho T \nabla P, \quad (2.50)$$

where we have $\nabla P = \rho \mathbf{g}$ under the lithostatic pressure approximation.

We integrate enthalpy (2.47) under constant pressure to relate enthalpy and temperature in the sense of thermodynamic state variables as

$$h - h_0 = c_p (T - T_0), \quad (2.51)$$

where the reference values h_0 and T_0 are set as zero for simplicity. Another important physical attribute in the context of melting is latent heat L. We write specific enthalpy in the solid and the liquid phase separately as

$$\begin{aligned} h_s &= c_p T, \\ h_l &= c_p T + L. \end{aligned} \quad (2.52)$$

Under the extended Boussinesq assumption, we rewrite the enthalpy equation with latent heat and temperature under constant pressure ($\partial P / \partial t = 0$) as

$$\frac{\partial \overline{\rho h}}{\partial t} + \rho c_p \nabla \cdot (\overline{\mathbf{v}} T) = \rho L \nabla \cdot (\phi_s \mathbf{v}_s) + \overline{K} \nabla^2 T + \rho \alpha_\rho T \overline{\mathbf{v}} \cdot \mathbf{g}, \quad (2.53)$$

since $\nabla \cdot (\phi_l \mathbf{v}_l) = -\nabla \cdot (\phi_s \mathbf{v}_s)$ using the continuous equation (2.17), i.e., $\nabla \cdot \overline{\mathbf{v}} = 0$. The last term including gravitational acceleration has previously been referred to as adiabatic cooling or decompression cooling by Mckenzie and Bickle (1988). Assuming no density difference between the solid and the liquid phase, we divide ρc_p on both sides of the equation to obtain a normalized form as

$$\frac{\partial \overline{H}}{\partial t} + \nabla \cdot (\overline{\mathbf{v}} T) = \frac{L}{c_p} \nabla \cdot (\phi_s \mathbf{v}_s) + \kappa \nabla^2 T + \frac{\alpha_\rho}{c_p} T \overline{\mathbf{v}} \cdot \mathbf{g}, \quad (2.54)$$

where $\overline{H} = \overline{\rho h} / \rho c_p = \overline{h} / c_p$ is the bulk specific enthalpy and $\kappa = \overline{K} / \rho c_p$ is the bulk heat diffusivity.

2.3 Eutectic Phase Behavior

With the transport equations defined, we can now proceed to redistribute chemical species and energy within the system. The next step is to determine the thermodynamic state of the system, assuming thermodynamic equilibrium is obtained at each time step. Recognizing that mid-ocean ridges form in regions where temperatures are generally lower, eutectic melting, where the presence of a secondary species significantly lowers the melting temperature, provides a suitable model for describing the melting behavior in these areas.

The binary eutectic system has been widely applied in studies involving chemical equilibrium, with detailed description can be found in textbooks such as Lupis (1983). In the context of mantle dynamics, eutectic melting was being incorporated by Ribe (1985). Moreover, eutectic phase behavior has been applied to solidification of alloys by Wheeler et al. (1996). More recently, binary eutectic melting has also been applied to water-salt-brine systems in glaciers Hesse et al. (2022).

We assume the two primary components in our partially molten system are olivine and orthopyroxene (opx), which are highly miscible in the liquid phase, but immiscible in the solid phase. Olivine is considered to dominate the system, while orthopyroxene is the minor component. Assuming thermodynamic equilibrium is obtained at each time step, we can determine the phase fraction of liquid, the volume fraction of each solid components and the temperature using a binary eutectic phase diagram with transported enthalpy and bulk composition.

A typical eutectic phase diagram is shown in Figure 2.2. There are two key concepts about the diagram:

- the eutectic melting point or eutectic temperature T_e, is the lowest temperature at which melting may occur. Melting at the eutectic temperature is called eutectic melting. The temperature of the system is fixed at the eutectic temperature when eutectic melting happens.

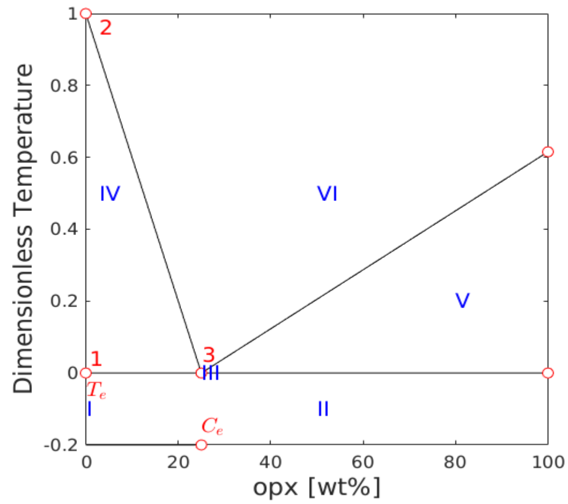


Figure 2.2: Eutectic phase diagram of a Olivine-Orthopyroxene system.

- The eutectic composition C_e is a fixed opx-olivine concentration ratio of the liquid produced by eutectic melting.

2.3.1 Clausius-Clapeyron Relation

In geological contexts, pressure can undergo significant variations during dynamic events such as mantle upwelling. As a result, the influence of pressure on the melting point cannot be neglected. To account for this, we approximate the solidus melting temperature T_m at a given pressure using the approximated Clausius–Clapeyron relationship. We also extend such approximation to eutectic temperature T_e as

$$\begin{aligned} T_m &= T_{m0} + \nu^{-1}(P - P_0), \\ T_e &= T_{e0} + \nu^{-1}(P - P_0), \end{aligned} \tag{2.55}$$

where ν is the Clausius-Clapeyron constant, and P_0 is the reference pressure usually set to be 0. The melting point T_{m0} and eutectic point T_{e0} refer to their respective values at standard atmospheric pressure.

2.3.2 Some Assumptions

In a eutectic system, the current thermodynamic state of the material can be identified using the bulk mass fractions of the components and bulk enthalpy of the system. Our bulk mass fraction will refer to opx in out binary system involving only two components so C is generally small. We begin by introducing some assumptions regarding densities and other material properties.

- We assume no density difference between solid phase olivine and opx, i.e., $\rho_{\text{opx}} = \rho_{\text{olivine}}$;
- We assume no density change during phase transition (extended Boussinesq approximation), i.e., $\rho_l = \rho_s$;
- We assume constant and uniform material thermodynamics properties like specific heat capacity c_p and thermal expansion coefficient α_p.

Following the assumptions, we relate the bulk composition with previously defined phase averaged concentration (2.39) as

$$\frac{\phi_l \rho_l c_l + \phi_s \rho_s c_s}{\phi_l \rho_l + \phi_s \rho_s} = \phi_l c_l + \phi_s c_s = \phi_2 + \phi_l c_l = C, \tag{2.56}$$

where ϕ_2 represents the phase fraction of solid opx. In the later discussion, ϕ_1 will be used to denote the phase fraction of solid olivine. We also express bulk specific enthalpy with temperature and latent heat as

$$\overline{H} = \phi_l h_l / c_p + \phi_s h_s / c_p = \phi_l T + \phi_s T + \phi_l L / c_p = T + \phi_l L / c_p. \tag{2.57}$$

According to equations (2.55), the difference between melting point and eutectic point, which is defined as $\Delta T = T_m - T_e = T_{m0} - T_{e0}$, remains constant regardless of pressure change. At any given pressure, we proceed by dividing by this ΔT to get a dimensionless formulation as

$$\frac{\overline{H}}{\Delta T} = \frac{T}{\Delta T} + \frac{\phi_l L}{c_p \Delta T}. \tag{2.58}$$

Let $\check{L} := \frac{L}{c_p \Delta T}$, $H := \frac{\overline{H}}{\Delta T}$ and $\check{T} := \frac{T}{\Delta T}$. We write the dimensionless enthalpy temperature relationship as

$$H = \check{T} + \phi_l \check{L}. \tag{2.59}$$

2.3.3 Phase Diagram

Binary eutectic melting is usually highly-nonlinear. For simplicity, we adopt a geometric parameterization of the binary eutectic diagram that can be analytically evaluated. We implement a linear eutectic phase boundary to distinguish between different phase regions within the diagram.

The eutectic composition of opx is set as $C_e \approx 0.25$. For illustration only, we plot bulk composition against a renormalized dimensionless temperature $\check{T} - T_e/\Delta T$, where $\check{T}_m - T_e/\Delta T = 1$ and $\check{T}_e - T_e/\Delta T = 0$.

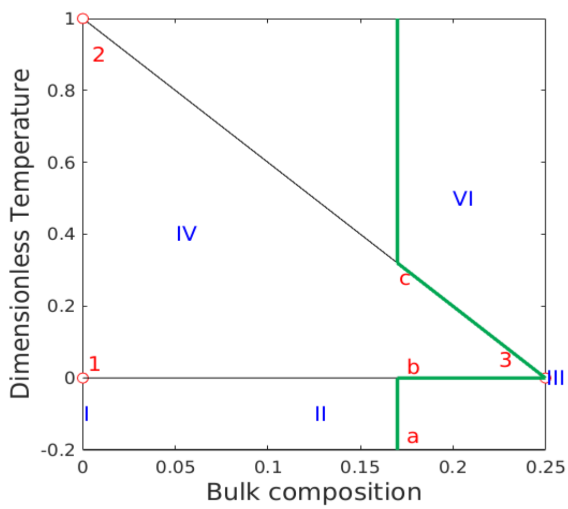


Figure 2.3: Phase diagram when the composition of opx is smaller than the eutectic composition. Region V does not appear.

We identify the following 6 regions with different phase assemblages:

I Single-phase sub-solidus : Olivine;

II Two-phase sub-solidus : Olivine + Orthopyroxene;

III Three-phase eutectic : Olivine + Orthopyroxene + melt;

IV Two-phase super-eutectic : Olivine + melt;

V Two-phase super-eutectic : Orthopyroxene + melt;

VI Single-phase super-liquidus : melt.

In a partial melting system, we assume the bulk composition of orthopyroxene (opx) remains below the eutectic composition. Consequently, we focus in the part of the phase diagram Figure 2.2, where the bulk composition is less than the eutectic composition, which is illustrated in the Figure 2.3.

Melting in the eutectic system is illustrated with the green line in the Figure 2.3. Now imagining a system containing 17% opx that starts to melting. Starting with all solid phase at Point a, the system temperature increase linearly in response to the enthalpy rise. When the temperature reaches the eutectic temperature, at Point b, both components of the system start melting with a fixed ratio of 25% opx and 75% olivine, which is the eutectic composition. Once all the solid phase opx is exhausted, the system leaves the eutectic points and solid olivine starts melting. The system climbs up along the line from Point 3 to Point c. After the remaining solid olivine has melted, the temperature of the system will continue to rise linearly in response to enthalpy input. It is worth noting that we can reverse the process to obtain a path of equilibrium crystallization.

2.3.4 Phase Split Calculation

A detailed description of the phase-splitting calculations is provided for the regions I, II, III, and IV in this chapter. Regions V and VI are not considered in this study, as we assume that the opx composition remains below the eutectic composition and that only partial melting occurs within the system. We draw a Enthalpy-Composition diagram including the effect of latent heat in the Figure 2.4. Again, we plot against the renormalized enthalpy $H - T_e/\Delta T$ for simplicity. The individual concentration and temperature are determined using equations (2.56) and (2.59). Follow the previous discussion, we denote the phase fraction of solid olivine as ϕ_1 and the phase fraction of solid opx as ϕ_2, such that $\phi_1 + \phi_2 = \phi_s$.

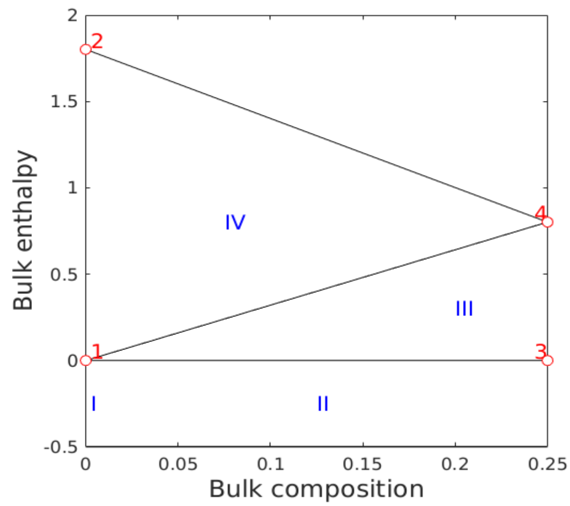


Figure 2.4: Enthalpy-Composition diagram

Region I. Pure olivine in the solid phase. The melting of pure olivine does not follow the eutectic melting path. The conditions to be in the region I are

$$C = 0, \quad H \leq \check{T}_m, \tag{2.60}$$

and we can determine the phase fractions and the temperature as

$$\phi_1 = 1, \quad \phi_2 = \phi_l = 0 \quad \text{and} \quad \check{T} = H. \tag{2.61}$$

We can determine partial derivatives of the temperature with respect to bulk enthalpy and bulk composition as

$$\frac{\partial \check{T}}{\partial H} = 1 \quad \text{and} \quad \frac{\partial \check{T}}{\partial C} = 0. \tag{2.62}$$

Region II. Sub-solidus: Olivine and opx coexist in the solid phase. In this phase region, temperature hasn't reached the eutectic melting temperature. The conditions to be in the region II are

$$C \neq 0, \quad H \leq \check{T}_e, \tag{2.63}$$

and we can determine the phase fractions and the temperature as

$$\phi_2 = C, \quad \phi_1 = 1 - \phi_2, \quad \phi_l = 0 \quad \text{and} \quad \check{T} = H. \tag{2.64}$$

We can determine partial derivatives of the temperature with respect to the bulk enthalpy and the bulk composition as

$$\frac{\partial \check{T}}{\partial H} = 1 \quad \text{and} \quad \frac{\partial \check{T}}{\partial C} = 0. \tag{2.65}$$

Region III. Two-phase eutectic: The system starts melting at eutectic temperature with a fixed liquid concentration. Both solid phase olivine and orthopyroxene and melt coexist at the same time. Temperature remains at the eutectic point until the opx is exhausted. Concentration of the opx in the melt is also fixed equal to the eutectic composition. The conditions to be in the region III are

$$C \leq C_e, \quad \check{T}_e < H \leq \text{line}_{1-4}, \tag{2.66}$$

where line_{1-4} is defined as $H(C) = \check{T}_e + \check{L}C/C_e$. We can determine the phase fractions and the current temperature as

$$\phi_l = \frac{H - \check{T}_e}{\check{L}}, \quad \phi_2 = C - \phi_l C_e, \quad \phi_1 = 1 - \phi_l - \phi_2 \quad \text{and} \quad \check{T} = \check{T}_e. \tag{2.67}$$

We can determine the partial derivatives of the temperature with respect to the bulk enthalpy and the bulk composition as

$$\frac{\partial \check{T}}{\partial H} = 0 \quad \text{and} \quad \frac{\partial \check{T}}{\partial C} = 0. \tag{2.68}$$

Region IV. Super-eutectic two phase region: All solid phase opx has been exhausted. The concentration of opx in liquid c_l is now approximated with a linear function. The conditions to be in the region IV are

$$C \leq C_e, \quad \text{line}_{1-4} < H \leq \text{line}_{2-4}, \tag{2.69}$$

where line$_{2-4}$ is defined as $H(C) = H_m - c_l/C_e$, and we can determine phase fractions as

$$\phi_2 = 0, \quad \phi_l = \frac{C}{c_l(\check{T})} \quad \text{and} \quad \phi_1 = 1 - \phi_l. \tag{2.70}$$

The corresponding temperature can be determined by solving the following equation

$$F(\check{T}) = H - \check{T} - \frac{C}{c_l(\check{T})}\check{L} = 0. \tag{2.71}$$

Let $c_l = C_e(\check{T}_m - \check{T})$, we can obtain a quadratic equation about temperature as

$$\check{T}^2 - (H + \check{T}_m)\check{T} + \check{T}_m H - CL/C_e = 0, \tag{2.72}$$

which can be solved with standard quadratic root finding formula. We omit the root that exceeding melting temperature, so

$$\begin{aligned} \check{T} &= \frac{H + \check{T}_m - \sqrt{(H + \check{T}_m)^2 - 4(\check{T}_m H - C\check{L}/C_e)}}{2} \\ &= \frac{H + \check{T}_m - \sqrt{(H - \check{T}_m)^2 + 4C\check{L}/C_e}}{2}. \end{aligned} \tag{2.73}$$

We can determine partial derivatives of the temperature with respect to the bulk enthalpy and the bulk composition as

$$\begin{aligned} \frac{\partial \check{T}}{\partial H} &= \frac{1}{2}\left(1 - \frac{H - \check{T}_m}{\sqrt{(H - \check{T}_m)^2 + 4C\check{L}/C_e}}\right), \\ \frac{\partial \check{T}}{\partial C} &= -\frac{\check{L}/C_e}{\sqrt{(H - \check{T}_m)^2 + 4C\check{L}/C_e}}. \end{aligned} \tag{2.74}$$

2.4 Nondimensionalization

Compaction and melting are the two most important aspects of mantle dynamics. We nondimensionlize the scaled formulation with a characteristic length scale related to compaction length and a scaled characteristic Darcy speed of the melt, i.e.,

$$l_0 = \left(\frac{k_0 \mu_s}{\mu_l}\right)^{1/2} \quad \text{and} \quad u_0 = \frac{k_0 \rho_r g}{\mu_l}. \tag{2.75}$$

We define a corresponding characteristic time scale as $t_0 = l_0/u_0$, and a characteristic pressure scale as $p_0 = \rho g l_0$. A set of nondimensional variables are defined as

$$\check{\mathbf{v}}_r = \frac{\mathbf{v}_r}{u_0}, \quad \check{\mathbf{v}}_s = \frac{\mathbf{v}_s}{u_0}, \quad \check{q} = \frac{q}{p_0}, \quad \check{q}_l = \frac{q_l}{p_0}. \tag{2.76}$$

The resulting dimensionless mechanics and transport equations are

$$\check{\mathbf{v}}_r + \phi^{1+\Theta}\nabla(\phi^{-1/2}\check{q}_l) = 0, \tag{2.77}$$

$$\phi^{-1/2}\nabla\cdot(\phi^{1+\Theta}\check{\mathbf{v}}_r) + \frac{1}{1-\phi}(\check{q}_l - \phi^{1/2}\check{q}) = 0, \tag{2.78}$$

$$\nabla\check{q} - \nabla\cdot\check{\tau}(\check{\mathbf{v}}_s) = -(1-\phi)\mathbf{n}, \tag{2.79}$$

$$\nabla\cdot\check{\mathbf{v}}_s - \frac{\phi^{1/2}}{1-\phi}(\check{q}_l - \phi^{1/2}\check{q}) = 0, \tag{2.80}$$

where dimensionless deviatoric stress of the solid phase is $\check{\tau} = 2(1-\phi)(\frac{1}{2}(\nabla\check{\mathbf{v}}_s + \nabla\check{\mathbf{v}}_s^T) - 1/3\nabla\cdot\check{\mathbf{v}}_s\mathbf{I})$, which maintains the same form as (2.20), and $\mathbf{n}$ is the unit direction vector pointing downwards. Now we rewrite the transport equations with dimensionless variables as

$$\begin{aligned} \frac{\partial C}{\partial \check{t}} + \nabla\cdot(\check{\mathbf{v}}_{\text{eff}}C - \frac{\phi\mathcal{D}_l}{u_0 l_0}\nabla c_l) = 0, \\ \frac{\partial H}{\partial \check{t}} + \nabla\cdot(\check{\bar{\mathbf{v}}}\check{T}) - \check{L}\nabla\cdot((1-\phi)\check{\mathbf{v}}_s) - \frac{\kappa}{u_0 l_0}\nabla^2\check{T} - \frac{\alpha_\rho l_0}{c_p}\check{T}\check{\bar{\mathbf{v}}}\cdot\mathbf{g} = 0. \end{aligned} \tag{2.81}$$

Parameters used in the flow mechanics and thermodynamic computations are listed in Table 2.1. The characteristic values are calculated as

$$l_0 = 3.16\times10^5\text{m}, \quad u_0 = 5.88\times10^{-5}\text{m/s}, \quad t_0 = 5.38\times10^6\text{s}, \quad \text{and} \quad p_0 = 9.30\times10^9\text{Pascal}, \tag{2.82}$$

which are

$$l_0 = 316\text{km}, \quad u_0 = 1.86\text{km/yr}, \quad t_0 = 0.1705\text{yrs}. \tag{2.83}$$

Quantity	Value	Units
ρ	3000	$kg \cdot m^{-3}$
ρ_r	600	$kg \cdot m^{-3}$
α_ρ	3×10^{-5}	K^{-1}
c_p	1200	$J \cdot kg^{-1} \cdot K^{-1}$
L	5×10^5	$J \cdot kg^{-1}$
μ_l	1	$\text{Pascal} \cdot s$
μ_s	10^{19}	$\text{Pascal} \cdot s$
k_0	10^{-8}	m^2
T_{e0}	1480	K
T_{m0}	2000	K
ν	6.5×10^6	$\text{Pascal} \cdot K^{-1}$

Table 2.1: Flow Mechanics and Thermodynamics Parameters

Chapter 3: Numerical Methods

In this chapter, we introduce a locally mass conserving numerical framework. A conforming mixed finite element method (MFEM) is proposed to solve the static Darcy-Stokes coupled methanics system. The proposed conforming MFEM captures the normal fluxes across element edges, ensuring compatibility with local mass conservation. The transport system is solved using a finite volume method (FVM), which inherently ensures local mass conservation.

We begin by introducing the mesh partition used for both the MFEM and the FVM schemes. Based on the mesh partition, two stable element pairs are presented, one for Darcy flow and one for Stokes flow. We then present a noval nonlinear WENO weighting scheme for the FVM. Error estimates are provided for both the MFEM and the FVM schemes.

3.1 Quadrilateral Mesh

We consider a bounded computational domain $\Omega \subset \mathbb{R}^2$, where $\partial\Omega$ denotes the Lipschitz-continuous boundary of the domain. For a given $h > 0$, let $\mathcal{T}_h$ be a *triangulation* of the closure of the domain $\overline{\Omega}$ made of quadrilaterals E, i.e.,

$$\overline{\Omega} = \cup_{E \in \mathcal{T}_h} E, \quad h_E < h, \tag{3.1}$$

where two quadrilaterals are either disjoint or share one edge or one vertex. We assume that the quadrilateral mesh is uniformly shape regular. A formal definition of uniform shape regularity of a quadrilateral mesh is given in Girault and Raviart (2011).

Definition 3.1. For any quadrilateral $E \in \mathcal{T}_h$, form the triangle $T_i, \quad i \in \{0, 1, 2, 3\}$ by excluding vertex v_i from the quadrilateral element E, and use the remaining three vertices. Let

$$\begin{aligned} h_E &= \text{diameter of Element } E, \\ \rho_E &= 2 \min_{1\le i\le 4}\{\text{diameter of largest circle inscribed in } T_i\}. \end{aligned} \tag{3.2}$$

A mesh partition $\mathcal{T}_h$ is considered to be uniformly shape regular if there exists a shape regularity parameter σ_* independent of $\mathcal{T}_h$, such that for all quadrilaterals $E \in \mathcal{T}_h$,

$$\rho_E/h_E \ge \sigma_* > 0. \tag{3.3}$$

Note that no subtriangle T_i of E has zero measure, i.e., each quadrilateral E is assumed to be strictly convex, not degenerating into a triangle, line or point.

The quadrilateral $E \in \mathcal{T}_h$ is referred to differently depending on the context. In the finite element method, a quadrilateral E is called an *element*, while in the finite volume method, it is typically referred to as a *cell*. For simplicity, we use the term *element* throughout this section.

In Figure 3.1, we show a reference element $\hat{E} = [-1,1]^2$, a local reference element $\tilde{E}$ and a general quadrilateral element E. The edges of element E are denoted by e_i, $i \in \{0,1,2,3\}$, and the vertices of the element E are denoted as v_i, $i \in \{0,1,2,3\}$, both numbered in the counterclockwise order. We begin counting from zero, first to align with the indexing convention of C++, and second to facilitate representation in terms of modulo arithmetic, which also starts from zero. The vertices can be understood by $v_i = e_i \cap e_{i+1}$, where $i \equiv i \pmod 4$. Here, $\boldsymbol{\nu}_i$ and $\boldsymbol{\tau}_i$ are used to represent the unit normal and the unit tangent vector respectively. The unit normal vector $\boldsymbol{\nu}_i$ points outwards from the element E. The corresponding unit tangent $\boldsymbol{\tau}_i$ is obtained by rotating $\boldsymbol{\nu}_i$ counterclockwise, following the right-hand rule.

We assume the general quadrilateral element E can be obtained by scaling from the local reference element $\tilde{E}$ such that $\tilde{E} = E/H$, where H is the maximal edge length of E, i.e., the length of edge e_1 in Figure 3.1. Here, $|e_1| = H$ and $|\tilde{e}_1| = 1$. There exists a non-affine bijective and smooth map $\mathbf{F}_{\hat{E}} : \hat{E} \to \tilde{E}$ mapping the vertices of $\hat{E}$ to the vertices of $\tilde{E}$. Note that the unit normal and the unit tangent vectors are invariant under the scaling mapping H.

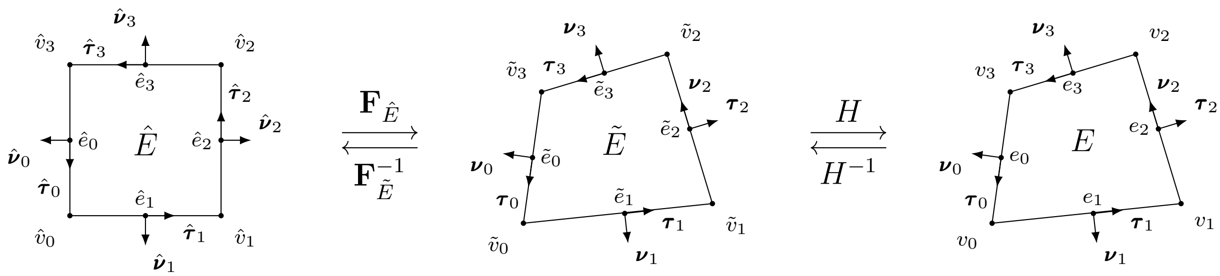


Figure 3.1: An exhibition of a reference square element, a local reference element, and a general non-degenerate quadrilateral element.

For the ease of implementation, we use a logically rectangular mesh, where each element is indexed using Cartesian coordinates $\{i, j\}$. In Figure 3.2, we illustrate a full logically rectangular, quasi-uniform quadrilateral mesh $\mathcal{T}_h$, which is generated by randomly distorting the interior vertices of a square mesh. The boundary vertices $v_i \in \partial\Omega$ remain fixed and are not subject to distortion.

3.2 Mathematical Preliminaries

In this section, we introduce some mathematical concepts that are relied upon in the remainder sections. The theory and notations are presented in a general dimension sense, but in practice, our discussion is confined to two dimension cases. In addition, for simplicity, we restrict our consideration to real-valued functions.

To begin with, we recall the definition of a Sobolev space.

Definition 3.2. Let $\Omega \subset \mathbb{R}^d$ be a domain, $1 \leq p \leq \infty$, and $m \geq 0$ be an integer. The Sobolev space of m derivatives in $L^p(\Omega)$ is

$$W_p^m(\Omega) = \{f \in L^p(\Omega) : D^\alpha f \in L^p(\Omega) \text{ for all multi-indices } \alpha \text{ such that } |\alpha| \leq m\}, \tag{3.4}$$

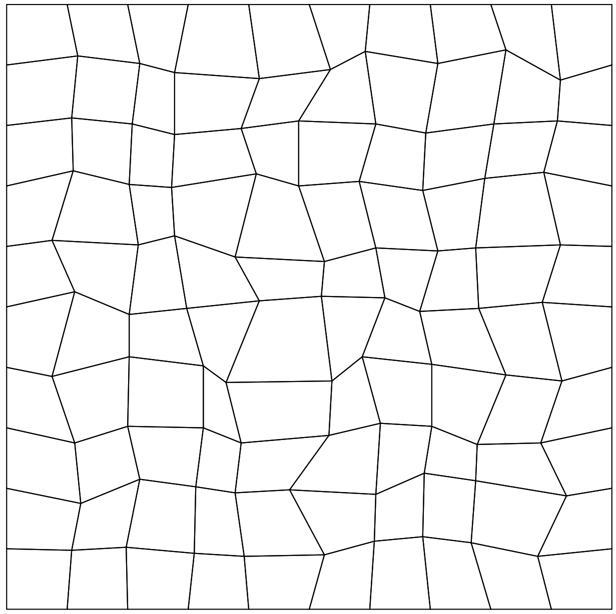


Figure 3.2: An exhibition of a full logically rectangular quasi-uniform quadrilateral mesh $\mathcal{T}_h$.

where the multi-index $\alpha = (\alpha_0, \alpha_1, \ldots, \alpha_{N-1}) \in \mathbb{N}^N$ and

$$|\alpha| = \sum_{i=0}^{N-1} \alpha_i.$$

The multi-index derivative is defined as

$$D^\alpha = \frac{\partial^{|\alpha|}}{\partial x_0^{\alpha_0} \ldots \partial x_{N-1}^{\alpha_{N-1}}}.$$

For $f \in L^p(\Omega)$, the $L^p(\Omega)$ norm is

$$\|f\|_{L^p(\Omega)} = \begin{cases} \left(\int_\Omega |f|^p \right)^{1/p}, & p < \infty, \\ \operatorname*{ess\,sup}_{x \in \Omega} |f(x)|, & p = \infty. \end{cases} \tag{3.5}$$

We proceed with the definition of the Sobolev norm and seminorm.

Definition 3.3. For $f \in W_p^m(\Omega)$, the norm $\|\cdot\|_{W_p^m}$ is

$$\|f\|_{W_p^m(\Omega)} = \left\{ \sum_{|\alpha| \le m} \|D^\alpha f\|_{L^p(\Omega)}^p \right\}^{1/p}, \quad p < \infty, \tag{3.6}$$

and

$$\|f\|_{W^m_\infty(\Omega)} = \max_{|\alpha|\le m} \|D^\alpha f\|_{L^\infty(\Omega)}, \quad p=\infty. \tag{3.7}$$

The seminorm $|\cdot|_{W^m_p}$ is

$$|f|_{W^m_p(\Omega)} = \left\{ \sum_{|\alpha|=m} \|D^\alpha f\|^p_{L^p(\Omega)} \right\}^{1/p}, \quad p<\infty. \tag{3.8}$$

In the special case when $p=2$, we denote $W^m_2(\Omega) = H^m(\Omega)$. The corresponding norm and seminorn are then written as $\|\cdot\|_{H^m(\Omega)}$ and $|\cdot|_{H^m(\Omega)}$ respectively. Recall the Trace Theorem in $H^m(\mathbb{R}^d)$.

Theorem 3.1. *(Trace Theorem)*

Let k and d be integers with $0<k<d$. The restriction map R extends to a bounded linear map from $H^m(\mathbb{R}^d)$ onto $H^{s-k/2}(\mathbb{R}^{d-k})$, provided that $s>k/2$.

Proof. See Arbogast and Bona (2025). □

Let $\mathbb{P}_r(\omega)$ denote the space of polynomials of degree up to r on $\omega \subset \mathbb{R}^d$. The dimension of $\mathbb{P}_r$ defined on $\mathbb{R}^d$ can be calculated as

$$\dim\mathbb{P}_r(\mathbb{R}^d) = \binom{r+d}{d} = \frac{(r+d)!}{r!d!}. \tag{3.9}$$

For simplicity, we momentarily restrict $d=2$. Let $\mathbb{Q}_{r,s}(\omega) :- \mathbb{P}_r(\omega) \otimes \mathbb{P}_s(\omega)$ denote the space of tensor product polynomials derived by polynomials of degree r in x and s in y on $\omega \subset \mathbb{R}^2$. The dimension of $\mathbb{Q}_{r,s}$ is

$$\dim\mathbb{Q}_{r,s}(\mathbb{R}^2) = \dim\mathbb{P}_r(\mathbb{R}) \times \dim\mathbb{P}_s(\mathbb{R}) = (r+1)\times(s+1). \tag{3.10}$$

We refer to the Bramble-Hilbert lemma for basic interpolation error estimation for polynomials of degree r.

Lemma 3.2. *(Bramble-Hilbert Lemma)*

Let $\Omega \in \mathbb{R}^d$ be a domain, and function $f \in W_p^{k+1}(\Omega)$, $k \geq 0$, $1 \leq p < \infty$. There exists a constant C such that

$$\inf_{p\in\mathbb{P}_k(E)} \|f - p\|_{W_p^m(E)} \leq Ch^{k+1-m}|f|_{W_p^{k+1}(E)}, \quad m = 0, 1, \tag{3.11}$$

where the constant C has an upper bound over all elements $E \in \mathcal{T}_h$ due to the shape regularity assumption.

Proof. See Bramble and Hilbert (1970) and Dupont and Scott (1980). □

3.3 Direct Serendipity Space

Serendipity finite elements have the same accuracy of approximation with classical tensor product elements while using fewer degrees of freedom. However, the classical serendipity spaces suffer from poor approximation on quadrilaterals. To address this issue, we follow the approach of Arbogast et al. (2022) by defining local shape functions directly on quadrilateral elements, thereby recovering the desired accuracy. In the next sections, H^1 and $H(\text{div})$ conforming direct mixed finite element spaces are constructed with the direct serendipity function spaces.

To begin with, we introduce some auxiliary distance functions. We define the linear polynomial $\lambda_i(\mathbf{x})$ giving the distance of a point at coordinate $\mathbf{x} \in \mathbb{R}^2$ to edge e_i as

$$\lambda_i(\mathbf{x}) = -(\mathbf{x} - \mathbf{x}_i^*) \cdot \boldsymbol{\nu}_i, \quad i = 0, 1, 2, 3, \tag{3.12}$$

where $\mathbf{x}_i^* \in e_i$ is any point on edge e_i. We pick $\mathbf{x}_i^* = \mathbf{x}_{v,i}$ for simplicity, where $\mathbf{x}_{v,i}$ denotes the coordinates of corner v_i. The distance function λ_i returns a positive value as long as the point is located in the interior of the element E.

We define two additional distance functions giving the distance of a point at coordinate $\mathbf{x} \in \mathbb{R}^2$ to the diagonal line d_i joining v_i with v_{i+2}, $i \in 0, 1$. Let $\boldsymbol{\nu}_{d,i}$ be the unit normal vector to the diagonal line d_i. The direction of the $\nu_{d,i}$ is defined by rotating counterclockwise of the unit vector pointing from v_i to v_{i+2}. The resulting diagonal distance function is defined as

$$\lambda_{d,i}(\mathbf{x}) = -(\mathbf{x} - \mathbf{x}^*_{d,i}) \cdot \boldsymbol{\nu}_{d,i}, \quad i = 0, 1, \tag{3.13}$$

where $\mathbf{x}^*_{d,i}$ is any point on the diagonal line d_i. Here, we use the corner point v_i for simplicity, i.e., $\mathbf{x}^*_{d,i} = \mathbf{x}_{v,i}$.

We now present a set of nodal basis functions for the first and second order direct serendipity spaces, $\mathcal{DS}_1$ and $\mathcal{DS}_2$.

3.3.1 Second-order Space $\mathcal{DS}_2$

The dimension of the second-order polynomials in two dimension is calculated by (3.9) as

$$\dim\mathbb{P}_2(\mathbb{R}^2) = \binom{2+2}{2} = \frac{(4)!}{2!2!} = 6, \tag{3.14}$$

while a quadrilateral element requires a minimum of 4 vertex dofs and 4 edge dofs. Therefore, we supplement polynomial space $\mathbb{P}_2(E)$ with two linearly independent functions, $\phi_{s,1}(\mathbf{x})$ and $\phi_{s,2}(\mathbf{x})$, spanning the supplemental function space $\mathbb{S}_r^{\mathcal{DS}}(E) = \operatorname{span}\{\phi_{s,1}, \phi_{s,2}\}, \ \ r \geq 2$. We present the second-order direct serendipity space as

$$\mathcal{DS}_2(E) = \mathbb{P}_2(E) \oplus \mathbb{S}_2^{\mathcal{DS}}(E). \tag{3.15}$$

The supplemental functions are defined as

$$\phi_{s,i}(\mathbf{x}) = \lambda_{i+1}(\mathbf{x})\lambda_{i+2}(\mathbf{x})R_{i,i+2}(\mathbf{x}), \quad i \in \{0, 1\}. \tag{3.16}$$

We use a simple definition for rational function $R_{i,i+2}$ as

$$R_{i,i+2}(\mathbf{x}) = \frac{\lambda_i(\mathbf{x}) - \lambda_{i+2}(\mathbf{x})}{\lambda_i(\mathbf{x}) + \lambda_{i+2}(\mathbf{x})}, \tag{3.17}$$

which satisfies the conditions

$$R_{i,i+2}|_{e_i} = -1, \quad R_{i,i+2}|_{e_{i+2}} = 1. \tag{3.18}$$

We continue to define

$$R_i(\mathbf{x}) = \frac{1}{2}\Big(1 - R_{i,i+2}(\mathbf{x})\Big) = \frac{\lambda_{i+2}(\mathbf{x})}{\lambda_i(\mathbf{x}) + \lambda_{i+2}(\mathbf{x})}, \quad i \in \{0,1,2,3\}, \tag{3.19}$$

where $i \equiv i \pmod 4$ as usual, and $R_{i,i+2} = -R_{i+2,i}$ according to the equation (3.17). Here, $R_i|_{e_i} = 1$, and $R_i|_{e_{i+2}} = 0$ by definition. In Figure 3.3, we draw R_0 on a general quadrilateral element, which evaluates to 1 on edge e_0, and evaluates to 0 on the opposite edge e_2.

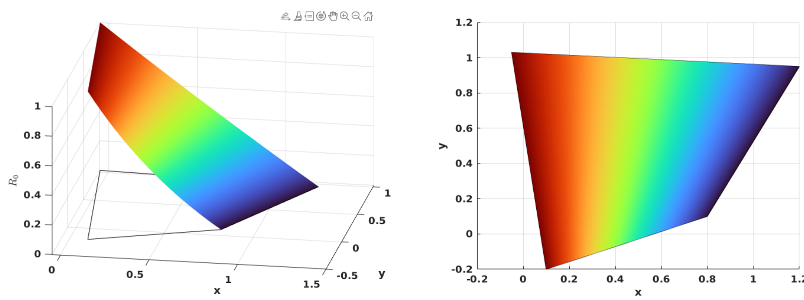


Figure 3.3: Rational function R_0.

The edge nodal basis functions for $\mathcal{DS}_2$ are constructed as

$$\varphi^{(2)}_{e,i}(\mathbf{x}) = \frac{\lambda_{i+1}(\mathbf{x})\lambda_{i+3}(\mathbf{x})R_i(\mathbf{x})}{\lambda_{i+1}(\mathbf{x}_{e,i})\lambda_{i+3}(\mathbf{x}_{e,i})R_i(\mathbf{x}_{e,i})}, \tag{3.20}$$

where $\mathbf{x}_{e,i}$ are the coordinates of the middle point of the edge e_i, again $i \equiv i \pmod 4$. We use superscript (2) to denote the order of the function space. We can restrict the edge basis function to edges as

$$\varphi^{(2)}_{e,i}|_{e,j} = \begin{cases} \dfrac{\lambda_{i+1}(\mathbf{x})\lambda_{i+3}(\mathbf{x})}{\lambda_{i+1}(\mathbf{x}_{e,i})\lambda_{i+3}(\mathbf{x}_{e,i})}, & i = j, \\ 0, & i \neq j. \end{cases} \tag{3.21}$$

In Figure 3.4, we present edge basis function $\varphi^{(2)}_{e,0}$ on a general quadrilateral element, where $\varphi^{(2)}_{e_0}$ evaluates 0 on edges $\{e_i, \quad i \neq 0\}$, and evaluates to a quadratic function on edge e_0.

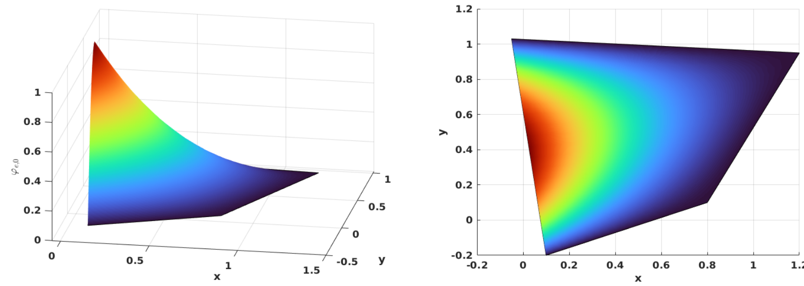


Figure 3.4: Edge nodal basis function $\varphi_{e,0}^{(2)}$ in $\mathcal{DS}_2$.

The vertex nodal basis functions for $\mathcal{DS}_2$ are constructed as

$$\varphi_{v,i}^{(2)}(\mathbf{x}) = \frac{\lambda_{i+2}(\mathbf{x})\lambda_{i+3}(\mathbf{x}) - \sum_{k=i}^{i+1} \lambda_{i+2}(\mathbf{x}_{e,i})\lambda_{i+3}(\mathbf{x}_{e,i})\varphi_{e,i}^{2}(\mathbf{x})}{\lambda_{i+2}(\mathbf{x}_{v,i})\lambda_{i+3}(\mathbf{x}_{v,i}) - \sum_{k=i}^{i+1} \lambda_{i+2}(\mathbf{x}_{e,i})\lambda_{i+3}(\mathbf{x}_{e,i})\varphi_{e,i}^{(2)}(\mathbf{x}_{v,i})}, \tag{3.22}$$

where $\mathbf{x}_{v,i}$ is the coordinates of the vertex node v_i, and $i \equiv i \pmod 4$ as well. At the nodal points

$$\varphi_{v,i}^{(2)}(\mathbf{x}) = \begin{cases} 1, & \mathbf{x} = \mathbf{x}_{v,i}, \\ 0, & \mathbf{x} = \mathbf{x}_{v,j} \text{ or } \mathbf{x}_{e,k}, \quad j \neq i, \ \forall k. \end{cases} \tag{3.23}$$

In Figure 3.5, we show $\varphi_{v,0}^{(2)}$ on a general quadrilateral.

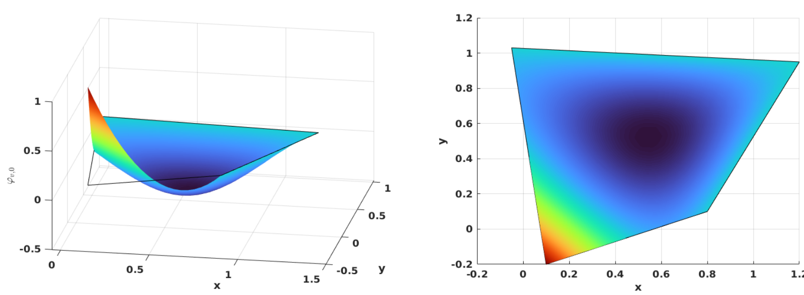


Figure 3.5: Vertex nodal basis function $\varphi_{v,0}^{2}$ in $\mathcal{DS}_2$.

We conclude that unisolvence follows directly from the constructed nodal basis functions.

3.3.2 First-order Space $\mathcal{DS}_1$

The dimension of the first-order polynomials in two dimension is also calculated by equation (3.9) as

$$\dim \mathbb{P}_1(\mathbb{R}^2) = \binom{1+2}{2} = \frac{(3)!}{2!1!} = 3, \tag{3.24}$$

while a quadrilateral element requires a minimum of 4 vertex dofs. Therefore, we supplement polynomial space $\mathbb{P}_1(E)$ with one additional function. The supplemental function for $\mathcal{DS}_1$ differs from that for $\mathcal{DS}_2$. We present the first order direct serendipity space as

$$\mathcal{DS}_1(E) = \mathbb{P}_1(E) \oplus \text{span}\{R\}, \tag{3.25}$$

where $R(\mathbf{x}_{v,0}) = R(\mathbf{x}_{v,2}) = -R(\mathbf{x}_{v,1}) = -R(\mathbf{x}_{v,3}) = 1$. The construction of the rational function R is not unique. We present one way to construct $R(\mathbf{x})$ using edge nodal basis function (3.20) in $\mathcal{DS}_2$ as

$$R(\mathbf{x}) = r(\mathbf{x}) - \sum_{i=0}^{3} r(\mathbf{x}_{e,i})\varphi_{e,i}^{(2)}(\mathbf{x}), \tag{3.26}$$

where the auxiliary function $r(\mathbf{x})$ is defined as

$$r(\mathbf{x}) = \frac{\lambda_2(\mathbf{x})\lambda_3(\mathbf{x})}{\lambda_2(\mathbf{x}_{v,0})\lambda_3(\mathbf{x}_{v,0})} - \frac{\lambda_0(\mathbf{x})\lambda_3(\mathbf{x})}{\lambda_0(\mathbf{x}_{v,1})\lambda_3(\mathbf{x}_{v,1})} + \frac{\lambda_0(\mathbf{x})\lambda_1(\mathbf{x})}{\lambda_0(\mathbf{x}_{v,2})\lambda_1(\mathbf{x}_{v,2})} - \frac{\lambda_1(\mathbf{x})\lambda_2(\mathbf{x})}{\lambda_1(\mathbf{x}_{v,3})\lambda_2(\mathbf{x}_{v,3})}. \tag{3.27}$$

In Figure 3.6, we plot the rational function $R(\mathbf{x})$ on a general quadrilateral.

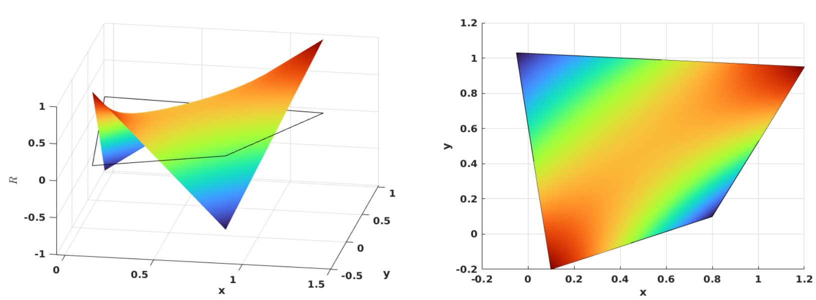


Figure 3.6: Supplemental rational function R in $\mathcal{DS}_1$.

We then define vertex nodal basis function accordingly as

$$\varphi_{v,i}^{(1)}(\mathbf{x}) = \frac{\lambda_{d,m}(\mathbf{x}) - \frac{1}{2}\lambda_{d,m}(\mathbf{x}_{v,i+2})(1 + (-1)^i R(\mathbf{x}))}{\lambda_{d,m}(\mathbf{x}_{v,i}) - \lambda_{d,m}(\mathbf{x}_{v,i+2})}, \tag{3.28}$$

where $m \equiv i+1 \pmod 2$, and $i \equiv i \pmod 4$. In Figure 3.7, we show the vertex nodal basis function $\varphi_{v,0}^{(1)}$ in $\mathcal{DS}_1$ on a general quadrilateral, which evaluates 1 at vertex v_0 and 0 at other vertices.

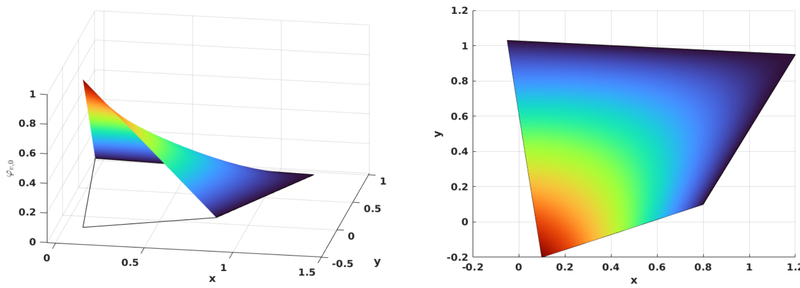


Figure 3.7: Vertex nodal basis function in $\mathcal{DS}_1$.

3.3.3 An Interpolation Operator

For completeness, we describe an interpolation operator mapping onto $\mathcal{DS}_r(K_i)$, $r = 1, 2$, following Arbogast et al. (2022). For each node a_i, the associated K_i has different meanings depending on the type of the nodal basis functions. If the basis function is associated with an edge node, we set $K_i = e_i$, where e_i is the corresponding edge. For the basis function associated with a vertex node, we set K_i to be any edge e containing node a_i, with the restriction that if $a_i \in \partial\Omega$, then $e \in \partial\Omega$.

An interpolation operator $\mathcal{I}_h^r : W_p^l(\Omega) \to \mathcal{DS}_r$ is defined as

$$\mathcal{I}_h^r v(\mathbf{x}) = \sum_{i=1}^{N_r} \varphi_i(\mathbf{x}) \int_{K_i} \psi_i(\mathbf{y}) v(\mathbf{y})\, d\mathbf{y} \in \mathcal{DS}_r, \tag{3.29}$$

where $1 \le p \le \infty$ and $l > 1/p$ (but $l \ge 1$ if $p = 1$), and $\psi(\mathbf{x})$ is the dual nodal basis such that

$$\int_{K_i} \psi_i(\mathbf{x})\varphi_j(\mathbf{x})\, d\mathbf{x} = \delta_{i,j}, \quad i, j \in \{0, 1, 2, \ldots, N_r\}, \tag{3.30}$$

where N_r is the number of nodal basis functions. We present the following two lemmas regarding the constructed interpolation operator.

Lemma 3.3. *(Boundedness of the interpolation operator.)*

Let $\mathfrak{T}_h$ be uniformly shape regular according to Definition 3.3 with shape regularity parameter σ_*. Let $v \in W^l_p(\Omega)$, with $1 \le p \le \infty$ and $l > q/p$. Moreover, assume that $R_{i,i+2}$ is a uniformly differentiable functions of the vertices of E up to order m. Then for $r = 1, 2$, $E \in \mathfrak{T}_h$, $1 \le q \le \infty$, and any non negative integer m,

$$\|\mathfrak{I}^r_h v\|_{W^m_q(E)} \le C(\sigma_*, m, q) \sum_{k=0}^{l} h_E^{k-m+\frac{2}{q}-\frac{2}{p}} |v|_{W^k_p(E^*)}, \tag{3.31}$$

where $E^* = \cup_{F\in\mathfrak{T}_h, F\cap E\neq\emptyset} F$ and $|\cdot|_{W^k_p}$ is the seminorm of k-th order derivatives.

Proof. See Arbogast et al. (2022). □

Combining Lemma 3.3 and the Bramble-Hilbert lemma 3.2, we obtain local and global error estimation.

Lemma 3.4. *(Approximability of the interpolation operator.)*

Under the assumption of Lemma 3.3, there exists a constant $C = C(r, \sigma_*) > 0$ such that for all functions $v \in W_{l,p}(E^*)$, with $1 \le p \le \infty$ and $l > 1/p$ or ($l \ge 1$ if $p = 1$),

$$\|v - \mathfrak{I}^r_h v\|_{W^m_p(E)} \le C h_E^{l-m} |v|_{W^l_p(E^*)}, \quad 0 \le m \le \min(l, r+1). \tag{3.32}$$

Moreover, there exists a constant $C = C(r, \sigma_*) > 0$, independent of $h = \max_{E\in\mathfrak{T}} h_E$, such that for all functions $v \in W^l_p(\Omega)$, $p < \infty$

$$\left(\sum_{E\in\mathfrak{T}_h} \|v - \mathfrak{I}^r_h v\|^p_{W^m_p(E)} \right)^{1/p} \le C h^{l-m} |v|_{W^l_p(\Omega)}, \quad 0 \le m \le \min(l, r+1). \tag{3.33}$$

Proof. See Arbogast et al. (2022). □

3.4 Mixed Finite Elements

In this section, we describe our mixed finite element approximation for the scaled coupled system (2.77)–(2.80). We begin by presenting the mixed variational formulations of the Stokes and Darcy model problems separately. Subsequently, we introduce H^1- and $H(\text{div})$-conforming mixed finite element spaces. We describe the construction of mixed finite elements from the aforementioned direct serendipity space. Recall Ciarlet's definition Ciarlet (2002) of a finite element.

Definition 3.4. A triple $(E, X(E), \{\phi_1, \ldots, \phi_n\})$ defines a *finite element*, where

- The element $E \in \mathcal{T}_h$;
- The space of shape functions is $X(E)$ with basis $\{\varphi_1, \ldots, \varphi_n\}$,where the dimension of the shape function space is $\dim(X(E)) = n$;
- A set of continuous and linear functionals is $\{\phi_1, \ldots, \phi_n\}$, defined on a subset $\mathcal{X}(E)$ of the energy space containing the finite element space $X(E)$ as
$$\phi_j : \mathcal{X}(E) \to \mathbb{R}, \tag{3.34}$$
such that the unisolvence condition is achieved as
$$\langle \phi_j, \varphi_i \rangle = \delta_{ij}, \quad i, j \in \{1, 2, 3, \ldots, n\}. \tag{3.35}$$

In later subsections, we give an explicit construction of the nodal basis functions and degrees of freedom (values of the $\phi_j,\ j = 1, 2, \ldots, n$) that is associated to them, thereby completing the definition of the finite element.

3.4.1 A Darcy Model Problem

Consider a model problem for Darcy flow equipped with homogeneous boundary condition as
$$\begin{aligned} \mathbf{u} + \nabla p &= \mathbf{f}, \text{ in } \Omega, \\ \nabla \cdot \mathbf{u} &= 0, \text{ in } \Omega, \\ \mathbf{u} \cdot \boldsymbol{\nu} &= 0, \text{ on } \partial\Omega. \end{aligned} \tag{3.36}$$

We seek the displacement and pressure solution pair $(\mathbf{u}, p)$ in the identical test and trial spaces

$$\begin{aligned}
\mathbb{V}_D &= \{\mathbf{v} \in (H(\text{div};\Omega))^2 : \mathbf{v}\cdot\boldsymbol{\nu} = \mathbf{0} \ \text{ on } \partial\Omega\}, \\
\mathbb{U}_D &= \{\mathbf{u} \in (H(\text{div};\Omega))^2 : \mathbf{u}\cdot\boldsymbol{\nu} = \mathbf{0} \ \text{ on } \partial\Omega\} = \mathbb{V}_D, \\
\mathbb{W}_D &= \left\{p \in L^2(\Omega) : \int_\Omega p\, d\mathbf{x} = 0\right\} = L^2(\Omega)/\mathbb{R}.
\end{aligned} \tag{3.37}$$

The standard mixed weak form reads as follows:

Find $\mathbf{u} \in \mathbb{U}_D$ and $p \in \mathbb{W}_D$ such that

$$\begin{aligned}
(\mathbf{u}, \mathbf{v}) - (p, \nabla\cdot\mathbf{v}) &= (\mathbf{f}, \mathbf{v}), \quad \forall \mathbf{v} \in \mathbb{V}_D, \\
(\nabla\cdot\mathbf{u}, \omega) &= 0, \qquad\quad \forall \omega \in \mathbb{W}_D.
\end{aligned} \tag{3.38}$$

We introduce two finite-dimensional subspaces $\mathbb{V}_{D,h} \subset \mathbb{V}_D$ and $\mathbb{W}_{D,h} \subset \mathbb{W}_D$, and consider the discretized problem:

Find $(\mathbf{u}_h, p_h) \in \mathbb{V}_{D,h} \times \mathbb{W}_{D,h}$ such that

$$\begin{aligned}
(\mathbf{u}_h, \mathbf{v}_h) - (p_h, \nabla\cdot\mathbf{v}_h) &= (\mathbf{f}, \mathbf{v}_h), \quad \forall \mathbf{v}_h \in \mathbb{V}_{D,h}, \\
(\nabla\cdot\mathbf{u}_h, \omega_h) &= 0, \qquad\quad \forall \omega_h \in \mathbb{W}_{D,h}.
\end{aligned} \tag{3.39}$$

In the next subsection, we present a construction of the discretized space pair $\mathbb{V}_{D,h} \times \mathbb{W}_{D,h}$ for this Darcy model problem.

3.4.2 A Reduced $H(\mathbf{div})$ Conforming Space $\mathbb{V}_1^0$

We have some choice to discretize the space $\mathbb{V}_D$. For example, we can choose the classical Ravier-Thomas element Raviart and Thomas (1977). The serendipity space is the natural precursor of the Brezzi-Douglas-Marini element Brezzi et al. (1985) related by a de Rham complex. We generate a reduced first-order $H(\text{div})$ approximation from the direct serendipity space, where $\nabla\cdot\mathbf{u}$ is approximated one less order than approximation of $\mathbf{u}$. We can also recover the Arbogast-Correa element Arbogast and Correa (2016) if we use the mapped serendipity space, as defined in that paper.

We write the de Rham exact sequence for the related energy spaces as

$$\begin{array}{ccccccc} \mathbb{R} \xhookrightarrow{\mathrm{i}} H^1 & \xrightarrow{\mathrm{curl}} & H(\mathrm{div}) & \xrightarrow{\mathrm{div}} & L^2 & \xrightarrow{0} 0 \\ \cup & & \cup & & \cup & \\ \mathcal{DS}_2 & \xrightarrow{\mathrm{curl}} & \mathbf{V}_1^0 & \xrightarrow{\mathrm{div}} & \mathbb{P}_0 & . \end{array} \tag{3.40}$$

We define the reduced first-order $H(\mathrm{div})$ conforming space as the first-order polynomial space supplemented with the curl of the second-order direct serendipity supplemental space of $\mathcal{DS}_2$, i.e.,

$$\mathbb{V}_1^0(E) = (\mathbb{P}_1(E))^2 \oplus \mathrm{curl}\, \mathbb{S}_2^{\mathcal{DS}}(E). \tag{3.41}$$

Here, $\mathbb{V}_1^0(E)$ can be given a global basis with the normal components of the shape functions merged continuously on the shared edge of two elements.

The dimension of this reduced first-order $H(\mathrm{div})$ conforming space is

$$\dim(\mathbb{V}_1^0(E)) = 2 \times 3 + 2 = 8. \tag{3.42}$$

On each edge of a quadrilateral element E, two independent basis functions are defined: one with vanishing average normal flux across the edge, and the other with non-vanishing average flux.

We state the degrees of freedom as normal fluxes on each edge as

$$DOF_{e_i}(\lambda_1) = \langle \mathbf{u}_h \cdot \boldsymbol{\nu}_i, \lambda_1 \rangle_{e_i}, \quad \lambda_1 \in \mathbb{P}_1, \tag{3.43}$$

where λ_1 is a linear function restricted to edge e_i, and $\dim(\mathbb{P}_1) = 2$. We then present a construction of two independent shape functions on each edge: $\boldsymbol{\varphi}_{l,i}$ and $\boldsymbol{\varphi}_{c,i}$, such that

$$\boldsymbol{\varphi}_{c,i} \cdot \boldsymbol{\nu}_j|_{e_j} = \begin{cases} 0, & j \neq i, \\ 1, & j = i. \end{cases} \tag{3.44}$$

$$\boldsymbol{\varphi}_{l,i} \cdot \boldsymbol{\nu}_j|_{e_j} = \begin{cases} 0, & j \neq i, \\ l_i(\mathbf{x}), & j = i, \end{cases} \tag{3.45}$$

where l_i is a non vanishing linear polynomial defined on edge e_i orthogonal to one, i.e., $\int_{e_i} l_i\, ds = 0$.

To begin with, we take the curl of the auxiliary rational function R_i (3.17) as

$$\operatorname{curl} R_i(\mathbf{x}) = \frac{\boldsymbol{\tau}_{i+2}(\mathbf{x})\lambda_i(\mathbf{x}) - \boldsymbol{\tau}_i(\mathbf{x})\lambda_{i+2}(\mathbf{x})}{(\lambda_i(\mathbf{x}) + \lambda_{i+2}(\mathbf{x}))^2}, \tag{3.46}$$

where $i \equiv i \pmod 4$. We present one way to construct the "linear" shape function $\boldsymbol{\varphi}_{l,i}$ in two steps as

$$\tilde{\boldsymbol{\varphi}}_{l,i} = \operatorname{curl}(\lambda_{i+1}\lambda_{i+3}R_i) = (\boldsymbol{\tau}_{i+1}\lambda_{i+3} + \lambda_{i+1}\boldsymbol{\tau}_{i+3})R_i + \lambda_{i+1}\lambda_{i+3}\operatorname{curl}R_i. \tag{3.47}$$

We then normalize it such that l_i varies from -1 to 1 when restricted to edge e_i as

$$\boldsymbol{\varphi}_{l,i} = \frac{\tilde{\boldsymbol{\varphi}}_{l,i}(\mathbf{x})|e_i|}{4\lambda_{i+1}(\mathbf{x}_{e,i})\lambda_{i+3}(\mathbf{x}_{e,i})R_i(\mathbf{x}_{e,i})}, \tag{3.48}$$

where the index $i \equiv i \pmod 4$, and the length of edge e_i is $|e_i| = |\mathbf{x}_{v,i} - \mathbf{x}_{v,i+3}|$. In Figure 3.8, we draw $\boldsymbol{\varphi}_{l,2}$ on the left element and $\boldsymbol{\varphi}_{l,0}$ on the right element sharing an edge, and show the normal flux, but not the tangential flux joining continuously as a linear function on the shared edge.

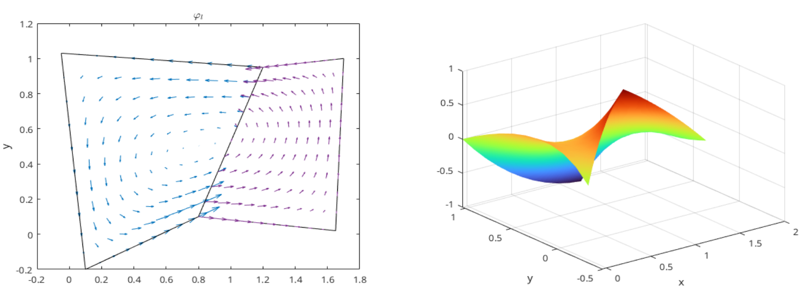


Figure 3.8: "Linear" shape function $\boldsymbol{\varphi}_l$.

The construction of the constant basis functions is a bit more complicated. We follow the process by Arbogast et al. (2022). We start by defining a linear nodal function $\phi^*_{v,i} \in \mathcal{DS}_2(E)$ such that $\phi^*_{v,i}(\mathbf{x}_{v,k}) = \delta_{ik}$. Its restriction to each edge of E is a linear function as

$$\phi^*_{v,i}(\mathbf{x}) = \varphi^2_{v,i}(\mathbf{x}) + \frac{1}{2}\big[\varphi^2_{e,i} + \varphi^2_{e,i+1}\big], \tag{3.49}$$

where $i \equiv i \pmod 4$. We then define $\boldsymbol{\varphi}_{v,i}^*(\mathbf{x}) = \operatorname{curl} \phi_{v,i}^*$ so that

$$\boldsymbol{\varphi}_{v,i}^* \cdot \boldsymbol{\nu}_j|_{e,j} = \begin{cases} 1/|e_i|, & j = i, \\ -1/|e_i|, & j = i+1, \\ 0, & j = i+2, i+3. \end{cases} \tag{3.50}$$

The "constant" basis function on the edge can now be defined as

$$\boldsymbol{\varphi}_{c,i} = \frac{(\mathbf{x}_{v,i} - \mathbf{x}_{v,i+2}) \cdot \boldsymbol{\nu}_{i+1}|e_{i+1}|\boldsymbol{\varphi}_{v,i}^* + \mathbf{x} - \mathbf{x}_{v,i+2}}{(\mathbf{x}_{v,i} - \mathbf{x}_{v,i+2}) \cdot \boldsymbol{\nu}_{i+1}|e_{i+1}|/|e_i| + (\mathbf{x}_{v,i} - \mathbf{x}_{v,i+2}) \cdot \boldsymbol{\nu}_i}, \tag{3.51}$$

where $i \equiv i \pmod 4$. The "constant" shape function $\boldsymbol{\varphi}_{c,i}$ has a normal flux of constant value 1 on edge e_i and evaluates to 0 normal flux on all the other edges. In Figure 3.9, we draw $\boldsymbol{\varphi}_{c,2}$ on the left element and $\boldsymbol{\varphi}_{c,0}$ on the right element sharing an edge, and show that the normal flux joins continuously as a constant 1 on the shared edge.

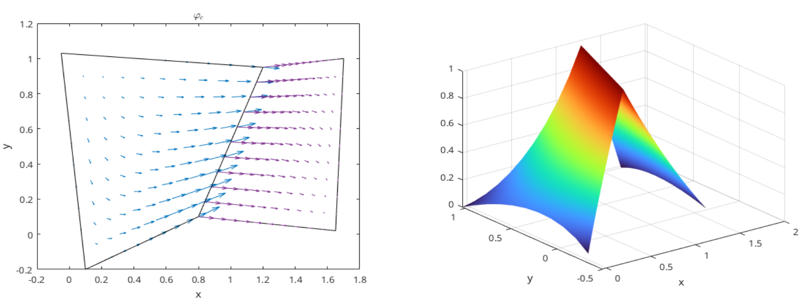


Figure 3.9: "Constant" shape function $\boldsymbol{\varphi}_c$.

3.4.3 Stability and Approximation Properties for $\mathbb{V}_1^0$

The accuracy and stability of $\mathbb{V}_1^0$ has been well studied and tested in the work by Arbogast et al. (2022). We state the result directly for completeness. They define a projection operation π that preserves element average divergence and average edge normal fluxes,

$$\pi : H(\mathrm{div}; \Omega) \cap (L^{2+\epsilon}(\Omega))^2 \to \mathbb{V}_1^0, \tag{3.52}$$

where $\epsilon > 0$. This global projection operator π is a combination of local operators π_E defined on each element E according to the degrees of freedom. The operator π

also satisfies the commuting diagram property

$$\mathcal{P}_{W_0} \nabla \cdot \mathbf{v} = \nabla \cdot \pi \mathbf{v}, \tag{3.53}$$

where $\mathcal{P}_{W_0}$ is the L^2-orthogonal projection operator onto $W_0 = \nabla \cdot \mathbb{V}_1^0$, which is a space of piecewise constant values.

Lemma 3.5. *Stability and approximation properties for $\mathbb{V}_1^0$.*

Under the assumptions of Lemma 3.3 and Lemma 3.2, there exists a constant $C > 0$, independent of $\mathcal{T}_h$ and $h > 0$, such that

$$\begin{aligned} \|\mathbf{u} - \pi_D \mathbf{u}\|_{0,\Omega} &\leq C \|\mathbf{u}\|_{k,\Omega} h^k, && k = 1, 2, \\ \|\nabla \cdot (\mathbf{u} - \pi_D \mathbf{u})\|_{0,\Omega} &\leq C \|\nabla \cdot \mathbf{u}\|_{k,\Omega} h^k, && k = 0, 1, \\ \|p - \mathcal{P}_{W_s} p\|_{0,\Omega} &\leq C \|p\|_{k,\Omega} h^k, && k = 0, 1. \end{aligned} \tag{3.54}$$

Moreover, the discrete inf-sup condition holds

$$\sup_{\mathbf{v}_h \in \mathbb{V}_1^0} \frac{(\omega_h, \nabla \cdot \mathbf{v}_h)}{\|\mathbf{v}_h\|_{H(div)}} \geq \gamma \|\omega_h\|_{0,\Omega}, \quad \forall \omega_h \in W_0, \tag{3.55}$$

for some $\gamma > 0$ independent of $\mathcal{T}_h$ and $h > 0$.

Proof. See Arbogast et al. (2022). □

3.4.4 A Stokes Model Problem

Consider a model problem of an incompressible isotropic equation equipped with homogeneous boundary condition as

$$\begin{aligned} -\Delta \mathbf{u} + \nabla p &= \mathbf{f} \quad \text{in } \Omega, \\ \nabla \cdot \mathbf{u} &= 0 \quad \text{in } \Omega, \\ \mathbf{u} &= \mathbf{0} \quad \text{on } \partial\Omega. \end{aligned} \tag{3.56}$$

We choose energy spaces for the displacement-pressure pair $(\mathbf{u}, p)$ in the identical test and trial spaces as

$$\begin{aligned} \mathbb{V}_S &= \{\mathbf{v} \in (H^1(\Omega))^2 : \mathbf{v} = \mathbf{0} \text{ on } \partial\Omega\} = (H_0^1(\Omega))^2, \\ \mathbb{U}_S &= \{\mathbf{u} \in (H^1(\Omega))^2 : \mathbf{u} = \mathbf{0} \text{ on } \partial\Omega\} = \mathbb{V}_S, \\ \mathbb{W}_S &= \left\{ p \in L^2(\Omega) : \int_\Omega p \, d\mathbf{x} = 0 \right\} = L^2(\Omega)/\mathbb{R}. \end{aligned} \tag{3.57}$$

We consider the standard mixed variational form as

Find $\mathbf{u} \in \mathbb{U}_S$ and $p \in \mathbb{W}_S$ such that

$$\begin{aligned} (\nabla \mathbf{u}, \nabla \mathbf{v}) - (p, \nabla \cdot \mathbf{v}) = (\mathbf{f}, \mathbf{v}), \quad & \forall \mathbf{v} \in \mathbb{V}_S, \\ (\nabla \cdot \mathbf{u}, \omega) = 0, \qquad\quad & \forall \omega \in \mathbb{W}_S. \end{aligned} \tag{3.58}$$

We introduce two finite-dimensional subspaces $\mathbb{V}_{S,h} \subset \mathbb{V}$ and $\mathbb{W}_{S,h} \subset \mathbb{W}$, and consider the discretized problem:

Find $(\mathbf{u}_h, p_h) \in \mathbb{V}_{S,h} \times \mathbb{W}_{S,h}$ such that

$$\begin{aligned} (\nabla \mathbf{u}_h, \nabla \mathbf{v}_h) - (p_h, \nabla \cdot \mathbf{v}_h) = (\mathbf{f}, \mathbf{v}_h), \quad & \forall \mathbf{v}_h \in \mathbb{V}_{S,h}, \\ (\nabla \cdot \mathbf{u}_h, \omega_h) = 0, \qquad\quad & \forall \omega_h \in \mathbb{W}_{S,h}. \end{aligned} \tag{3.59}$$

Now we construct a pair of discretized space $\mathbb{V}_{S,h} \times \mathbb{W}_{S,h}$ for this Stokes model problem in the next subsection.

3.4.5 Modified Bernardi-Raugel Element

Stable mixed finite element pairs for the Stokes problem have been widely and thoroughly studied on triangles and rectangles. One notable construction on rectangles was discussed in Bernardi and Raugel (1985) and Fortin (1981), where polynomial spaces are supplemented with bubble functions. Consider the standard type of Bernardi-Raugel (BR) element using the space of tensor product polynomials

$$(\mathbb{Q}_{1,1}(E))^2 \oplus \operatorname{span}\{\mathbf{b}_0, \mathbf{b}_1, \mathbf{b}_2, \mathbf{b}_3\} \subset (\mathbb{Q}_{2,2}(E))^2, \tag{3.60}$$

where $\mathbf{b}_i$ is a quadratic bubble function defined for each edge e_i. A similar but more general type of element was discussed by Arbogast and Wheeler (2005).

We now modify the original BR element to extend it to a new finite element suited to quadrilaterals using the direct serendipity space as

$$\mathbb{V}_1^{\mathcal{BR}}(E) = (\mathcal{DS}_1(E))^2 \oplus \operatorname{span}\{\mathbf{b}_0, \mathbf{b}_1, \mathbf{b}_2, \mathbf{b}_3\} \subset (\mathcal{DS}_2(E))^2, \tag{3.61}$$

where $\dim \mathbb{V}_1^{\mathcal{BR}}(E) = 12$. One natural choice for the supplemental bubble function $\mathbf{b}_i$ is the previously defined edge nodal basis function $\varphi_{e,i}^{(2)} \in \mathcal{DS}_2$ times the unit normal, i.e.,

$$\mathbf{b}_i(\mathbf{x}) = \varphi_{e,i}^{(2)}(\mathbf{x})\boldsymbol{\nu}_i, \tag{3.62}$$

which is quadratic when restricted to edge e_i according to (3.21). The quadratic bubble is guaranteed to be conforming since it evaluates to 0 on two vertices and 1 at the middle point of the edge due to normalization, which is uniquely determined by these three points. The integral of this quadratic bubble can be accurately computed by a three-point Gaussian quadrature rule.

We state the degrees of freedom of this type of element as follows.

Definition 3.5. For an element $E \in \mathcal{T}_h$, $\mathbf{u}_h$ is uniquely defined by the following 12 degrees of freedom:

(i) for corner point v_i and Cartesian direction unit normal vectors $\{\mathbf{i}, \mathbf{j}\}$,

$$DOF^{(\mathrm{i})}_{(v_i,\mathbf{i})}(\mathbf{u}_h) = \mathbf{u}_h(v_i) \cdot \mathbf{i}, \quad DOF^{(\mathrm{i})}_{(v_i,\mathbf{j})}(\mathbf{u}_h) = \mathbf{u}_h(v_i) \cdot \mathbf{j};$$

(ii) for edge e_i,

$$DOF^{(ii)}_{e_i} = \langle \mathbf{u}_h \cdot \boldsymbol{\nu}_i, 1 \rangle.$$

We give an explicit construction of the shape functions. We choose the nodal basis function on vertex v_i in two Cartesian directions as

$$N_{v_i,x} = \begin{bmatrix} \varphi_{v,i}^{(1)}(\mathbf{x}) \\ 0 \end{bmatrix}, \quad N_{v_i,y} = \begin{bmatrix} 0 \\ \varphi_{v,i}^{(1)}(\mathbf{x}) \end{bmatrix}, \tag{3.63}$$

where $\varphi_{v,i}^{(1)}$ is the vertex nodal shape function in $\mathcal{DS}_1(E)$. We choose the edge basis function on edge e_i as (3.62), i.e.,

$$N_{e,i} = \mathbf{b}_i(\mathbf{x}) = \varphi_{e,i}^{(2)}\boldsymbol{\nu}_i \tag{3.64}$$

Note that bubble functions are quadratic when restricted to edge e_i, while nodal basis functions are linear when restricted to edge e_i. Therefore the bubble functions cannot be in the same space as the nodal basis functions. We conclude that this element is well-defined, i.e., the degrees of freedom uniquely determine the function.

We pair it with the space

$$\mathbb{W}_0 = \{\omega \in L^2(\Omega);\quad \omega|_E \in P_0\}/\mathbb{R}, \tag{3.65}$$

which has piecewise constant values for each element E. We provide stability and approximation analysis for this space pair $\mathbb{V}_1^{\mathcal{BR}} \times \mathbb{W}_0$ in the next subsection.

3.4.6 Stability and Approximation Properties for $\mathbb{V}_1^{\mathcal{BR}} \times \mathbb{W}_0$

Before analyzing the stability and approximation properties for the new modified BR element, we state one useful lemma.

Lemma 3.6. *Let E be a nondengenerate convex quadrilateral element that is shape regular as in Definition 3.1. For each integer $m \geq 0$ and real number $p \in [1, \infty]$ there exist positive constants C_1 and C_2, possibly depending on σ^* but otherwise independent of the geometry of E, such that the following upper bounds hold for all $v \in W_p^m(E)$:*

$$\begin{cases} |v|_{W_p^0(E)} \leq C_1 h_E^{2/p} |\tilde{v}|_{W_p^0(\tilde{E})}, \\ |v|_{W_p^1(E)} \leq C_2(\sigma^*, p) h^{2/p-1} |\tilde{v}|_{W_p^1(\tilde{E})}, \end{cases} \tag{3.66}$$

where $\tilde{E}$ represents the local reference quadrilateral element $\tilde{E}$ illustrated in Figure 3.1, and $\tilde{v}$ is the function defined on the local reference element $\tilde{E}$.

Proof. See Girault and Raviart (2011). □

We reuse the interpolation operator $\mathcal{I}_h^1$ defined in equation (3.29). Define the projection operator $\pi \in (H_0^1(\Omega))^2 \to \mathbb{V}_1^{\mathcal{BR}} \cap (H_0^1(\Omega))^2$ as a combination of local projection operators π_E as:

(i) For each vertex $v \in \partial E$

$$\pi_E \mathbf{v}(\mathbf{x}_v) = \mathcal{I}_h^1 \mathbf{v}(\mathbf{x}_v); \tag{3.67}$$

(ii) For each edge $e \subset \partial E$,

$$\int_e (\pi_E \mathbf{v} - \mathbf{v}) \cdot \boldsymbol{\nu} \, ds = 0. \tag{3.68}$$

Lemma 3.7. *For all* $\mathbf{v} \in (H^k(\Omega))^2 \cap (H_0^1(\Omega))^2, \quad k = 1, 2$*, the operator* π_E *defined by* (3.67)–(3.68) *satisfies*

$$(q_h, \nabla \cdot (\mathbf{v} - \pi_E \mathbf{v})) = 0, \quad \forall q_h \in \mathbb{W}_0 \tag{3.69}$$

Assuming shape regularity as defined in Definition 3.1*, the global projection operator* π *satisfies the error bound*

$$\|\mathbf{v} - \pi \mathbf{v}\|_{H^1(\Omega)} \leq C h^m \|\mathbf{v}\|_{H^{m+1}(\Omega)}, \quad \forall \mathbf{v} \in (H^{m+1}(\Omega))^2, \; m = 0, 1. \tag{3.70}$$

Proof. According to the divergence theorem, we have

$$\int_E \nabla \cdot (\mathbf{v} - \pi_E \mathbf{v}) \, d\mathbf{x} = \sum_{i=0}^{3} \int_{e_i} (\mathbf{v} - \pi_E \mathbf{v}) \cdot \boldsymbol{\nu} \, ds = 0. \tag{3.71}$$

Given that q_h is piecewise constant on the element E, it is clear that we have

$$(q_h, \nabla \cdot (\mathbf{v} - \pi_E \mathbf{v})) = 0, \quad \forall q_h \in \mathbb{W}_0. \tag{3.72}$$

From the definition of the degrees of freedom (3.67) and (3.68), we decompose the projection operator π_E as

$$\pi_E \mathbf{v} = \mathcal{I}_h^1 \mathbf{v} + \sum_{i=0}^{3} \alpha_i \mathbf{b}_i, \tag{3.73}$$

where the coefficient

$$\alpha_i = \frac{\int_{e_i} (\mathbf{v} - \mathcal{I}_h^1 \mathbf{v}) \cdot \boldsymbol{\nu}_i \, ds}{\int_{e_i} \mathbf{b}_i \cdot \boldsymbol{\nu}_i \, ds}. \tag{3.74}$$

The approximation property of the interpolation operator $\mathcal{I}_h^1$ has been discussed in Lemma 3.4,

$$\left\|\mathbf{v} - \mathcal{I}_h^1 \mathbf{v}\right\|_{H^m(E)} \leq h^{n-m} |\mathbf{v}|_{H^n(E^*)}, \quad m = 0, 1, \quad n = 1, 2. \tag{3.75}$$

where $E^* = \cup_{F\in\mathcal{T}_h, F\cap E\neq\emptyset}$ defined in Lemma 3.3.

Lemma 3.6 indicates an estimation of the bubble function as

$$|\mathbf{b}_i|_{H^m(E)} \le C(\sigma^*)h^{1-m}|\tilde{\mathbf{b}}_i|_{H^1(\tilde{E})}, \quad m=0,1. \tag{3.76}$$

Because of the shape regularity, the vertices of the local reference elements $\tilde{E}$ vary in a compact set. Moveover, as in Arbogast et al. (2022), the definition of $\tilde{\mathbf{b}}_i$ depends only on the vertices in a C^1 manner. Thus the maximum over the elements is finite, and so

$$|\tilde{\mathbf{b}}_i|_{H^1(\tilde{E})} \le C(\sigma^*), \quad \forall E\in\mathcal{T}_h. \tag{3.77}$$

Therefore

$$|\tilde{\mathbf{b}}_i|_{H^m(E)} \le C(\sigma^*)h^{1-m}, \quad m=0,1. \tag{3.78}$$

We continue to estimate $|\alpha_i|$. Note

$$\int_{e,i} \mathbf{b}_i\cdot\boldsymbol{\nu}_i\,ds = \int_{e,i}\varphi_{e,i}^{(2)}\,ds = |e_i|\int_{\tilde{e}_i}\tilde{\varphi}_{e,i}^{(2)}\,d\tilde{s} = C_1|e_i|. \tag{3.79}$$

Then

$$\begin{aligned}|\alpha_i| = \left|\frac{\int_{e_i}(\mathbf{v}-\mathcal{I}_h^1\mathbf{v})\cdot\boldsymbol{\nu}_i\,ds}{\int_{e_i}\mathbf{b}_i\cdot\boldsymbol{\nu}_i\,ds}\right| &\le C\int_{\tilde{e},i}|\tilde{\mathbf{v}}-\tilde{\mathcal{I}}_h^1\tilde{\mathbf{v}}|\,ds \\ &\le C\left\|\tilde{\mathbf{v}}-\tilde{\mathcal{I}}_h^1\mathbf{v}\right\|_{L^2(\tilde{e}_i)}.\end{aligned} \tag{3.80}$$

As the trace mapping is contimuous from $H^1(\tilde{E})$ into $L^2(\tilde{e})$, according to the Trace Theorem 3.1,

$$\begin{aligned}|\alpha_i| &\le C\left\|\tilde{\mathbf{v}}-\tilde{\mathcal{I}}_h^1\mathbf{v}\right\|_{H^1(\tilde{E})} \\ &\le C\bigg(\left\|\tilde{\mathbf{v}}-\tilde{\mathcal{I}}_h^1\mathbf{v}\right\|_{L^2(E)} + h^{-1}\left|\tilde{\mathbf{v}}-\tilde{\mathcal{I}}_h^1\mathbf{v}\right|_{H^1(E)}\bigg).\end{aligned} \tag{3.81}$$

According to Lemma 3.4,

$$|\alpha_i| \le Ch^{k-1}\left\|\mathbf{v}\right\|_{H^k(E^*)}, \quad k=1,2. \tag{3.82}$$

The overall estimate is then

$$|\alpha_i||\mathbf{b}_i|_{H^m(E)} \le Ch^{k-m}\left\|\mathbf{v}\right\|_{H^k(E^*)}, \quad m=0,1, \quad k=1,2. \tag{3.83}$$

Combining with the interpolation error estimate result in equation (3.75), we get a local error estimate as

$$\|\mathbf{v} - \pi_E \mathbf{v}\|_{H^1(E)} \leq Ch \|\mathbf{v}\|_{H^2(E)} . \tag{3.84}$$

The global error estimate is obtained by combining local results together. $\square$

In addition to the approximability lemma 3.4, we also need a stability result.

Lemma 3.8. *Assuming that* $\mathbf{v} \in (H^1(\Omega))^2$, *the linear operator* π *is bounded on* $(H^1(\Omega))^2$ *independently of* h, *i.e.,*

$$\|\pi \mathbf{v}\|_{H^1(\Omega)} \leq C \|\mathbf{v}\|_{H^1(\Omega)} . \tag{3.85}$$

Proof. According to the triangle inequality,

$$\|\pi_E \mathbf{v}\|_{H^1(E)} \leq \|\mathbf{v}\|_{H^1(E)} + \|\mathbf{v} - \pi_E \mathbf{v}\|_{H^1(E)} \leq (1 + C) \|\mathbf{v}\|_{H^1(E)} , \tag{3.86}$$

where C comes from Lemma 3.7. The global stability result can be obtained by combining local results. $\square$

Theorem 3.9. *With the established approximability result stated in Lemma* 3.7, *and the stability result stated in Lemma* 3.8, *for a convex domain* Ω, *there exists a constant* $\gamma > 0$ *such that*

$$\inf_{q_h \in \mathbb{W}_0} \sup_{\mathbf{v}_h \in \mathbb{V}_1^{\mathcal{BR}}} \frac{(\nabla \cdot \mathbf{v}_h, q_h)}{\|\mathbf{v}_h\|_{H^1(\Omega)} \|q_h\|_{L^2(\Omega)}} \geq \gamma > 0. \tag{3.87}$$

Proof. Note that

$$\sup_{\mathbf{v}_h \in \mathbb{V}_1^{\mathcal{BR}}} \frac{(\nabla \cdot \mathbf{v}_h, q_h)}{\|\mathbf{v}_h\|_{H^1(\Omega)} \|q_h\|_{L^2(\Omega)}} \geq \frac{(\nabla \cdot \pi \mathbf{v}, q_h)}{\|\pi \mathbf{v}\|_{H^1(\Omega)} \|q_h\|_{L^2(\Omega)}}. \tag{3.88}$$

Recall that the inf-sup condition holds for the continuous space as

$$\inf_{q \in L^2(\Omega)} \sup_{\mathbf{v} \in (H^1(\Omega))^2} \frac{(\nabla \cdot \mathbf{v}, q)}{\|\mathbf{v}\|_{H^1(\Omega)} \|q\|_{L^2(\Omega)}} \geq \gamma > 0. \tag{3.89}$$

With the condition $(q_h, \nabla \cdot (\mathbf{v} - \pi_E \mathbf{v})) = 0$ proved in Lemma 3.7 we have

$$\frac{(\nabla \cdot \pi\mathbf{v}, q_h)}{\|\pi\mathbf{v}\|_{H^1(\Omega)} \|q_h\|_{L^2(\Omega)}} = \frac{(\nabla \cdot \mathbf{v}, q_h)}{\|\pi\mathbf{v}\|_{H^1(\Omega)} \|q_h\|_{L^2(\Omega)}} \geq \frac{\gamma}{C} > 0, \tag{3.90}$$

where C here is the constant in (3.85). □

We conclude that our space $\mathbb{V}_1^{\mathcal{BR}} \times \mathbb{W}_0$ forms a inf-sup stable pair with first order overall accuracy.

3.4.7 Numerical Test

We test our modified Bernardi-Raugel elements on the incompressible isotropic elasticity problem but with a non-homogenous Dirichlet boundary condition, i.e., on the problem:

Find the displacement $\mathbf{u}$ and pressure p such that

$$\begin{aligned} -\Delta\mathbf{u} + \nabla p &= \mathbf{f} \quad \text{in } \Omega, \\ \nabla \cdot \mathbf{u} &= 0 \quad \text{in } \Omega, \\ \mathbf{u} &= \mathbf{g} \quad \text{on } \partial\Omega. \end{aligned} \tag{3.91}$$

We assume that $\mathbf{g} \in (H^{1/2}(\partial\Omega))^2$ and understand $\tilde{\mathbf{g}} \in (H^1(\Omega))^2$ as a continuous extension from the boundary into the domain. We define a simple choice of energy spaces

$$\begin{aligned} \mathbb{V} &= \{\mathbf{v} \in (H^1(\Omega))^2 : \mathbf{v} = \mathbf{0} \quad \text{on } \partial\Omega\}, \\ \mathbb{U} &= \{\mathbf{u} \in (H^1(\Omega))^2 : \mathbf{u} = \mathbf{g} \quad \text{on } \partial\Omega\} = \tilde{\mathbf{g}} + \mathbb{V}, \\ \mathbb{W} &= \{p \in L^2(\Omega)\}/\mathbb{R}. \end{aligned} \tag{3.92}$$

We write the problem in weak form as:

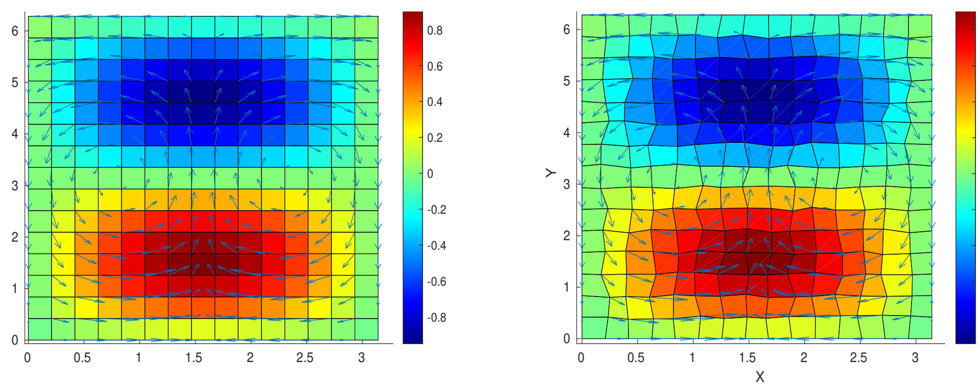
Find $\mathbf{u} \in \mathbb{U}$ and $p \in L^2(\Omega)$ such that

$$\begin{aligned} (\nabla\mathbf{u}, \nabla\mathbf{v}) - (p, \nabla \cdot \mathbf{v}) &= (\mathbf{f}, \mathbf{v}), \quad \forall \mathbf{v} \in \mathbb{V}, \\ (\nabla \cdot \mathbf{u}, \omega) &= 0, \qquad\quad \forall \omega \in \mathbb{W}. \end{aligned} \tag{3.93}$$

We consider the true solution of test problem (3.93) defined on a square region $\Omega = [0, \pi] \times [0, 2\pi]$. We manufacture a displacement and pressure as

$$\mathbf{u} = \begin{bmatrix} \cos(x)\sin(y) \\ -\sin(x)\cos(y) \end{bmatrix}, \quad p = \sin(x)\sin(y), \tag{3.94}$$

which has 0 normal flux on the boundary. We deal with the constant kernal by enforcing $\int_\Omega p\, d\mathbf{x} = 0$ and solve the linear system directly. The numerical results are computed on rectangular and logically rectangular quadrilateral meshes as depicted in Figures 3.10a and 3.10b, respectively. We calculate the L^2 norm of the residual (i.e., the difference) of the true and approximate pressure, velocity, and derivative of velocity.



(a) Rectangular Grids. Shown are pressure (color) and velocity (arrows).

(b) Distorted Grids. Shown are pressure (color) and velocity (arrows).

Figure 3.10: Numerical results on rectangular and quadrilateral mesh.

In Table 3.1 and Table 3.2 we display the result of a calculation of convergence on $n \times n$ rectangular and distorted grids, respectively, for $n = 8, 16, 32, 64$. A second order convergence rate is achieved by velocity. Pressure achieves some superconvergence due to the use of the mid point integration rule. We see second order convergence on the rectangular grid, and some rate better than the first on distorted grid. A first order convergence rate is achieved by derivative of velocity. These meet our expectations for optimal convergence order.

Grid size	8×8	16×16	32×32	64×64
$L^2(\mathrm{res}_u)$	4.502E−1	1.052E−1	2.513E−2	6.130E−3
$L^2(\mathrm{res}_{du})$	1.039E+0	4.463E−1	2.093E−1	1.019E−1
$L^2(\mathrm{res}_p)$	2.632E+0	5.845E−1	1.374E−1	3.330E−2

Table 3.1: Rectangular Grids. L^2-error in the residual of the solution u (res_u), the pressure p (res_p), and the of derivatives of the solution Du (res_{du}).

Grid size	8×8	16×16	32×32	64×64
$L^2(\mathrm{res}_u)$	4.603E−1	1.074E−2	2.567E−2	6.266E−3
$L^2(\mathrm{res}_{du})$	1.070E+0	4.670E−1	2.207E−1	1.078E−1
$L^2(\mathrm{res}_p)$	2.685E+0	5.981E−1	1.525E−1	4.580E−2

Table 3.2: Distorted Grids. L^2-error in the residual of the solution u (res_u), the pressure p (res_p), and the of derivatives of the solution Du (res_{du}).

3.5 Coupled Degenerate Darcy-Stokes System

We consider a degenerate Darcy-Stokes coupled boundary-value problem in the dimensionless form:

Find two velocity-pressure potential pairs $(\check{\mathbf{v}}_r, \check{q}_l)$ and $(\check{\mathbf{v}}_s, \check{q})$ such that

$$\check{\mathbf{v}}_r + \phi^{1+\Theta}\nabla(\phi^{-1/2}\check{q}_l) = 0, \tag{3.95}$$

$$\phi^{-1/2}\nabla\cdot(\phi^{1+\Theta}\check{\mathbf{v}}_r) + \frac{1}{1-\phi}(\check{q}_l - \phi^{1/2}\check{q}) = 0, \tag{3.96}$$

$$\nabla\check{q} - \nabla\cdot\check{\tau}(\check{\mathbf{v}}_s) = -(1-\phi)\mathbf{n}, \tag{3.97}$$

$$\nabla\cdot\check{\mathbf{v}}_s - \frac{\phi^{1/2}}{1-\phi}(\check{q}_l - \phi^{1/2}\check{q}) = 0, \tag{3.98}$$

and

$$\begin{aligned}
\phi^{1+\Theta}\check{\mathbf{v}}_r\cdot\boldsymbol{\nu} &= g_r && \text{on } \Gamma_{u,D},\\
\check{\mathbf{v}}_s &= \mathbf{g}_s && \text{on } \Gamma_{u,S},\\
\check{q}_l &= \phi^{1/2}q_{l,0} && \text{on } \Gamma_{i,D},\\
\boldsymbol{\tau}\cdot\boldsymbol{\nu} - q &= t_0 && \text{on } \Gamma_{i,S}.
\end{aligned} \tag{3.99}$$

Here, $\Gamma_{u,D}$ and $\Gamma_{u,S}$ denote the parts of the boundary, where essential boundary conditions are prescribed for the Darcy and the Stokes problems, respectively, while $\Gamma_{i,D}$ and $\Gamma_{i,S}$ represent the portions of the boundary, where natural boundary conditions are imposed. We assume the essential and natural boundary conditions are non overlapping and covering the entire boundary, i.e., $\Gamma_{u,D} \cup \Gamma_{i,D} = \partial\Omega$, $\Gamma_{u,D} \cap \Gamma_{i,D} = \emptyset$ and $\Gamma_{u,S} \cup \Gamma_{i,S} = \partial\Omega$, $\Gamma_{u,S} \cap \Gamma_{i,S} = \emptyset$.

To ensure well-posedness in the presence of degeneracy ($\phi = 0$), we must verity that all terms scaled by $\phi^{-1/2}$ remain well-defined. We expand the gradient term in equation (3.95) and the divergence term in equation (3.96) as

$$\begin{aligned} \phi^{1+\Theta}\nabla(\phi^{-1/2}\check{q}_l) &= \phi^{1/2+\Theta}\nabla\check{q}_l + \phi^{\Theta-1/2}\check{q}_l\nabla\phi, \\ \phi^{-1/2}\nabla\cdot(\phi^{1+\Theta}\check{\mathbf{v}}_r) &= \phi^{1/2+\Theta}\nabla\cdot\check{\mathbf{v}}_r + \phi^{\Theta-1/2}\nabla\phi\cdot\check{\mathbf{v}}_r. \end{aligned} \tag{3.100}$$

They are well-defined provided that

$$\phi^{\Theta-1/2}\nabla\phi \in (L^\infty(\Omega))^2. \tag{3.101}$$

We define a scaled Hilbert space tailored to the present formulation, within which the scaled velocity $\check{\mathbf{v}}_r$ should lie

Definition 3.6. Let

$$H_\phi(\mathrm{div};\Omega) = \Big\{\mathbf{v} \in (L^2(\Omega))^2 : \phi^{-1/2}\nabla\cdot(\phi^{1+\Theta}\mathbf{v}) \in L^2(\Omega)\Big\}. \tag{3.102}$$

This is a Hilbert space equipped with the inner product

$$(\mathbf{u},\mathbf{v})_{H_\phi(\mathrm{div};\Omega)} = (\mathbf{u},\mathbf{v}) + (\phi^{-1/2}\nabla\cdot(\phi^{1+\Theta}\mathbf{u}), \phi^{-1/2}\nabla\cdot(\phi^{1+\Theta}\mathbf{v})). \tag{3.103}$$

The normal trace on $\partial\Omega$ is well-defined by

$$\langle\phi^{1/2+\Theta}\mathbf{v}\cdot\boldsymbol{\nu},\psi\rangle = \big(\mathbf{v},\phi^{1+\Theta}\nabla(\phi^{-1/2}\psi)\big) + \big(\phi^{-1/2}\nabla\cdot(\phi^{1+\Theta}\mathbf{v}),\psi\big), \quad \forall\psi \in H^1(\Omega), \tag{3.104}$$

using (3.101), moreover,

$$\big\|\phi^{1/2+\Theta}\mathbf{v}\cdot\boldsymbol{\nu}\big\|_{H^{-1/2}(\partial\Omega)} < C_\Omega\Big\{\,\|\mathbf{v}\|_{L^2(\Omega)} + \big\|\phi^{-1/2}\nabla\cdot(\phi^{1+\Theta}\mathbf{v})\big\|_{L^2(\Omega)}\Big\}. \tag{3.105}$$

Subsequently, we have the space $H_\phi^{-1/2}(\mathrm{div};\partial\Omega)$ understood as the image of the normal trace operator on $H_\phi(\mathrm{div};\Omega)$.

3.5.1 The Weak Formulation

We assume that the Dirichlet data $\mathbf{g}_s \in (H^{1/2}(\Gamma_{u,S}))^2$, and understand $\tilde{\mathbf{g}}_s \in (H^1(\Omega))^2$ as a continuous extension from the boundary into the domain. Similarly, we assume that $\phi^{-1/2} g_r \in H_\phi^{-1/2}(\partial\Omega)$ understood as the image of the scaled normal trace operator (3.104). Let

$$\phi^{1/2+\Theta}\mathbf{g}_r \cdot \boldsymbol{\nu} = \phi^{-1/2} g_r \quad \text{on} \quad \partial\Omega. \tag{3.106}$$

We view $\tilde{\mathbf{g}}_r$ as a continuous extension from the boundary into the domain.

The following energy spaces are defined for this scaled Darcy-Stokes coupled system

$$\begin{aligned}
\mathbb{V}_D^\phi &= \Big\{ \mathbf{v} \in H_\phi(\mathrm{div};\Omega) : \phi^{1/2+\Theta}\mathbf{v}\cdot\boldsymbol{\nu} = 0 \text{ on } \Gamma_{u,D} \Big\}, \\
\mathbb{U}_D^\phi &= \Big\{ \mathbf{u} \in H_\phi(\mathrm{div};\Omega) : \phi^{1/2+\Theta}\mathbf{u}\cdot\boldsymbol{\nu} = \phi^{-1/2} g_r \text{ on } \Gamma_{u,D} \Big\} = \tilde{\mathbf{g}}_r + \mathbb{V}_D^\phi, \\
\mathbb{W}_D &= L^2(\Omega), \\
\mathbb{V}_S &= \Big\{ \mathbf{v} \in (H^1(\Omega))^2 : \mathbf{v} = \mathbf{0} \text{ on } \Gamma_{u,S} \Big\}, \\
\mathbb{U}_S &= \Big\{ \mathbf{u} \in (H^1(\Omega))^2 : \mathbf{u} = \mathbf{g}_s \text{ on } \Gamma_{u,S} \Big\} = \tilde{\mathbf{g}}_s + \mathbb{V}_S, \\
\mathbb{W}_S &= L^2(\Omega),
\end{aligned} \tag{3.107}$$

where each space is equipped with its natural norm.

A compatibility condition is required to ensure overall mass conservation when essential boundary conditions are prescribed on the entire boundary $\partial\Omega$, i.e., when $\Gamma_{u,D} = \Gamma_{u,S} = \partial\Omega$. In this case, the prescribed Dirichlet data must satisfy the compatibility condition

$$\int_{\partial\Omega} (\phi^{1+\Theta}\mathbf{g}_r + \mathbf{g}_s)\cdot\boldsymbol{\nu}\, ds = 0. \tag{3.108}$$

The standard mixed form for the proposed coupled system reads as follows:

Find $\check{\mathbf{v}}_r \in \mathbb{U}_D^\phi$, $\check{q}_l \in \mathbb{W}_D$, $\check{\mathbf{v}}_s \in \mathbb{U}_S$, and $\check{q} \in \mathbb{W}_S$ such that

$$(\check{\boldsymbol{\tau}}(\mathbf{v}_s), \nabla \boldsymbol{\psi}_s) - (\check{q}, \nabla \cdot \boldsymbol{\psi}_s) = -((1-\phi)\mathbf{n}, \boldsymbol{\psi}_s) \qquad \forall \boldsymbol{\psi}_s \in \mathbb{V}_S, \tag{3.109}$$

$$(\nabla \cdot \check{\mathbf{v}}_s, \omega) + \Big(\frac{\phi^{1/2}}{1-\phi}(\check{q}_l - \phi^{1/2}\check{q}), \omega\Big) = 0 \qquad \forall \omega \in \mathbb{W}_S, \tag{3.110}$$

$$(\check{\mathbf{v}}_r, \boldsymbol{\psi}_r) - (\check{q}_l, \phi^{-1/2}\nabla \cdot (\phi^{1+\Theta}\boldsymbol{\psi}_r)) = 0, \qquad \forall \boldsymbol{\psi}_r \in \mathbb{V}_D^\phi, \tag{3.111}$$

$$(\phi^{-1/2}\nabla \cdot (\phi^{1+\Theta}\check{\mathbf{v}}_r), \omega_l) + \Big(\frac{1}{1-\phi}(\check{q}_l - \phi^{1/2}\check{q}), \omega_l\Big) = 0 \qquad \forall \omega_f \in \mathbb{W}_D. \tag{3.112}$$

We state the well-posedness result of the scaled formulation directly as follows.

Lemma 3.10. *(Existence and Uniqueness)*

Assume the condition (3.101) *holds,* $0 \le \phi \le \phi^* < 1$. *There exists a unique solution to the scaled formulation* (3.109)–(3.112), *and it satisfies*

$$\begin{aligned}\|\check{\mathbf{v}}_r\|_{(L^2(\Omega))^2} + \big\|\phi^{-1/2}\nabla \cdot (\phi^{1+\Theta}\check{\mathbf{v}}_r)\big\|_{(L^2(\Omega))^2} + \|\check{q}_l\|_{(L^2(\Omega))^2} + \|\check{\mathbf{v}}_s\|_{(H^1(\Omega))^2} + \|\check{q}\|_{(L^2(\Omega))^2} \\ \le C\big\{|\rho_r| + \|\tilde{\mathbf{g}}_r\|_{(L^2(\Omega))^2} + \big\|\phi^{-1/2}\nabla \cdot (\phi^{1+\Theta}\tilde{\mathbf{g}}_r)\big\|_{(L^2(\Omega))^2} + \|\tilde{\mathbf{g}}_s\|_{H^1(\Omega)}\big\}.\end{aligned} \tag{3.113}$$

Proof. See Arbogast et al. (2017). □

3.5.2 A Scaled Mixed Finite Element Method

We discretize the scaled mixed formulation using two pairs of finite element spaces: $\mathbb{V}_{D,h} \times \mathbb{W}_{D,h} \subset H(\mathrm{div};\Omega) \times L^2(\Omega)$ for the Darcy part of the system and $\mathbb{V}_{S,h} \times \mathbb{W}_{S,h} \subset (H^1(\Omega))^2 \times L^2(\Omega)$ for the Stokes part of the system. We recall the aforementioned inf-sup stable mixed finite element space pairs on each element $E \in \mathcal{T}_h$:

- Reduced $H(\mathrm{div})$ conforming space $\mathbb{V}_1^0$ in Section 3.4.2,

$$\begin{aligned}\mathbb{V}_{D,h}(E) &= \mathbb{V}_1^0(E) = (\mathbb{P}_1(E))^2 \oplus \mathrm{curl}\,\mathbb{S}_2^{\mathcal{DS}}(E), \\ \mathbb{W}_{D,h}(E) &= \mathbb{W}(E) = \mathbb{P}_0(E);\end{aligned} \tag{3.114}$$

- Modified Bernardi-Raugel element in Section 3.4.5,

$$\begin{aligned} \mathbb{V}_{S,h}(E) &= \mathbb{V}_1^{\mathcal{BR}}(E) = (\mathcal{DS}_1(E))^2 \oplus \mathrm{span}\{\mathbf{b}_0, \mathbf{b}_1, \mathbf{b}_2, \mathbf{b}_3\}, \\ \mathbb{W}_{S,h}(E) &= \mathbb{W}(E) = \mathbb{P}_0(E). \end{aligned} \tag{3.115}$$

We impose essential boundary conditions by projecting extensions $\tilde{\mathbf{g}}_r$ and $\tilde{\mathbf{g}}_s$ into the finite element spaces as $\hat{\mathbf{g}}_r \in \mathbb{V}_{D,h}$ and $\hat{\mathbf{g}}_s \in \mathbb{V}_{S,h}$.

We also define

$$\begin{aligned} \mathbb{V}_{D,h,0} &= \{\mathbf{v} \in \mathbb{V}_{D,h} : \mathbf{v} \cdot \boldsymbol{\nu} = 0 \;\text{ on }\; \Gamma_{u,D}\} \\ \mathbb{V}_{S,h,\mathbf{0}} &= \{\mathbf{v} \in \mathbb{V}_{S,h} : \mathbf{v} = \mathbf{0} \;\text{ on }\; \Gamma_{u,S}\} \end{aligned} \tag{3.116}$$

The scaled mixed finite element method: Find $\check{\mathbf{v}}_{r,h} \in \mathbb{V}_{D,h,0} + \hat{\mathbf{g}}_r$, $\check{q}_{r,h} \in \mathbb{W}_{D,h}$, $\check{\mathbf{v}}_{s,h} \in \mathbf{V}_{S,h,\mathbf{0}} + \hat{\mathbf{g}}_s$ and $\check{q}_h \in \mathbb{W}_{S,h}$ such that

$$(\boldsymbol{\tau}(\check{\mathbf{v}}_{s,h}), \nabla \boldsymbol{\psi}_{s,h}) - (\check{q}_h, \nabla \cdot \boldsymbol{\psi}_{s,h}) = -((1-\phi)\mathbf{n}, \boldsymbol{\psi}_{s,h}), \qquad \forall \boldsymbol{\psi}_{s,h} \in \mathbb{V}_{S,h,\mathbf{0}}, \tag{3.117}$$

$$(\nabla \cdot \check{\mathbf{v}}_{s,h}, \omega_h) - \Big(\frac{\phi^{1/2}}{1-\phi}(\check{q}_{l,h} - \phi^{1/2}\check{q}_h), \omega_h\Big) = 0, \qquad \forall \omega_h \in \mathbb{W}_{S,h}, \tag{3.118}$$

$$(\check{\mathbf{v}}_{r,h}, \boldsymbol{\psi}_{r,h}) - (\check{q}_{l,h}, \phi^{-1/2}\nabla \cdot (\phi^{1+\Theta}\boldsymbol{\psi}_{r,h})) = 0, \qquad \forall \boldsymbol{\psi}_{r,h} \in \mathbb{V}_{D,h,0}, \tag{3.119}$$

$$(\phi^{-1/2}\nabla \cdot (\phi^{1+\Theta}\check{\mathbf{v}}_{r,h}), \omega_{l,h}) + \Big(\frac{1}{1-\phi}(\check{q}_{l,h} - \phi^{1/2}\check{q}_h), \omega_{l,h}\Big) = 0, \qquad \forall \omega_{l,h} \in \mathbb{W}_{D,h}. \tag{3.120}$$

The existence and convergence results of the proposed new mixed finite element discretization can be established in complete analogy with the proof in Arbogast et al. (2017). We state the results below

Lemma 3.11. *If* (3.101) *holds, then there exists a unique solution to the scaled mixed finite element method* (3.117)–(3.120).

Proof. See Arbogast et al. (2017). □

Lemma 3.12. *Assume that* (3.101) *holds on the porosity,* $0 \leq \phi \leq \phi^* < 1$*, and the mesh is quasiuniform. Then the difference of the solution to the scaled formula-*

tion (3.117)–(3.120)*, and its finite elment approximation satisfy*

$$
\begin{aligned}
&\|\tilde{\mathbf{v}}_r - \tilde{\mathbf{v}}_{r,h}\|_{L^2(\Omega)} + \left\|\mathcal{P}_{\mathbb{W}_{D,h}}\left[\phi^{-1/2}\nabla\cdot(\phi^{1+\Theta}(\tilde{\mathbf{v}}_r - \tilde{\mathbf{v}}_{r,h}))\right]\right\|_{L^2(\Omega)} + \|\mathbf{v}_s - \mathbf{v}_{s,h}\|_{H^1(\Omega)} \\
&+ \|\tilde{q}_l - \tilde{q}_{l,h}\|_{L^2(\Omega)} + \|q - q_h\|_{L^2(\Omega)} \\
&\leq C\Bigg\{ \left\|\tilde{\mathbf{v}}_r - \pi_{\mathbb{V}_1^0}\tilde{\mathbf{v}}_r\right\|_{L^2(\Omega)} + \left\|\mathcal{P}_{\mathbb{W}_{D.h}}[\phi^{-1/2}\nabla\cdot(\phi^{1+\Theta}(\tilde{\mathbf{v}}_r - \pi_{\mathbb{V}_1^0}\tilde{\mathbf{v}}_r))]\right\|_{L^2(\Omega)} \\
&+ \left\|\pi_{\mathbb{V}_1^0}\tilde{\mathbf{g}}_r - \hat{\mathbf{g}}_r\right\|_{L^2(\Omega)} + \left\|\tilde{q}_l - \mathcal{P}_{\mathbb{W}_{D,h}}\tilde{q}_l\right\|_{L^2(\Omega)} + \left\|q - \mathcal{P}_{\mathbb{W}_{S,h}}q\right\|_{L^2(\Omega)} \\
&+ \|\mathbf{v}_s - \pi_{\mathbb{V}_{\mathcal{BR}}}\mathbf{v}_s\|_{H^1(\Omega)} + \|\pi_{\mathbb{V}_{\mathcal{BR}}}\tilde{\mathbf{g}}_s - \hat{\mathbf{g}}_s\|_{H^1(\Omega)}\Bigg\},
\end{aligned}
\tag{3.121}
$$

where $\pi_{\mathbb{V}_1^0}$ *and* $\pi_{\mathbb{V}_{\mathcal{BR}}}$ *are projection operators defined in Section* 3.4.3 *and Section* 3.4.6 *respectively.*

Proof. See Arbogast et al. (2017). □

3.5.3 Local Mass Conservation

Before proceeding, we ensure that the formulation is well-defined—specifically, that no division by zero occurs. The term $\phi^{-1/2}$ in the symmetric equations (3.119) and (3.120) causes trouble when there is no melting, i.e., $\phi = 0$. To address this, we utilize an element-averaged variable ϕ_E as

$$\phi_E = \frac{1}{|E|}\int_E \phi(\mathbf{x})\,d\mathbf{x}, \tag{3.122}$$

where $|E|$ is the area of element E and $\phi_E = \mathcal{P}_{\mathbb{W}_{D,h}}\phi|_E \in \mathbb{W}_{D,h}$. Then the two terms involving divergence are modified as

$$\phi^{-1/2}\nabla\cdot(\phi^{1+\Theta}\mathbf{v})|_E = \begin{cases} \phi_E^{-1/2}\nabla\cdot(\phi^{1+\Theta}\mathbf{v}) & \text{if } \phi_E \neq 0, \\ 0 & \text{if } \phi_E = 0, \end{cases} \tag{3.123}$$

wherein the point-wise term $\phi^{-1/2}$ was modified to be a cell-averaged term $\phi_E^{-1/2}$. We apply the divergence theorem to expand the integral term involving the divergence operator in equation (3.120) as

$$\int_E \phi_E^{-1/2}\nabla\cdot(\phi^{1+\Theta}\boldsymbol{\psi}_{r,h})\omega_{l,h}\,d\mathbf{x} = \int_{\partial E}\phi_E^{-1/2}(\phi^{1+\Theta}\boldsymbol{\psi}_{r,h}\cdot\boldsymbol{\nu})\omega_{l,h}\,ds, \tag{3.124}$$

where $\omega_{l,h} \in \mathbb{W}_{D,h}$ is constant on the element $E \in \mathcal{T}_h$. We now propose our local mass conserved version with $\phi_E^{1/2}$ as

$$(\boldsymbol{\tau}(\check{\mathbf{v}}_{s,h}), \nabla \boldsymbol{\psi}_{s,h}) - (\check{q}_h, \nabla \cdot \boldsymbol{\psi}_{s,h}) = -((1-\phi)\mathbf{n}, \boldsymbol{\psi}_{s,h}) \qquad \forall \boldsymbol{\psi}_{s,h}, \in \mathbb{V}_{S,h,\mathbf{0}}, \tag{3.125}$$

$$(\nabla \cdot \check{\mathbf{v}}_{s,h}, \omega_h) - \Big(\frac{\phi\phi_E^{-1/2}}{1-\phi}(\check{q}_{l,h} - \phi_E^{1/2}\check{q}_h), \omega_h\Big) = 0, \qquad \forall \omega_h \in \mathbb{W}_{S,h}, \tag{3.126}$$

$$(\check{\mathbf{v}}_{r,h}, \boldsymbol{\psi}_{r,h}) - (\check{q}_{l,h}, \phi_E^{-1/2}\nabla \cdot (\phi^{1+\Theta}\boldsymbol{\psi}_{r,h})) = 0, \qquad \forall \boldsymbol{\psi}_{r,h} \in \mathbb{V}_{D,h,0}, \tag{3.127}$$

$$(\phi_E^{-1/2}\nabla \cdot (\phi^{1+\Theta}\check{\mathbf{v}}_{r,h}), \omega_{l,h}) + \Big(\frac{\phi\phi_E^{-1}}{1-\phi}(\check{q}_{l,h} - \phi_E^{1/2}\check{q}_h), \omega_{l,h}\Big) = 0, \qquad \forall \omega_{l,h} \in \mathbb{W}_{D,h}. \tag{3.128}$$

Here, the term $\phi\phi_E^{-1}$ in equation (3.128) is viewed as 1 when $\phi_E = 0$. We sum equations (3.126) and (3.128) together assuming that $\omega_h = \omega_E = 1 \in \mathbb{W}_{S,h}$ and $\omega_{l,h} = \phi_E^{-1/2}\omega_E \in \mathbb{W}_{D,h}$. To verify local mass conservation of the fluid, consider any element $E \in \mathcal{T}_h$. Equation (3.128) gives

$$\int_E \nabla \cdot (\phi^{1+\Theta}\check{\mathbf{v}}_{r,h})\, d\mathbf{x} + \int_E \frac{\phi\phi_E^{-1/2}}{1-\phi}(\check{q}_{l,h} - \phi_E^{1/2}\check{q}_h)\, dx = 0, \tag{3.129}$$

and equation (3.126) gives

$$\int_E \nabla \cdot \check{\mathbf{v}}_{s,h}\, d\mathbf{x} - \int_E \frac{\phi\phi_E^{-1/2}}{1-\phi}(\check{q}_{l,h} - \phi_E^{1/2}\check{q}_h)\, dx = 0. \tag{3.130}$$

The local mass conservation of the phase-averaged mixture is now retrieved as

$$\int_E \nabla \cdot (\phi^{1+\Theta}\check{\mathbf{v}}_r + \check{\mathbf{v}}_s)\, d\mathbf{x} = 0. \tag{3.131}$$

Lemma 3.13. *Assuming* (3.101) *holds, there exists a unique solution to the modified local mass conservation modified scaled mixed finite element method* (3.125)–(3.128).

Proof. See Arbogast et al. (2017). The local mass conservation modification replaces the term $1/(1-\phi)$ with $(\phi\phi_E^{-1})/(1-\phi)$,which makes it completely analogous to the previous proof. Hence, we do not repeat it here. □

3.6 Saddle Point System

Saddle point systems arise frequently in scientific and engineering applications, including constrained optimization problems, image reconstruction, and optimal control. Saddle point systems have been extensively and systematically studied, notably by Benzi et al. (2005). In our case, the saddle point system arises naturally from the mixed finite element discretization.

In this section, we give a brief overview of the properties of a saddle point system and present an iteration method tailored to our problem. We express the abstract saddle point system as

$$\begin{bmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{B}^T & -\mathbf{C} \end{bmatrix} \begin{bmatrix} \mathbf{x} \\ \mathbf{y} \end{bmatrix} = \begin{bmatrix} \mathbf{F} \\ \mathbf{G} \end{bmatrix}, \tag{3.132}$$

where we assume the following properties:

- $\mathbf{A}$ is symmetric: $\mathbf{A} = \mathbf{A}^T$;
- $\mathbf{A}$ is nonsingular: $\mathbf{A}^{-1}$ exists;
- $\mathbf{C}$ is symmetric and positive semidefinite;
- $\mathbf{B}$ is rank deficient.

3.6.1 Reduced System

We now present a detailed discussion of the saddle point system (SPS) arising from the mixed finite element method (MFEM) including the treatment of boundary conditions.

As a first step, we regroup the degrees of freedom (dofs) ($i \in I$) into three distinct sets: the set of the interior dofs I_c, the set of the boundary dofs assigned with essential boundary condition I_{Γ_u} and the set of the boundary dofs assigned with natural boundary condition I_{Γ_i}. We have $I_c \cap I_{\Gamma_u} = I_c \cap I_{\Gamma_i} = I_{\Gamma_u} \cap I_{\Gamma_i} = \emptyset$ and

$I_c \cup I_{\Gamma_i} \cup I_{\Gamma_u} = I$. We then reorganize the $\mathbf{A}$ matrix accordingly as

$$\mathbf{A} = \begin{bmatrix} \mathbf{A}_K & \mathbf{A}_M \\ \mathbf{A}_M^T & \mathbf{M} \end{bmatrix}, \tag{3.133}$$

where $\mathbf{A}_K = \{a_{k_1,k_2}\}$ with $k_1, k_2 \in I_c + I_{\Gamma_i}$, consists of all the "free" dofs. The matrix $\mathbf{A}_M = \{a_{m_1,m_2}\}$ with $m_1 \in I_c + I_{\Gamma_i}$, $m_2 \in I_{\Gamma_u}$, represents the interaction between "free" and prescribed dofs. Finally, $\mathbf{M} = \{m_{i,j}\}$, $i, j \in I_{\Gamma_u}$ accounts for the contributions from the fixed dofs associated with essential boundary conditions.

We apply the same treatment to the matrix $\mathbf{B}$ and the right-hand side vector $\mathbf{F}$ as

$$\mathbf{B} = \begin{bmatrix} \mathbf{B}_K \\ \mathbf{B}_M \end{bmatrix} \quad \text{and} \quad \mathbf{F} = \begin{bmatrix} \mathbf{F}_K \\ \mathbf{F}_M \end{bmatrix}. \tag{3.134}$$

We reformulate the linear system using the regrouped dofs and derive a reduced system of the form

$$\begin{bmatrix} \mathbf{A}_K & \mathbf{B}_K \\ \mathbf{B}_K^T & -\mathbf{C} \end{bmatrix} \begin{bmatrix} \mathbf{x} \\ \mathbf{y} \end{bmatrix} = \begin{bmatrix} \mathbf{F}_K - \mathbf{A}_M \mathbf{g} - \mathbf{g}_N \\ \mathbf{G} - \mathbf{B}_M^T \mathbf{g} \end{bmatrix}, \tag{3.135}$$

where $\mathbf{g}$ denotes the values prescribed by the essential boundary condition, and $\mathbf{g}_N$ represents the contribution from the natural boundary conditions. The reformed reduced system remains the structure of a saddle point system, and can be represented using the same abstract form (3.132).

3.6.2 Linear System for Coupled Darcy-Stokes Equations

The linear system formed from the equations (3.125)–(3.128) is shown as

$$\begin{bmatrix} \mathbf{A}_S & -\mathbf{B}_S & \mathbf{0} & \mathbf{0} \\ \mathbf{B}_S^T & \mathbf{C}_S & \mathbf{0} & \mathbf{K} \\ \mathbf{0} & \mathbf{0} & \mathbf{A}_D & -\mathbf{B}_D \\ \mathbf{0} & \mathbf{K} & \mathbf{B}_D^T & \mathbf{C}_D \end{bmatrix} \begin{bmatrix} \mathbf{x}_S \\ \mathbf{y}_S \\ \mathbf{x}_D \\ \mathbf{y}_D \end{bmatrix} = \begin{bmatrix} \mathbf{F}_S \\ \mathbf{G}_S \\ \mathbf{F}_D \\ \mathbf{G}_D \end{bmatrix}, \tag{3.136}$$

wherein all the matrices are reduced matrices excluding boundary dofs associated with essential boundary conditions, and all the right-hand side vectors are modified by boundary conditions. Most of these matrics can be computed directly from the definition of the inner product over each element E. However, the $\mathbf{B}_D$ matrix should be computed using the divergence theorem (3.124) to avoid evaluating the derivatives of the porosity ϕ. For any element $E \in \mathcal{T}_h$,

$$\begin{aligned}\mathbf{B}_D &= \left(\phi_E^{-1/2}\nabla\cdot(\phi^{1+\Theta}\check{\mathbf{v}}_{r,h}), \omega_{l,h}\right) \\ &= \int_e (\phi_E^{-1/2}\phi^{1+\Theta})\check{\mathbf{v}}_{r,h}\cdot\boldsymbol{\nu}\omega_{l,h}\, ds,\end{aligned} \tag{3.137}$$

where the term $\phi_E^{-1/2}\phi^{1+\Theta}$ is well-defined and evaluates 0 if $\phi_E = 0$.

We then form a double saddle point system by symmetrically permutating the original coupled linear system as

$$\begin{bmatrix}\mathbf{A}_S & \mathbf{0} & -\mathbf{B}_S & \mathbf{0} \\ \mathbf{0} & \mathbf{A}_D & \mathbf{0} & -\mathbf{B}_D \\ \mathbf{B}_S^T & \mathbf{0} & \mathbf{C}_S & \mathbf{K} \\ \mathbf{0} & \mathbf{B}_D^T & \mathbf{K} & \mathbf{C}_D\end{bmatrix}\begin{bmatrix}\mathbf{x}_S \\ \mathbf{x}_D \\ \mathbf{y}_S \\ \mathbf{y}_D\end{bmatrix} = \begin{bmatrix}\mathbf{F}_S \\ \mathbf{F}_D \\ \mathbf{G}_S \\ \mathbf{G}_D\end{bmatrix}. \tag{3.138}$$

We group these block matrices accordingly and retrieve the simple abstract saddle point system given in equation (3.132) with

$$\begin{aligned}&\mathbf{A} = \begin{bmatrix}\mathbf{A}_S & \mathbf{0} \\ \mathbf{0} & \mathbf{A}_D\end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix}-\mathbf{B}_S & \mathbf{0} \\ \mathbf{0} & -\mathbf{B}_D\end{bmatrix}, \quad \mathbf{C} = \begin{bmatrix}\mathbf{C}_S & \mathbf{K} \\ \mathbf{K} & \mathbf{C}_D\end{bmatrix}, \\ &\mathbf{F} = \begin{bmatrix}\mathbf{F}_S \\ \mathbf{F}_D\end{bmatrix}, \quad \mathbf{G} = \begin{bmatrix}-\mathbf{G}_S \\ -\mathbf{G}_D\end{bmatrix}, \quad \mathbf{x} = \begin{bmatrix}\mathbf{x}_S \\ \mathbf{x}_D\end{bmatrix}, \quad \mathbf{y} = \begin{bmatrix}\mathbf{y}_S \\ \mathbf{y}_D\end{bmatrix}.\end{aligned} \tag{3.139}$$

We now examine the structure of the resulting saddle point system. The scaled formulation (3.111) guarantees that the matrix $\mathbf{A}_D$ is nonsingular, therefore the coupled matrix $\mathbf{A}$ is nonsingular. The saddle point system can be factored as

$$\begin{bmatrix}\mathbf{A} & \mathbf{B} \\ \mathbf{B}^T & \mathbf{C}\end{bmatrix} = \begin{bmatrix}I & \mathbf{0} \\ \mathbf{B}^T\mathbf{A}^{-1} & I\end{bmatrix}\begin{bmatrix}\mathbf{A} & \mathbf{0} \\ \mathbf{0} & \mathbf{S}\end{bmatrix}\begin{bmatrix}I & \mathbf{A}^{-1}\mathbf{B} \\ \mathbf{0} & I\end{bmatrix}, \tag{3.140}$$

where $\mathbf{S}$ is the Schur complement

$$\mathbf{S} = -(\mathbf{C} + \mathbf{B}\mathbf{A}^{-1}\mathbf{B}^T). \tag{3.141}$$

The presence of the evolving no-melting region, where the porosity $\phi = 0$, affects the kernal space of the matrix $\mathbf{B}_D$. On one hand, the rank deficiency in matrix

$\mathbf{B}$ makes the Schur complement singular. On the other hand, the changing kernal space poses some difficulties applying classical null space methods to our system. Taking into consideration that the solver will be implemented in parallel in the future, an iterative solver is more suitable than a direct solver.

Rather than applying a general-purpose solver such as the generalized minimal residual method (GMRES) to the full linear system, we focus on iterative methods that exploit the structural properties of the block matrices. We start with a "direct" way to solve this derived saddle point system. Consider the block LU matrix factorization of (3.132) equivalent to (3.140),

$$\begin{bmatrix} I & \mathbf{0} \\ -\mathbf{B}^T\mathbf{A}^{-1} & I \end{bmatrix} \begin{bmatrix} I & \mathbf{0} \\ \mathbf{B}^T\mathbf{A}^{-1} & I \end{bmatrix} \begin{bmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{0} & \mathbf{S} \end{bmatrix} \begin{bmatrix} \mathbf{x} \\ \mathbf{y} \end{bmatrix} = \begin{bmatrix} I & \mathbf{0} \\ -\mathbf{B}^T\mathbf{A}^{-1} & I \end{bmatrix} \begin{bmatrix} \mathbf{F} \\ \mathbf{G} \end{bmatrix}. \tag{3.142}$$

We can solve for $\mathbf{x}$ and $\mathbf{y}$ with block backsubstitution,

$$\begin{aligned} \mathbf{x} &= \mathbf{A}^{-1}(\mathbf{F} - \mathbf{B}\mathbf{y}), \\ \mathbf{y} &= \mathbf{S}^{-1}(\mathbf{G} - \mathbf{B}^T\mathbf{A}^{-1}\mathbf{F}), \end{aligned} \tag{3.143}$$

where $\mathbf{A}^{-1}$ can be computed either directly using the Cholesky factorization or iteratively via the Conjugate Gradient method. The Schur complement $\mathbf{S}$ is usually non-invertible. Its "inverse" $\mathbf{S}^{-1}$ can be approximated with the Minimal Resigual algorithm following Paige and Saunders (1975).

3.6.3 Uzawa Iteration

The Uzawa algorithm, originally introduced by Arrow et al. (1958), later gained popularity in computational fluid dynamics. It was later applied to the discretization of the Stokes problem using Mixed Finite Element methods. Notable variants incorporating an inexact inner solver to accelerate the Uzawa iteration was discussed by Elman and Golub (1994) and Bramble et al. (1997).

We begin with a brief overview of the inexact Uzawa iteration, which generates a sequence of approximations $\{\mathbf{x}_k\}$ and $\{\mathbf{y}_k\}$ as

$$\begin{aligned} \mathbf{x}_k &= \mathbf{x}_{k-1} + \mathbf{Q}_A^{-1}\big[\mathbf{F} - (\mathbf{A}\mathbf{x}_{k-1} + \mathbf{B}\mathbf{y}_{k-1})\big], \\ \mathbf{y}_k &= \mathbf{y}_{k-1} + \mathbf{Q}_B^{-1}\big(\mathbf{B}^T\mathbf{x}_k - \mathbf{C}\mathbf{y}_{k-1} - \mathbf{G}\big), \end{aligned} \tag{3.144}$$

where the initial conditions $\mathbf{x}_0$ and $\mathbf{y}_0$ are typically set to zero for simplicity. The matrices $\mathbf{Q}_A$ and $\mathbf{Q}_B$ are user-defined preconditioners. The inexact Uzawa iteration is formally presented in Algorithm 1.

The choice of $\mathbf{Q}_A^{-1}$ can be straightforward, we use $\mathbf{A}^{-1}$ to obtain an accurate $\mathbf{x}_k$. For $\mathbf{Q}_B$, some approaches choose a simple scaled diagonal matrix $\alpha\mathbf{I}$. However, such a simplistic choice does not guarantee convergence of our coupled system.

Algorithm 1 Inexact Uzawa Iteration

Initialize $\mathbf{x}_0 = \mathbf{0}$, $\mathbf{y}_0 = \mathbf{0}$, tolence tol $= 1$ and residual $res = \infty$.
while res $>$ tol **do**
  Obtain $\tilde{\mathbf{x}} = \mathbf{Q}_A^{-1}(\mathbf{F} - (\mathbf{A}\mathbf{x}_i + \mathbf{B}\mathbf{y}_i))$
  Update $\mathbf{x}_{i+1} = \mathbf{x}_i + \tilde{\mathbf{x}}$
  Obtain $\tilde{\mathbf{y}} = \mathbf{Q}_B^{-1}(\mathbf{B}^T\mathbf{x}_{i+1} - \mathbf{C}\mathbf{y}_i - \mathbf{G})$
  Update $\mathbf{y}_{i+1} = \mathbf{y}_i + \tilde{\mathbf{y}}$
  Update res $= \|\tilde{\mathbf{x}}\|_{l^2} + \|\tilde{\mathbf{y}}\|_{l^2}$
end while

We now seek an alternative for the preconditioner $\mathbf{Q}_B$. For the remainder of the discussion, we set $\mathbf{Q}_A = \mathbf{A}$,

$$\mathbf{x}_k = \mathbf{A}^{-1}\big(\mathbf{F} - \mathbf{B}\mathbf{y}_k\big). \tag{3.145}$$

Substituting equation (3.145) back into the system (3.144) yields a first order Richardson iteration for $\mathbf{y}$ as

$$\mathbf{y}_k = \mathbf{y}_{k-1} + \mathbf{Q}_B^{-1}\big[\mathbf{B}^T\mathbf{A}^{-1}\mathbf{F} - \mathbf{G} - (\mathbf{B}^T\mathbf{A}^{-1}\mathbf{B} + \mathbf{C})\mathbf{y}_{k-1}\big]. \tag{3.146}$$

The residual for this formulation would be when $y_k = y_{k-1}$, so let

$$\mathbf{r} = \mathbf{B}^T\mathbf{A}^{-1}\mathbf{F} - \mathbf{G} - (\mathbf{B}^T\mathbf{A}^{-1}\mathbf{B} + \mathbf{C})\mathbf{y}_{k-1}, \tag{3.147}$$

and this can be decomposed as $[\mathbf{r}_S, \mathbf{r}_D]^T$ according to the two subsystems. It is evident from (3.141) and (3.143) that the closer the preconditioner $\mathbf{Q}_B$ approximates the Schur complement for the coupled system, the better the algorithm behaves. Therefore, we can construct an alternative preconditioner $\mathbf{Q}_B$ by approximating the Schur complement separately for the Stokes and the Darcy subsystems as

$$\mathbf{Q}_B = \begin{bmatrix} (\mathbf{B}_S^T \mathbf{A}_S^{-1} \mathbf{B}_S + \mathbf{C}_S) & \mathbf{0} \\ \mathbf{0} & (\mathbf{B}_D^T \mathbf{A}_D^{-1} \mathbf{B}_D + \mathbf{C}_D) \end{bmatrix} = \begin{bmatrix} -\mathbf{S}_S & \mathbf{0} \\ \mathbf{0} & -\mathbf{S}_D \end{bmatrix}. \tag{3.148}$$

In practice, the Schur complement is not formed explicitly. Instead we compute an approximate increment $\tilde{\mathbf{y}}_i$, $i \in \{S, D\}$ within the Krylov subspace $\mathbf{V}_{k,i} = \mathcal{K}_r(-\mathbf{S}_i, \mathbf{r}_i)$, which minimizes the l^2 norm of the residual, i.e.,

$$\tilde{\mathbf{y}}_i = \operatorname{argmin}_{\tilde{\mathbf{y}} \in \mathbf{V}_{k,i}} \|-\mathbf{S}_i \tilde{\mathbf{y}} - \mathbf{r}_i\|, \quad i \in \{S, D\}. \tag{3.149}$$

Note that we may relax the minimization to accelerate the inner solver. The modified algorithm is presented formally in Algorithm 2.

Algorithm 2 Modified Uzawa Iteration

1: Initialize $\mathbf{x}_0 = \mathbf{0}$, $\mathbf{y}_0 = \mathbf{0}$, tolence tol > 0 and residual $res = \infty$.

2: **while** res $>$ tol **do**

3:     Obtain $\tilde{\mathbf{x}} = \mathbf{A}^{-1}(\mathbf{F} - (\mathbf{A}\mathbf{x}_i + \mathbf{B}\mathbf{y}_i))$, where $\tilde{\mathbf{x}} = [\tilde{\mathbf{x}}_S, \tilde{\mathbf{x}}_D]^T$

4:     Update $\mathbf{x}_{i+1} = \mathbf{x}_i + \tilde{\mathbf{x}}$

5:     Obtain $\mathbf{r} = \mathbf{B}^T \mathbf{x}_{i+1} + \mathbf{C}\mathbf{y}_i - \mathbf{G}$, where $\mathbf{r} = [\mathbf{r}_S, \mathbf{r}_D]^T$

6:     Obtain

$$\tilde{\mathbf{y}}_S = \operatorname{argmin}_{\tilde{\mathbf{y}} \in \mathbf{V}_{k,S}} \|-\mathbf{S}_S \tilde{\mathbf{y}} - \mathbf{r}_S\|_{l^2}$$
$$\tilde{\mathbf{y}}_D = \operatorname{argmin}_{\tilde{\mathbf{y}} \in \mathbf{V}_{k,D}} \|-\mathbf{S}_D \tilde{\mathbf{y}} - \mathbf{r}_D\|_{l^2}$$

7:     Update $\mathbf{y}_{i+1} = \mathbf{y}_i + [\tilde{\mathbf{y}}_S, \tilde{\mathbf{y}}_D]^T$

8:     Update res $= \|\tilde{\mathbf{x}}\|_{l^2} + \|\tilde{\mathbf{y}}\|_{l^2}$

9: **end while**

The Uzawa iteration, being a fixed point Richardson iteration, lends itself naturally to acceleration techniques, such as Anderson Acceleration Walker and Ni

(2011). We compare the result of our modified Uzawa iteration with the reference algorithm (3.143) and get identical results. Further investigation into its convergence behavior will be pursued in future work.

3.7 Convergence Test

In this section, we perform a convergence test on a 2D $M \times N$ grid with a pseudo 1D problem. We study a compacting column aligned along the y-direction, defined over the domain $y \in [-H, H]$ and $x \in [-L, L]$ with no flow boundary conditions imposed at $y = \pm H$.

For all tests presented below, we take $H = 2$ and $L = 0.2$, and the entire domain spans four compaction lengths. We fix the parameter $\Theta = 0$. Each problem is solved using four different levels of refinement in the y-direction, i.e., $N \in \{20, 40, 80, 160\}$. The code is implemented in PETSc/C++, with implementation details discussed in the appendix.

3.7.1 1D Simplification

To begin with, we assume that both the solid and the liquid velocities have nonzero components only in the y-direction, i.e., $\check{\mathbf{v}}_s = (0, v(y))$ and $\phi\check{\mathbf{v}}_r = (0, u(y))$. The 2D equations are then simplified as

$$
\begin{aligned}
u + \phi^2 q_l' &= 0, \\
u' + \phi(q_l - q_s) &= 0, \\
[q_s - \frac{1}{3}(1 - 4\phi)v']' &= 1 - \phi, \\
v' - \phi(q_l - q_s) &= 0.
\end{aligned}
\tag{3.150}
$$

The no flow boundary conditions for this one-dimensional problem are specified as

$$
u(\pm H) = -v(\pm H) = 0. \tag{3.151}
$$

When there is no melting present, $\phi = 0$, the equations reduce to a simple solution as

$$u = -v = 0, \quad q_s' = 1, \quad \text{and} \quad q_l = 0. \tag{3.152}$$

In the presence of melt $\phi > 0$, the equations can be reorganized as

$$\phi^2 \left(\frac{3 + \phi - 4\phi^2}{3\phi} u' \right)' - u = \phi^2 (1 - \phi), \tag{3.153}$$

where all other quantities are expressed in terms of u as

$$v = -u, \quad \text{and} \quad q_l' = -\phi^{-2} u, \quad \text{and} \quad q_s = u'/\phi + q_l. \tag{3.154}$$

We define an auxiliary function R for the case where ϕ is constant:

$$R = \left(\phi^2 \frac{3 + \phi - 4\phi^2}{3\phi} \right)^{-1/2}. \tag{3.155}$$

3.7.2 2D Boundary Conditions

Despite the problem being reduced to one dimension, boundary conditions are still specified for the full 2D computational domain. To begin with, we list all the degrees of freedom (dofs) that must be prescribed on edges of each element $e \in \partial E \subset \partial \Omega$.

- Stokes velocity $\check{\mathbf{v}}_s$ and bubble function $\mathbf{b}$;
  - Normal part: $\check{\mathbf{v}}_s \cdot \boldsymbol{\nu}$ and supplemental edge averaged flux $|e|^{-1} \int_e \mathbf{b} \cdot \boldsymbol{\nu} \, ds$ are prescribed compatibly together, i.e., they should be prescribed with same type of boundary conditions;
  - Tangential part: $\check{\mathbf{v}}_s \cdot \boldsymbol{\tau}$.
- Edge averaged Darcy normal flux $|e|^{-1} \int_e \phi \check{\mathbf{v}}_l \cdot \boldsymbol{\nu} \, ds = |e|^{-1} \int_e \mathbf{u} \cdot \boldsymbol{\nu} \, ds$.

The following boundary conditions are designed to replicate the behavior of the one-dimensional problem within the two-dimensional grid.

- Top and Bottom Edge $y = \pm H$;
  - Normal Stokes velocity and supplemental edge averaged flux are prescribed with essential boundary conditions:
  $$\check{\mathbf{v}}_s \cdot \boldsymbol{\nu} = v = 0,\ |e|^{-1} \int_e \mathbf{b} \cdot \boldsymbol{\nu}\, ds = 0.$$
  - Tangential Stokes velocity is prescribed with essential boundary condition:
  $$\check{\mathbf{v}}_s \cdot \boldsymbol{\tau} = 0.$$
  - Normal Darcy flux is prescribed with essential boundary condition:
  $$|e|^{-1} \int_e \phi \check{\mathbf{v}}_r \cdot \boldsymbol{\nu}\, ds = |e|^{-1} \int_e u\, ds = 0.$$
- Left and Right Edges $x = \pm L$;
  - Normal Stokes velocity and supplemental edge averaged flux are prescribed with essential boundary conditions:
  $$\check{\mathbf{v}}_s \cdot \boldsymbol{\nu} = v = 0,\ |e|^{-1} \int_e \mathbf{b} \cdot \boldsymbol{\nu}\, ds = 0;$$
  - Tangential Stokes velocity is prescribed with the natural boundary condition: free traction
  $$t_0 = 0;$$
  - Normal Darcy flux is prescribed with essential boundary condition:
  $$|e|^{-1} \int_e \phi \check{\mathbf{v}}_r \cdot \boldsymbol{\nu}\, ds = |e|^{-1} \int_e u\, ds = 0.$$

We then test our MFEM implementation using three different porosity configurations, and compare the results against some of the closed-form solutions derived in Arbogast et al. (2017).

3.7.3 Column Compaction with Constant Porosity

First, we start with a constant porosity:

$$\phi(y) = \phi_0. \tag{3.156}$$

We solve the ODE (3.153) and write the closed form solution in terms of the auxiliary constant (3.155).

$$\begin{aligned}
u &= -\phi_0^2(1-\phi_0)\left[1 - \frac{\cosh(Ry)}{\cosh(RL)}\right],\\
v &= -u,\\
q_l &= (1-\phi_0)\left[y - \frac{1}{R}\frac{\sinh(Ry)}{\cosh(RL)}\right] + c,\\
q_s &= (1-\phi_0)\left[y - \frac{1-4\phi_0}{3+\phi_0-4\phi_0^2}\frac{\phi_0}{R}\frac{\sinh(Ry)}{\cosh(RL)}\right] + c,
\end{aligned} \tag{3.157}$$

where c is a constant.

In Table 3.3, we confirm an optimal second-order convergence rate in L^2-norm of errors for the scalar velocities u and v. Note that the L^2-norm of errors for the pressure potentials q_s and q_l are computed with the mid-point quadrature rule, which results in a second order superconvergence behavior for these quantities. In Figure 3.11, we demonstrates that the numerical solutions (dashed lines) closely match the closed-form solutions (solid lines).

	u		v		q_l		q_s	
$M \times N$	L^2 error	rate	L^2 error	rate	L^2 error	rate	L^2 error	rate
2×20	3.100e-05	-	3.100e-05	-	3.285e-03	-	3.638e-05	-
2×40	8.047e-06	1.95	8.047e-06	1.95	8.731e-04	1.91	9.671e-06	1.91
2×80	2.031e-06	1.99	2.031e-06	1.99	2.218e-04	1.98	2.457e-06	1.98
2×160	5.091e-07	2.00	5.091e-07	2.00	5.567e-05	1.99	6.166e-07	1.99

Table 3.3: Constant porosity (3.156), with $\phi_0 = 0.04$. L^2 norm of errors and convergence rates for the scalar velocities u and v, the solid and the liquid pressure potentials q_s and q_l.

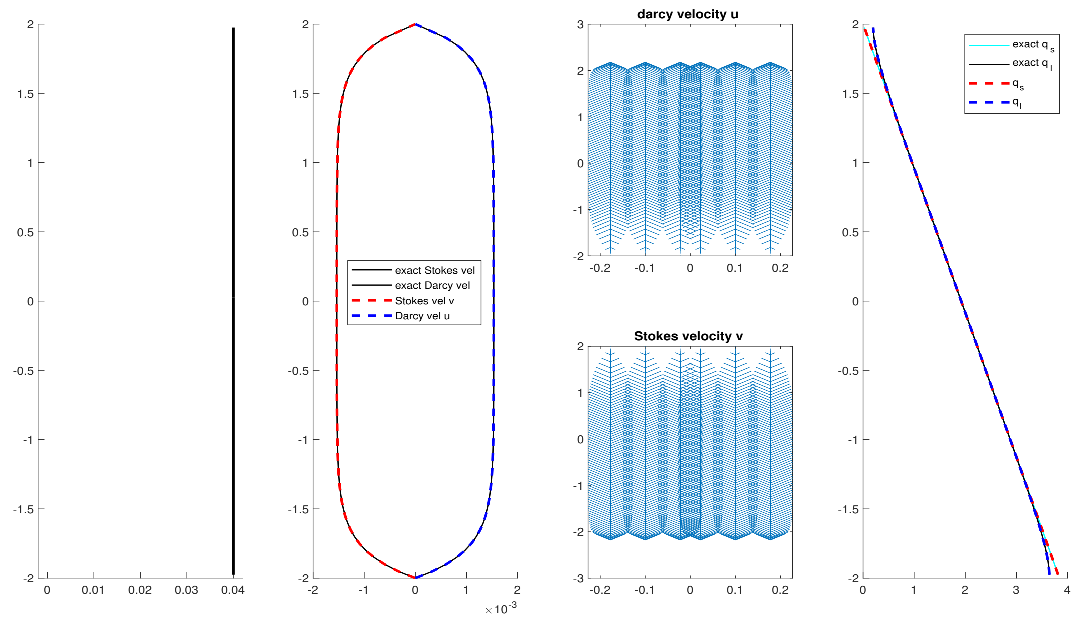


Figure 3.11: Constant porosity (3.156) with $\phi_0 = 0.04$. Shown are the porosity ϕ, scalar velocities u and v along with closed form solution, velocity vectors $\check{\mathbf{v}}_s$ and $\phi\check{\mathbf{v}}_r$ reconstructed on the edge quadrature points, q_l and q_s together.

3.7.4 Column Compaction with Discontinuous Porosity

We proceed to test our implementation with a discontinuous porosity distribution, featuring a "jump" at $y = 0$. The region $y > 0$ part is considered as degenerate $\phi = 0$.

$$\phi(z) = \begin{cases} 0 & \text{if } z > 0, \\ \phi_0 & \text{if } z \leq 0, \end{cases} \tag{3.158}$$

The even number of mesh grid in y-direction ensures that the discontinuity lands on the edge between two elements. The closed form solution is also given in terms of the auxiliary constant (3.155) as:

- When $y > 0$,

$$u = -v = 0,\ q_l = 0,\ \text{and } q_s = z + c;$$

$M \times N$	u L^2 error	rate	v L^2 error	rate	q_l L^2 error	rate	q_s L^2 error	rate
2×20	2.980e-05	-	2.980e-05	-	3.794e-03	-	4.202e-05	-
2×40	7.681e-06	1.96	7.681e-06	1.96	1.005e-03	1.92	1.114e-05	1.92
2×80	1.958e-06	1.97	1.958e-06	1.97	2.472e-04	2.02	2.738e-06	2.02
2×160	4.974e-07	1.98	4.974e-07	1.98	5.972e-05	2.04	6.615e-07	2.05

Table 3.4: Discontinuous porosity (3.158), with $\phi_0 = 0.04$. L^2 norm of errors and convergence rates for the scalar velocities u and v, the solid and the liquid pressure potentials q_s and q_l.

- When $y < 0$ and $\phi_0 \neq 0$,

$$
\begin{aligned}
u &= \phi_0^2(1-\phi_0)\left[1-\cosh(Ry)+\frac{1-\cosh(RL)}{\sinh(RL)}\sinh(Ry)\right],\\
v &= -u,\\
q_l &= -(1-\phi_0)\left[z-\frac{1}{R}\sinh(Ry)+\frac{1-\cosh(RL)}{R\sinh(RL)}\cosh(Ry)\right]+c,\\
q_s &= q_l+\frac{u'}{\phi_0}+c,
\end{aligned}
$$

  where c is a constant.

In Table 3.4, we also confirm an optimal second-order convergence rate in L^2-norm of errors for the scalar velocities u and v and a second-order superconvergence rate of the pressure potentials q_s and q_l due to using the mid-point quadrature rule. In Figure 3.12, we show that the numerical solutions (dashed lines) closely match the closed-form solutions (solid lines) with a degenerate region.

3.7.5 Column Compaction With Quadratic Porosity

For the final test, we consider our implementation with the continuous quadratic porosity

$$
\phi(y) = \begin{cases} 0 & \text{if } y > 0, \\ \phi_0 y^2 & \text{if } y \leq 0. \end{cases} \tag{3.159}
$$

We focus on the case where $\phi_0 = 0.01$, which is not accurately captured by the closed-form solution derived in Arbogast et al. (2017) due to the breakdown of the

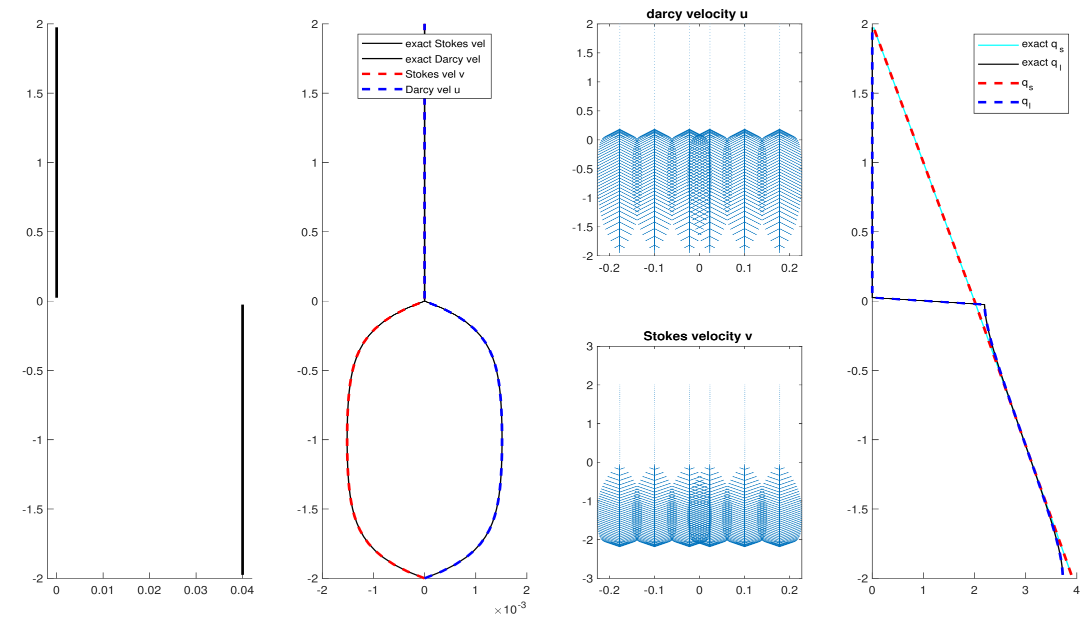


Figure 3.12: Discontinuous porosity (3.158) with $\phi_0 = 0.04$. Shown are the porosity ϕ, scalar velocities u and v along with closed form solution, velocity vectors $\check{\mathbf{v}}_s$ and $\phi\check{\mathbf{v}}_r$ reconstructed on the edge quadrature points, q_l and q_s together.

assumption $\phi \ll 1$. To address this, we present an alternative approach of constructing an approximation solution to the second-order ordinary differential equation (ODE) (3.153).

To begin with, we rewrite our second order ODE into

$$f(u'', u', u, z) = P(z)u'' + Q(z)u' - u = \phi^2(1-\phi), \tag{3.160}$$

where $z = |y|$ for simplicity and

$$\begin{aligned} P(z) &= \phi_0 z^2 + \frac{\phi_0^2 z^4}{3} - \frac{4\phi_0^3 z^6}{3}, \\ Q(z) &= -2\phi_0 z - \frac{8\phi_0^3 z^5}{3}. \end{aligned} \tag{3.161}$$

We conclude that the singular point at $z = 0$ is a regular singular point given

$$\lim_{z\to 0} z\frac{Q(z)}{P(z)} < \infty, \text{ and } \lim_{z\to 0} z^2\frac{1}{P(z)} < \infty, \tag{3.162}$$

which suggests that this second-order ODE can be approximated using a series solution via the Frobenius method. The solution can be interpreted as the sum of a particular solution solving $f(u'', u', u, z) = \phi^2(1-\phi)$ and a homogeneous power-series solution solving $f(u'', u', u, z) = 0$. We assume that the series solution u and its first and second order derivatives take the form

$$u = \sum_{n=0}^{\infty} c_n z^{n+r},\ u' = \sum_{n=0}^{\infty}(n+r)c_n z^{n-1+r},\ \text{and}\ u'' = \sum_{n=0}^{\infty}(n+r)(n-1+r)c_n z^{n-2+r}. \tag{3.163}$$

For the homogenous soution, substitute these series (3.163) in the homogeneous equation. We have the recurrence relationship of the coefficients as

$$\begin{aligned}
&\left\{\phi_0\Big[(n+r-3)(n+r)\Big]-1\right\}c_n+\\
&\left\{\frac{\phi_0^2}{3}\Big[(n+r-2)(n+r-3)\Big]\right\}c_{n-2}+\\
&\left\{\frac{-4\phi_0^3}{3}\Big[(n+r-4)(n+r-3)\Big]\right\}c_{n-4}=0,
\end{aligned} \tag{3.164}$$

where $c_m = 0$ if $m < 0$. We obtain the indicial solution by equating the coefficient associated to the c_0 term to zero, i.e.,

$$\phi_0\big[(0+r-3)(0+r)\big]-1=0. \tag{3.165}$$

The associated quadratic equation yields two roots, which can be written explicitly with the quadratic formula as

$$r_1 = \frac{3+\sqrt{9+4/\phi_0}}{2}\ \text{and}\ r_2 = \frac{3-\sqrt{9+4/\phi_0}}{2}. \tag{3.166}$$

Here, we denote u_{r_1} and u_{r_2} the two homogenous solutions with different powers r_1 and r_2.

For the particular solution, since the right hand side of (3.160) is an even polynomial, we can find a solutin with $r = 0$, and all odd indexed coefficients being zero. Moreover, $c_0 = c_2 = 0$ and the first three non-zero coefficients of the particular solution are

$$\begin{aligned} c_4 &= \frac{\phi_0^2}{4\phi_0 - 1}, \\ c_6 &= \frac{-\phi_0^3 - 4\phi_0^2 c_4}{18\phi_0 - 1}, \\ c_8 &= \frac{80/3\phi_0^3 c_4 - 10\phi_0^2 c_6}{40\phi_0 - 1}. \end{aligned} \tag{3.167}$$

The remaining coefficients can be determined through the recurrence relationship (3.163). We assume the solution takes the form as a linear combination of particular solution and two homogeneous solutions:

$$u = u_p + Au_{r_1} + Bu_{r_2}, \tag{3.168}$$

where u_p is the particular solution, u_{r_1} and u_{r_2} are the two homogeneous solutions with different powers r_1 and r_2. The linear coefficients A and B can be determined by solving a linear system arising from the imposed boundary conditions. In our case, we drop the negative root r_2 when the singular point $z = 0$ is included. The coefficient A can therefore be solved with the boundary condition at $z = 2$ as

$$A = -u_p(z = 2)/u_{r_1}(z = 2). \tag{3.169}$$

Using the full series solution (3.168), we now perform the convergence test on two different porosity scales $\phi_0 = 0.01$ and $\phi_0 = 0.001$.

In Table 3.5 and Table 3.6, we observe second order convergence rate of the L^2-norm of errors for scalar velocities u and v without degradation. However, it is surprising that the solid and the liquid pressure potentials q_s and q_l do not exhibit the correct convergence rate towards the new closed-form solution. We notice that the numerical solution and the close-form solution no longer agree with each other close to the bottom boundary $y = -2$. Figure 3.13 and Figure 3.14 illustrate that the numerical solutions (dashed lines) closely match the closed-form solutions (solid lines), despite the pressure potentials not converging to the closed-form solution. Further investigation is needed to understand this discrepency.

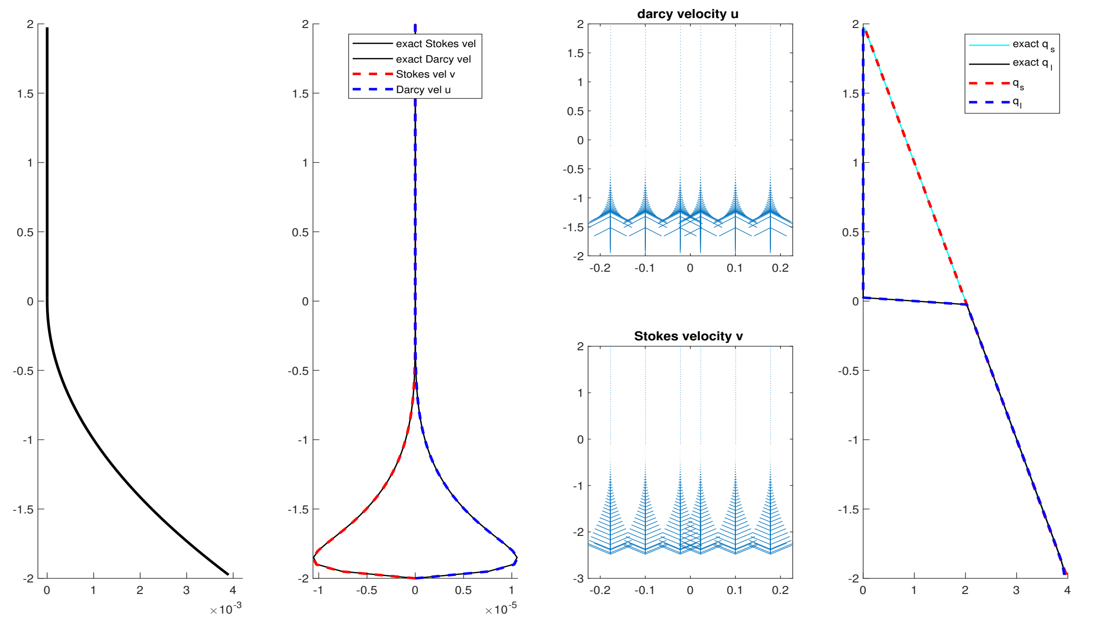


Figure 3.13: Quadratic porosity (3.159) with $\phi_0 = 0.001$. Shown are the porosity ϕ, scalar velocities u and v along with closed form solution, velocity vectors $\check{\mathbf{v}}_s$ and $\phi\check{\mathbf{v}}_r$ reconstructed on the edge quadrature points, q_l and q_s together.

	u		v		q_l		q_s	
$M \times N$	L^2 error	rate	L^2 error	rate	L^2 error	rate	L^2 error	rate
2×20	9.558e-07	-	9.558e-07	-	1.210e-03	-	1.895e-06	-
2×40	3.176e-07	1.59	3.176e-07	1.59	3.618e-04	-	6.278e-07	-
2×80	8.686e-08	1.87	8.686e-08	1.87	3.844e-04	-	5.518e-07	-
2×160	2.222e-08	1.97	2.222e-08	1.97	5.099e-04	-	7.0145e-07	-

Table 3.5: Quadratic porosity (3.159) with $\phi_0 = 0.001$. L^2 norm of errors and convergence rates for the scalar velocities u and v, the solid and the liquid pressure potentials q_s and q_l.

3.8 Simulation of a Simple Mid-Ocean Ridge Problem

We now revisit the Mid-Ocean ridge test case in Arbogast et al. (2017). The full system of equations is solved on a dimensionless rectangular domain defined by $x \in [-0.5, 0.5]$ and $y \in [-0.5, 0.0]$. Boundary conditions are prescribed by imposing classical corner flow solution for the Stokes velocity as described by Spiegelman and

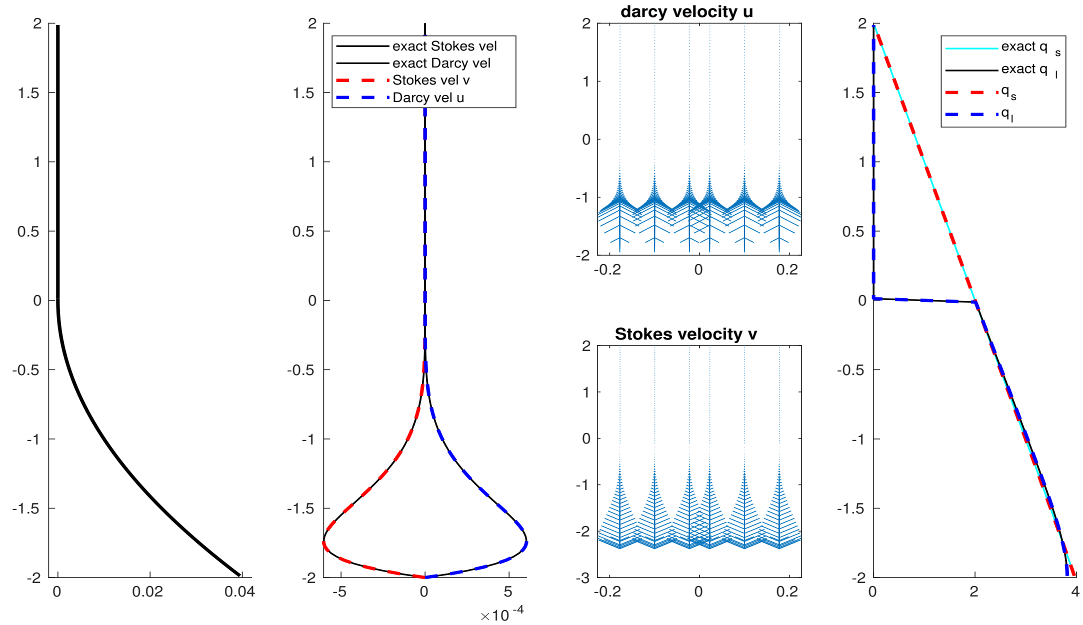


Figure 3.14: Quadratic porosity (3.159) with $\phi_0 = 0.01$. Shown are the porosity ϕ, scalar velocities u and v along with closed form solution, velocity vectors $\check{\mathbf{v}}_s$ and $\phi\check{\mathbf{v}}_r$ reconstructed on the edge quadrature points, q_l and q_s together.

	u		v		q_l		q_s	
$M \times N$	L^2 error	rate	L^2 error	rate	L^2 error	rate	L^2 error	rate
2×20	2.438e-05	-	2.438e-05	-	5.571e-03	-	7.842e-05	-
2×40	6.249e-06	1.96	6.249e-06	1.96	7.416e-03	-	9.870e-05	
2×80	1.486e-06	2.07	1.486e-06	2.07	9.317e-03	-	1.254e-04	
2×160	3.908e-07	1.96	3.908e-07	1.96	1.056e-02	-	1.439e-04	

Table 3.6: Quadratic porosity (3.159) with $\phi_0 = 0.01$. L^2 norm of errors and convergence rates for the scalar velocities u and v, the solid and the liquid pressure potentials q_s and q_l.

McKenzie (1987) as essential boundary condition along the entire boundary of the computational domain

$$\mathbf{v}_s = \frac{2U_0}{\pi(x^2+y^2)} \begin{pmatrix} \tan^{-1}(\frac{x}{-y})(x^2+y^2) + xy \\ z^2 \end{pmatrix}, \tag{3.170}$$

where $U_0 = 10^{-9}$m/s $\approx$ 3.2cm/yr is the constant upwelling velocity of mantle. In accordance with the mass conservation condition, we impose zero Darcy flux across the boundary. We then set a not identically vanishing porosity by the formula:

$$\phi(x,y) = \begin{cases} 0.05(\dfrac{0.375+y}{0.375})^2(1-\dfrac{|x|}{|y|+l}), & \text{if } |y| \leq 0.375 \text{ and } |x| \leq |z|+l, \\ 0, & \text{otherwise,} \end{cases} \tag{3.171}$$

where $l = 20\,\text{m}$ is the offset used to avoid a singularity at the center $x = 0$, i.e., we translate the coordinates x to $x - l$ if $x < 0$, and $x + l$ when $x > 0$. In Figure 3.15, we show the porosity distribution within the computational domain. We perform the current numerical simulation over the full domain.

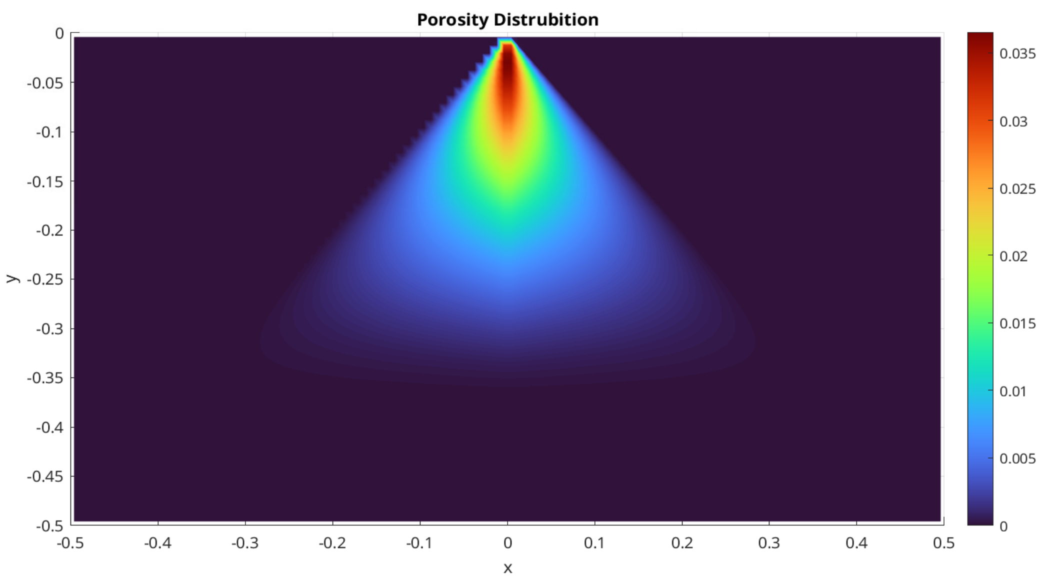


Figure 3.15: Porosity distribution.

In Figure 3.16 and Figure 3.17, we exhibit the dimensionless Stokes velocity $(\check{\mathbf{v}}_s, \check{q}_s)$ and Darcy velocity $(\phi\check{\mathbf{v}}_r, \check{q}_l)$ respectively. We observe the molten magma rising toward the surface and concentrating in regions where most porosity is presents. The solid matrix upwells from the bottom of the domain. We observe compaction occurs in the partially molten region where the buoyancy can no longer support overlaying the solid matrix due to the density difference between the solid and the liquid phases. Despite the presence of an extensive no melting region, the system is solved successfully using the modified Uzawa iteration

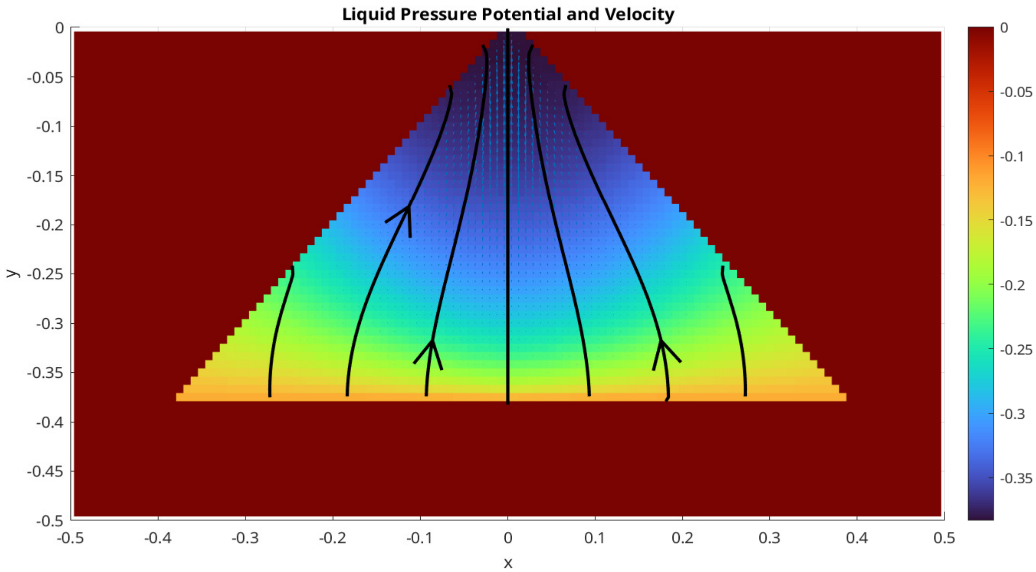


Figure 3.16: Liquid pressure potential and velocity.

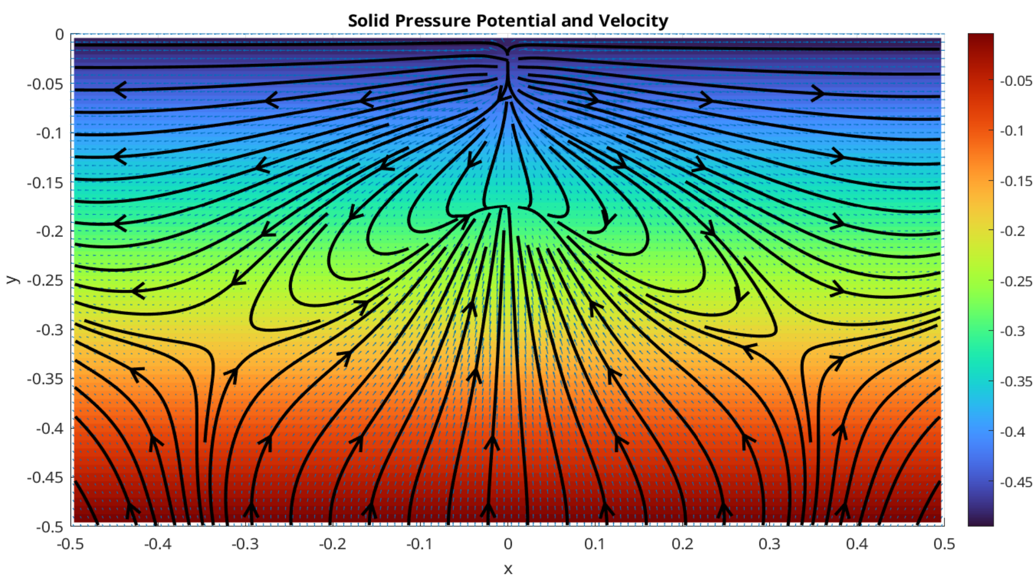


Figure 3.17: Solid pressure potential and velocity.

3.9 Finite Volume Method

The finite volume method (FVM) is a widely used discretization method for various problems governed by conservation laws. The FVM is particularly robust and well-suited for arbitrary complex geometry. A key advantage of FVM that benefits this simulation is its natural enforcement of local mass conservation.

To begin with, we consider a general hyperbolic conservation law regarding a scalar variable u as

$$\begin{cases} \dfrac{\partial u(\mathbf{x},t)}{\partial t} + \nabla \cdot \mathbf{f}(u;\mathbf{x},t) = 0, & t > 0, \mathbf{x} \in \Omega, \\ u(\mathbf{x},0) = u_0(\mathbf{x}), & \mathbf{x} \in \Omega, \end{cases} \tag{3.172}$$

where we assume the domain $\Omega \subset \mathbb{R}^2$. The term $\mathbf{f}$ represents the flux transporting the scalar variable u. The basic semi-discretized formulation of the finite volume method can be written as

$$\frac{\partial \overline{u}_E}{\partial t} + \frac{1}{|E|} \int_{\partial E} F(\mathbf{x},t)\, d\mathbf{x} = 0, \tag{3.173}$$

where the cell-averaged value $\overline{u}$ is calculated as

$$\overline{u}_E = \frac{1}{|E|} \int_E u(\mathbf{x},t)\, d\mathbf{x}, \tag{3.174}$$

where $|E|$ denote the area of the cell E. Here, $F(\mathbf{x},t)$ denotes the approximated numerical flux. We employ the Lax-Friedrichs numerical flux

$$F(u^+,u^-) = \frac{1}{2}\Big[\big(\mathbf{f}(u^+) + \mathbf{f}(u^-)\big) \cdot \boldsymbol{\nu} - \alpha_{LF}(u^+ - u^-)\Big], \tag{3.175}$$

where u^+ and u^- denote the values approximated from the perspective of element $E^+ \in \mathcal{T}_h$ and $E^- \in \mathcal{T}_h$ respectively, where $E^+ \cap E^-$ is an edge of the mesh. The unit normal vector $\boldsymbol{\nu}$ points from E^- to E^+, and is orthogonal to the edge. The parameter α_{LF} is the Lax-Friedrichs stabilizer factor. Depending on the available information about the solution, there are two possible approaches for computing α_{LF}:

- The global Lax-Friedrichs Stabilizer:

$$\alpha_{LF} = \max_{\mathbf{x}\in\Omega,\, t>0,\, u\in\mathbb{R}} \left| \frac{\partial \mathbf{f}(u;\mathbf{x},t) \cdot \boldsymbol{\nu}}{\partial u} \right|;$$

- The local Lax-Friedrichs Stabilizer:

$$\alpha_{LF} = \max_{\text{edges}\, e} \left(\left| \frac{\partial \mathbf{f}(u^+; \mathbf{x}, t) \cdot \boldsymbol{\nu}}{\partial u} \Big|_e \right|, \left| \frac{\partial \mathbf{f}(u^-; \mathbf{x}, t) \cdot \boldsymbol{\nu}}{\partial u} \Big|_e \right| \right).$$

The numerical flux F, as defined in equation (3.175), is consistent and conservative across element interfaces. We now replace the u^+ and u^- with reconstructed values $\mathcal{R}^+$ and $\mathcal{R}^-$ and express the numerical flux as

$$F(\mathcal{R}^+, \mathcal{R}-) = \frac{1}{2}[(\mathbf{f}(\mathcal{R}^+) + \mathbf{f}(\mathcal{R}^-) \cdot \boldsymbol{\nu} - \alpha_{LF}(\mathcal{R}^+ - \mathcal{R}^-)]. \tag{3.176}$$

Here, the reconstructed value $\mathcal{R}$ is of higher-order accuracy. We evaluate the integrals over each edge $e_i \in \partial E$ with a suitable quadrature rule. Let $|e_i|$ denote the length of edge e_i. Assuming a quadrature rule with G point per edge, the semi-discretized formulation becomes:

$$\frac{\partial \overline{u}_E}{\partial t} + \sum_{i=0}^{4} \frac{|e_i|}{|E|} \sum_{g=0}^{G} \omega_{i,g} F(\mathcal{R}^+_{i,g}, \mathcal{R}^-_{i,g}) = 0, \tag{3.177}$$

where $\omega_{i,g}$ represents the weight associated to the $g - th$ quadrature points on edge e_i.

3.10 The Multi-level Weighted Essentially Non-Oscillatory (ML-WENO) Method

We developed an accurate, versatile, robust and efficient finite volume multi level weighted essentially non oscillatory (ML-WENO) reconstruction in Arbogast et al. (2024). ML-WENO is primarily used to reconstruct values on both sides of the element interfaces. It can also be applied to a general reconstruction of discontinuous variables, e.g., piece-wise constant discontinuous Darcy pressure.

3.10.1 Stencil Polynomials

Consider rectangular stencils $\mathcal{S} = \{E_{i,j},\, i \leq s,\, j \leq t\} \subset \mathcal{T}_h$ on a quasi uniform logically rectangular mesh, which covers the area $\Omega_{\mathcal{S}} = \cup_{E \in \mathcal{S}} E$. For each stencil, we construct a tensor product stencil polynomial $Q^{\mathcal{S}}_{s,t} \in \mathbb{Q}_{s-1,t-1}(\Omega_{\mathcal{S}})$. The scaled tensor product stencil polynomial is expressed as

$$Q^{\mathcal{S}}_{s,t}(x,y) = \sum_{p<s,\ q<t} c_{p,q}\Big(\frac{x-x_{\mathcal{S}}}{h_{\mathcal{S}}}\Big)^p\Big(\frac{y-y_{\mathcal{S}}}{h_{\mathcal{S}}}\Big)^q, \tag{3.178}$$

where $(x_{\mathcal{S}}, y_{\mathcal{S}})$ is any point in the stencil $\mathcal{S}$. We choose the stencil "center" point: the intersection point of two diagonal lines joining top left, bottom right and top right, bottom left corners. Assuming that each cell $E_{i,j} \in \mathcal{S} \subset \mathcal{T}_h$ is shape regular as in Definition 3.1, we define the stencil scale parameter $h_{\mathcal{S}} = \sum_{i\leq s, j\leq t} E_{i,j}/(st)$ as the average diameter of all cells in the stencil. Introducing the normalized coordinates

$$\xi = \frac{x-x_{\mathcal{S}}}{h_{\mathcal{S}}}, \quad \text{and} \quad \eta = \frac{y-y_{\mathcal{S}}}{h_{\mathcal{S}}}. \tag{3.179}$$

We can rewrite equation (3.178) in normalized coordinates as

$$\hat{Q}^{\mathcal{S}}_{s,t}(\xi,\eta) = \sum_{p<s,\ q<t} c_{p,q}\xi^p\eta^q. \tag{3.180}$$

The tensor product stencil polynomial satisfies the condition that the cell-averaged value (3.174) is preserved through integration over each cell as

$$\frac{1}{|E_{i,j}|}\iint_{E_{i,j}} Q^{\mathcal{S}}_{s,t}(x,y)\,dxdy = \overline{u}_{i,j}, \quad \forall E_{i,j} \in \mathcal{S}, \tag{3.181}$$

where $\overline{u}_{E_{i,j}}$ represents the cell-averaged scalar variable over cell $E_{i,j}$.

One way to construct a stencil polynomial $Q^{\mathcal{S}}_{s,t}$ is using basis polynomials. Let $Q^{\mathcal{S}}_{s,t} = \sum_{i,j} \overline{u}_{i,j}\phi^{i,j}_{s,t}$, where the basis polynomials $\phi^{i,j}_{s,t}$ satisfy

$$\frac{1}{|E_{i',j'}|}\iint_{E_{i',j'}} \phi^{i,j}_{s,t}(x,y)\,dxdy = \begin{cases} 1 & (i',j') = (i,j) \\ 0 & (i',j') \neq (i,j) \end{cases}, \quad \forall E_{i',j'} \in \mathcal{S}, \tag{3.182}$$

which always leads to an $(st)\times(st)$ non-singular linear system. We can write the basis polynomials in terms of normalized coordinates ξ and η as well, and rewrite them as

$$\hat{Q}^{\mathcal{S}}_{s,t}(\xi,\eta) = \sum_{i,j} \overline{u}_k \hat{\phi}^{i,j}_{s,t}(\xi,\eta). \tag{3.183}$$

The approxmatibility of a tensor product polynomial is related to the associated complete polynomial , as discussed by Dupont and Scott (1980). Given a bounded interpolation projection operator $\mathcal{Q}$, for all $\mathbf{x} \in E \in \mathcal{S}$, we have

$$|D^\alpha u(\mathbf{x}) - D^\alpha\big(\mathcal{Q}u(\mathbf{x})\big)| \leq \begin{cases} Ch^{r-|\alpha|} & \text{if } u \text{ is smooth on the stencil,} \\ C & \text{otherwise,} \end{cases} \tag{3.184}$$

where $\alpha = (\alpha_1, \alpha_2)$ is a two-dimensional multi-index, and $|\alpha| = \alpha_1 + \alpha_2 \leq r$. This theorem applies to our stencil polynomials (3.178).

In most two-dimensional cases, we implement a square stencil with $s = t = r$. Uneven stencils ($s \neq t$) are used in the following two situations.

- When some a prior knowledge of the solution is acquired before simulation, e.g., the transport velocity is known to be fixed in a single direction.
- When derivatives are approximated, e.g., diffusive flux is computed across the element edge, where a stretched stencil is necessary to compensate for the loss of accuracy associated with derivative approximation.

For simplicity, we restrict the following discussions to the square case $s = t = r$.

3.10.2 Stencil Polynomial Smoothness Indicators

A smoothness indicator σ is defined to quantify the smoothness of the tensor product polynomial $Q_{r,r}^{\mathcal{S}}$ on the stencil $\mathcal{S}$. The classical approach by Jiang and Shu (1996) and Friedrich (1998) defines σ in analogue to a Sobolev seminorm as

$$\sigma_{JS} = \sum_{1 \leq |\alpha| \leq r} \int\int_{E_0} \frac{h_{\mathcal{S}}^{2|\alpha|}}{|E_0|} (\mathcal{D}^\alpha Q(x,y))^2 dxdy, \tag{3.185}$$

where E_0 can be any cell $E \in \mathcal{S}$, and is usually referred to as the target cell, i.e., the cell E form witch (x, y) is taken in $Q_{r,r}^{\mathcal{S}}(x, y)$.

The Jiang-Shu smoothness indicator can be computed utilizing the predefined basis polynomials (3.182):

$$\begin{aligned}
\sigma_{JS} &= \sum_{1\le|\alpha|\le r} \frac{h_{\mathcal{S}}^{2|\alpha|}}{|E_0|} \iint_{E_0} (\mathcal{D}^\alpha Q(x,y))^2 dxdy \\
&= \frac{h_{\mathcal{S}}^2}{|E_0|} \sum_{1\le|\alpha|\le r} \iint_{\hat{E}_0} (\mathcal{D}^\alpha \hat{Q}(\xi,\eta))^2 d\xi d\eta \\
&= \frac{h_{\mathcal{S}}^2}{|E_0|} \sum_{1\le|\alpha|\le r} \iint_{\hat{E}_0} \big(\sum_{ij} \overline{u}_{ij} \mathcal{D}^\alpha \hat{\phi}_{i,j}(\xi,\eta)\big)^2 d\xi d\eta \\
&= \frac{h_{\mathcal{S}}^2}{|E_0|} \sum_{ij}\sum_{i'j'} \overline{u}_{ij}\overline{u}_{i'j'} \sum_{1\le|\alpha|\le r} \iint_{\hat{E}_{ij}} \mathcal{D}^\alpha \hat{\phi}_{ij}(\xi,\eta) \mathcal{D}^\alpha \hat{\phi}_{i'j'}(\xi,\eta) d\xi d\eta \\
&= \frac{h_{\mathcal{S}}^2}{|E_0|} \sum_{ij}\sum_{i'j'} \overline{u}_{ij}\overline{u}_{i'j'} \sigma_{ij}^{i'j'},
\end{aligned} \tag{3.186}$$

where the coefficient tensor $\sigma_{ij}^{i'j'}$ can be precomputed before the time-stepping loop, after the mesh is established.

Alternatively, we can compute the Jiang-Shu smoothness indicator directly using the polynomial coefficients formulation (3.180) as

$$\begin{aligned}
\sigma_{JS} &= \frac{h_{\mathcal{S}}^2}{|E_0|} \sum_{1\le|\alpha|\le r} \iint_{\hat{E}_0} (\mathcal{D}^\alpha \hat{Q}(\xi,\eta))^2 d\xi d\eta \\
&= \sum_{|\gamma|\le r}\sum_{|\beta|\le r} c_\gamma c_\beta \Big(\sum_{1\le|\alpha|\le r} \frac{h_{\mathcal{S}}^2}{|E_0|} \iint_{E_0} \mathcal{D}^\alpha \xi^\gamma \mathcal{D}^\alpha \eta^\beta \, d\xi d\eta \Big) \\
&= \sum_{|\gamma|\le r}\sum_{|\beta|\le r} c_\gamma c_\beta \zeta_{\gamma,\beta},
\end{aligned} \tag{3.187}$$

which paves the way for further simplification in (3.190) below.

The smoothness indicator exhibits the following convergence property: there exists a constant $C > 0$ such that when $h \to 0^+$,

$$\sigma = \begin{cases} Ch^2 + \mathcal{O}(h^3) & \text{u is smooth on } \Omega_{\mathcal{S}}, \\ \Theta(1) & \text{u is not smooth on } \Omega_{\mathcal{S}}. \end{cases} \tag{3.188}$$

Actually $\sigma = O(1)$ in the nonsmooth case, but in practice we find that $\sigma = \Theta(1)$. Being $\Theta(1)$ means that σ is bounded above and velow by a constant.

3.10.3 Two Alternative Smoothness Indicators

We propose two alternative ways of defining a smoothness indicator

- *Simple smoothness indicator* Huang et al. (2023)

$$\sigma_{\text{simp}} = \frac{1}{N_{\mathcal{S}} - 1} \sum_k (\overline{u}_k - \overline{u}_0)^2, \tag{3.189}$$

  where $\overline{u}_0$ is the cell averaged solution in the target cell. This simple smoothness indicator is designed to reduce the computational cost and to deal with distorted stencils on unstructured meshes. However, this simple smoothness indicator is not as robust as the classic Jiang-Shu smoothness indicator.

- *Polynomial smoothness indicator* Arbogast et al. (2025)

$$\sigma_P = \sum_\alpha \zeta_{\alpha,\alpha} c_\alpha^2 \tag{3.190}$$

  which is obtained by dropping all the cross terms appeared in the formulation (3.187). This alternative serves as a good approximation to the classic Jiang-Shu smoothness indicator while the computing efficiency is greatly improved.

3.10.4 ML-WENO Weighting

Given a target cell $E_{i,j}$, we can create a nonempty set of stencils $\{\mathcal{S}_l \ni E_{i,j}, l = 0, 1, \ldots, L, L \geq 0\}$.(See Figure 3.18 for an example.) The reconstruction of u on the target cell $E_{i,j}$ is a convex combination of the stencil polynomials defined in equation (3.183)

$$\mathcal{R}(\mathbf{x}) = \sum_l \tilde{\omega}_l Q^{\mathcal{S}_l}(\mathbf{x}), \quad \mathbf{x} \in E_{i,j}, \tag{3.191}$$

where $\sum_l \tilde{\omega}_l = 1$ and $\tilde{\omega}_l$ are the normalized nonlinear weights associated with stencils. We begin by arbitrarily choosing a set of linear weights $\{\omega_l\}$ and scale them with

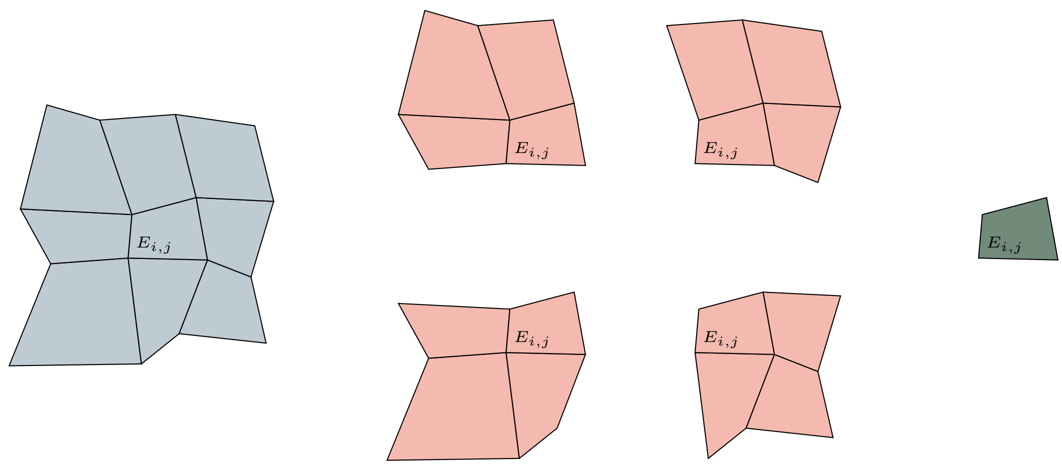


Figure 3.18: An example of a (3,2,1) multi-level stencils combination on a logically rectangular quadrilateral mesh.

stencil smoothness indicator as

$$\hat{\omega}_l = \frac{\omega_l}{(\sigma_l + \epsilon h_0^2)^{ar_l + n_l}}, \tag{3.192}$$

where $r_l = \max\{s, t\} = r$ in the cases of using square stencils, and the constants ϵ = 1e-4 and $a > 0.5$ are chosen accordingly. The biasing factor $n_l = n(r_l)$ is used to biasing further away from the constant level, whose smoothness indicator is always zero. We choose

$$n(r_l) = n_l = \begin{cases} 1, & \text{if} \quad r_l = 1, \\ 3, & \text{if} \quad r_l = 2, \\ 4, & \text{if} \quad r_l \geq 3. \end{cases} \tag{3.193}$$

We finally renormalise the scaled weights as

$$\tilde{\omega}_l = \frac{\hat{\omega}_l}{\sum_{l'} \hat{\omega}_{l'}}. \tag{3.194}$$

3.10.5 The ML-WENO Reconstruction Error

In this subsection, we analyze the accuracy of the new multi-level weighting scheme. Consider the set containing indices of stencils $l \in \mathcal{L} = \{0, 1, 2, \ldots L\}$. Specifically, we denote the set of indices of stencils of size $s \times t$ as $\mathcal{L}_{s,t} \subset \mathcal{L}$. We assume the mesh resolution is sufficiently fine such that at least one smooth stencil exists in the presence of shocks. Accordingly, we define two index sets: one containing the stencils identified as smooth, and the other containing those that exhibit discontinuities as

$$\begin{aligned} \text{Smooth} &= \{l: \ \ u \text{ is smooth on } \mathcal{S}_l\}, \\ \text{Jump} &= \{l: \ \ u \text{ has a jump on } \mathcal{S}_l\}, \end{aligned} \tag{3.195}$$

where $\text{Smooth} \cup \text{Jump} = \mathcal{L}$ and $\text{Smooth} \cap \text{Jump} = \emptyset$. Moreover, we assume $\text{Smooth} \neq \emptyset$ We show that the reconstruction is accurate with respect to the target cell E_0 to order

$$r_{\max} = \max\{r : \text{Smooth} \cap \mathcal{L}_{r,r} \neq \emptyset\}. \tag{3.196}$$

Figure 3.19 illustrates a (3,2,1) combination of stencils in the presence of two shocks. In this example, the set containing indices of smooth stencils is $\text{Smooth} = \{0, 1\}$, indicating that u on stencils $\mathcal{S}_0$ and $\mathcal{S}_1$ does not have a jump. Consequently, we expect the accuracy of the reconstruction is $r_{\max} = \text{Smooth} \cap \mathcal{L}_{2,2} = 2$.

We follow the estimation of reconstruction error in Arbogast et al. (2024). We start by estimating the error of the nonlinear scaling (3.192) associated with some stencils $\mathcal{S}_l$ for some $m > 0$.

$$\hat{\omega}_l = \frac{\omega_l}{(\sigma_l + \epsilon h_0^2)^m}. \tag{3.197}$$

We estimate the reconstruction error under two distinct situtaions:

- $l \in \text{Smooth}$, the smoothness indicator $\sigma_l = Ch^2 + \mathcal{O}(h^3)$;
- $l \in \text{Jump}$, the smoothness indicator $\sigma_l = \Theta(1)$.

We employ the asymptotic estimation result from Olver (1997). Assuming $n \geq m$, $a > 0$, and $b > 0$, then

$$\begin{aligned} \frac{a}{bh^m + \mathcal{O}(h^n)} &= \frac{ah^{-m}}{b}\big(1 + \mathcal{O}(h^{n-m})\big), \\ \frac{a}{bh^{-n} + \mathcal{O}(h^{-m})} &= \frac{ah^n}{b}\big(1 + \mathcal{O}(h^{n-m})\big). \end{aligned} \tag{3.198}$$

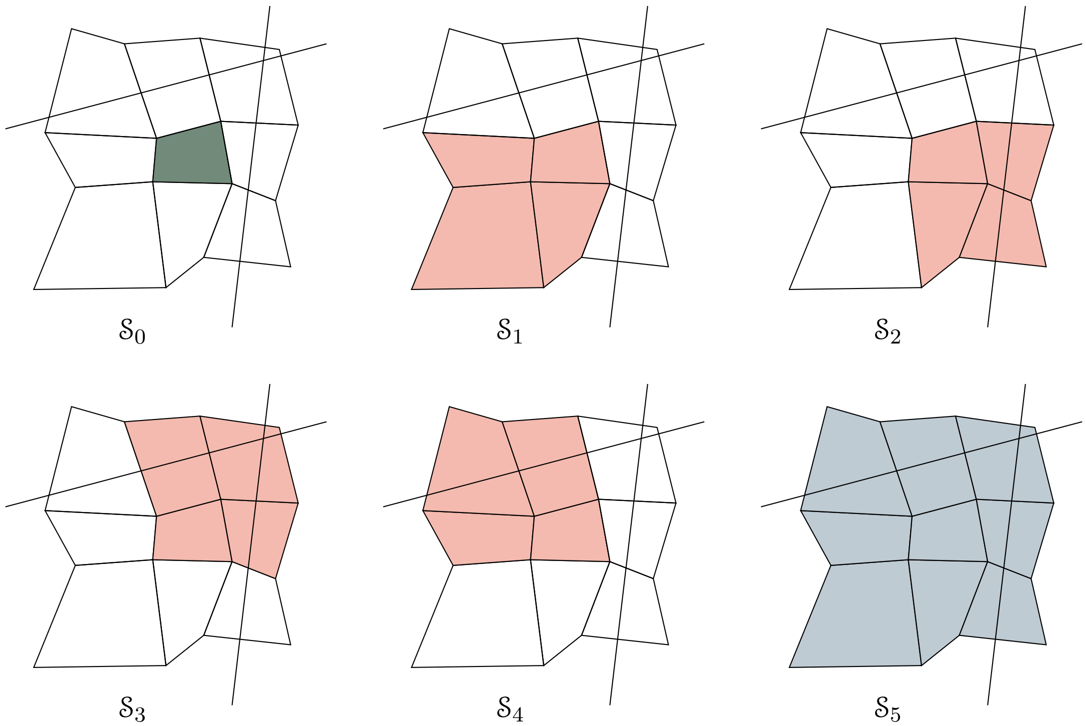


Figure 3.19: An illustration of a (3,2,1) multi-level stencils combination on a logically rectangular quadrilateral mesh with two shock present.

For the smooth stencil $\mathcal{S}_l$, $l \in \text{Smooth}$

$$\begin{aligned}\hat{\omega}_l = \frac{\omega_l}{(\sigma_l + \epsilon h_0^2)^m} &= \frac{\omega_l}{((C+\epsilon)h_0^2 + \mathcal{O}(h^3))^m} \\ &= \omega_l \Big(\frac{h_0^{-2}}{C+\epsilon}(1+\mathcal{O}(h)) \Big)^m \\ &= \frac{\omega_l}{(D+\epsilon)^m} h_0^{-2m}(1+\mathcal{O}(h)).\end{aligned} \tag{3.199}$$

In conclusion, we estimate the nonlinear scaling as

$$\hat{\omega}_l = \begin{cases} \dfrac{\omega_l}{(C+\epsilon)^m} h_0^{-2m}(1+\mathcal{O}(h)), & \text{if} \quad l \in \text{Smooth}, \\ \Theta(1), & \text{if} \quad l \in \text{Jump}. \end{cases} \tag{3.200}$$

Then we estimate the error for the normalized nonlinear weighting (3.194). We expand the sum of nonlinear scaling as

$$\sum_l \hat{\omega}_l = \sum_r \sum_{\substack{l \in \\ \text{Smooth} \cap \mathcal{L}_{r,r}}} \hat{\omega}_l + \sum_{l \in \text{Jump}} \hat{\omega}_l. \tag{3.201}$$

We estimate the smooth part of the nonlinear weighting as

$$\sum_r \sum_{\substack{l\in \\ \text{Smooth}\cap\mathcal{L}_{r,r}}} \hat{\omega}_l = \sum_r \sum_{\substack{l\in \\ \text{Smooth}\cap\mathcal{L}_{r,r}}} \frac{\omega_l}{(C+\epsilon)^m} h_0^{-2(ar+\eta)}(1+\mathcal{O}(h)). \tag{3.202}$$

Assume $h<1$, we keep the largest $ar+\eta(r)$ number as our estimator

$$\begin{aligned}\sum_r \sum_{\substack{l\in \\ \text{Smooth}\cap\mathcal{L}_{r,r}}} \hat{\omega}_l &= \sum_{\substack{l\in \\ \text{Smooth}\cap\mathcal{L}_{r_{\max},r_{\max}}}} \frac{\omega_l}{(C+\epsilon)^{ar_{\max}+\eta(r_{\max})}} h_0^{-2(ar_{\max}+\eta(r_{\max}))}(1+\mathcal{O}(h)) \\ &\qquad + \Theta(h^{-2(ar+\eta(r))}) \\ &= \sum_{\substack{l\in \\ \text{Smooth}\cap\mathcal{L}_{r_{\max},r_{\max}}}} \frac{\omega_l}{(C+\epsilon)^{ar_{\max}+\eta(r_{\max})}} h_0^{-2(ar_{\max}+\eta(r_{\max}))} \\ &\qquad + \mathcal{O}(h^{1-2(ar_{\max}+\eta(r_{\max}))}).\end{aligned} \tag{3.203}$$

later, we denote the set Smooth$\cap\mathcal{L}_{r,r}$ as Smooth$_r$ for simplicity. The overall accuracy estimation of the nonlinear weighting (3.194) is presented as

- $l \in$ Smooth

$$\tilde{\omega}_l = \frac{\hat{\omega}_l}{\sum_{l'}\hat{\omega}_{l'}} = \frac{\omega_l}{\sum_{l'\in\text{Smooth}_r}\omega_{l'}} h^{2[a(r_{\max}-r)+\eta(r_{\max})-\eta(r)]}(1+\mathcal{O}(h)); \tag{3.204}$$

- $l \in$ Jump

$$\tilde{\omega}_l = \frac{\hat{\omega}_l}{\sum_{l'}\hat{\omega}_{l'}} = \Theta(h^{2[a(r_{\max})+\eta(r_{\max})]}). \tag{3.205}$$

Now we can estimate the reconstruction error as

$$\begin{aligned}|u(\mathbf{x})-\mathcal{R}(\mathbf{x})| &= |\sum_l \tilde{\omega}_l(u(\mathbf{x}-Q_l(\mathbf{x})))| \\ &\le \sum_{l\in\text{Smooth}} \tilde{\omega}_l|u(\mathbf{x})-Q_l(\mathbf{x})| + \sum_{l\in\text{Jump}} \tilde{\omega}_l|u(\mathbf{x})-Q_l(\mathbf{x})| \\ &\le \sum_r \sum_{l\in\text{Smooth}_r} \Theta(h^{2[a(r_{\max}-r)+(\eta(r_{\max})-\eta(r))]})\mathcal{O}(h^r) \\ &\qquad + \sum_{l\in\text{Jump}} \Theta(h^{2(ar_{\max}+\eta(r_{\max}))})\mathcal{O}(1) \\ &= \mathcal{O}(h^{r_{\max}})\end{aligned} \tag{3.206}$$

Combining everything above and (3.191), we arrive at the formal error estimate of the reconstruction.

Lemma 3.14. *Let $\mathcal{R}(\mathbf{x})$ be the reconstruction. Assuming a qusi-uniform mesh, $h_0 = \Theta(h)$, there exists a constant $C > 0$ such that for any $\mathbf{x} \in E_0$,*

$$|u(\mathbf{x}) - \mathcal{R}(\mathbf{x})| \leq C h^{r_{\max}}, \tag{3.207}$$

where $r_{\max}$ is defined by (3.196)*.*

3.10.6 Time stepping

In this subsection, we briefly discuss some suitable time integration techniques. We focus on explicit schemes mainly because the phase package cannot be accurately coupled over long time intervals.

While the forward Euler method is sufficient for stability analysis, it only provides first-order accuracy. Therefore, we employ the strong-stability preservation (SSP) scheme introduced in Gottlieb et al. (2001). This scheme achieves higher order time accuracy while theoretically preserving the stability properties of the forward Euler method.

To begin with, the first-order forward Euler method is written as

$$u^{n+1} = u^n + \Delta t \tilde{\mathbf{F}}(u^n), \tag{3.208}$$

where $\tilde{\mathbf{F}}(u^n)$ represents the numerical integration of the numerical flux $\mathbf{F}(u^n)$ in Equation (4.11). A multi-stage explicit SSP Runge-Kutta method is

$$\begin{aligned} u^0 &= u^n, \\ u^{(i)} &= \sum_{j=0}^{i-1} \alpha_{i,j} u^{(j)} + \Delta t \beta_{i,j} \tilde{\mathbf{F}}(u^{(j)}), \quad 1 \leq i \leq m \\ u^{n+1} &= u^m, \end{aligned} \tag{3.209}$$

where $\sum_j^{i-1} \alpha_{i,j} = 1$ for consistency. To be consistent with the MFEM solver's second-order accuracy in velocity, we omit further discussion of time-integration schemes beyond third-order accuracy. The second- and third-order explicit SSP Runge-Kutta schemes can be expressed in the multi-stage form (3.209) as, for SSP2,

$$\begin{aligned} u^0 &= u^n, \\ u^{(1)} &= u^{(0)} + \Delta t\tilde{\mathbf{F}}(u^{(0)}), \\ u^{n+1} &= u^{(0)} + \frac{1}{2}\Delta t\tilde{\mathbf{F}}(u^{(0)}) + \frac{1}{2}\Delta t\tilde{\mathbf{F}}(u^{(1)}), \end{aligned} \tag{3.210}$$

and, for SSP3,

$$\begin{aligned} u^0 &= u^n, \\ u^{(1)} &= u^{(0)} + \Delta t\tilde{\mathbf{F}}(u^{(0)}), \\ u^{(2)} &= \frac{3}{4}u^n + \frac{1}{4}u^{(1)} + \frac{1}{4}\Delta t\tilde{\mathbf{F}}(u^{(1)}), \\ u^{n+1} &= \frac{1}{3}u^{(0)} + \frac{2}{3}u^{(2)} + \frac{2}{3}\Delta t\tilde{\mathbf{F}}(u^{(2)}). \end{aligned} \tag{3.211}$$

3.11 Handling Boundary Conditions

We couple MFEM and FVM solvers on a single mesh in our simulation. Therefore, accurate and compatible boundary conditions need to be assigned accordingly. However, it is usually difficult to manage boundary conditions for higher order finite volume and finite difference methods, which usually require wide stencils.

Some methods address this difficulty by using ghost cells or ghost points outside the computational domain, allowing the use of interior-like stencils near boundaries. However, the extrapolation or estimation of the manufactured values of ghost cells or ghost points are not straightforward. A notable method, the inverse Lax-Wendroff procedure introduced in Tan and Shu (2010), iteratively substitute normal derivatives of the solution with tangential and time derivatives using the PDE relation to impose more precise values for the ghost cells or ghost points.

We focus on utilizing the flexibility of the ML-WENO reconstruction to tackle the difficulties in a direct way without the use of ghost cells. This approach is independent of the complexity of the governing partial differential equations and intricacies of the geometry of the boundary. Similar discussion using the adaptive order reconstruction can be found in Semplice and Visconti (2020) and Semplice et al. (2023).

In Finite Volume Methods, we must use the boundary condition to determine the numerical flux on the boundary. To assign proper boundary conditions to a FVM discretization, it is necessary to first identify whether the flow is inflow or outflow first. Let u_B denote the value of u on the boundary. We classify the types of boundary conditions based on the magnitude and direction of the numerical flux $\mathbf{f}(u_B) \cdot \boldsymbol{\nu}$.

- If $\mathbf{f}(u_B) \cdot \boldsymbol{\nu} > 0$, it is classified as an *outflow* boundary, and no fixed boundary condition should be assigned. Instead a *free outflow* boundary condition is used, where the numerical flux is simply given by $\mathbf{f}(u_B) \cdot \boldsymbol{\nu}$. The outflow boundary may lead to degradation in the accuracy of the numerical solution, as high-order stencils that would extend beyond the computational domain must be truncated. ML-WENO offers the flexibility to add stencils of the desired size that span back into the domain, thereby preserving accuracy near outflow boundaries.

- If $\mathbf{f}(u_B) \cdot \boldsymbol{\nu} < 0$, it is classified as an *inflow* boundary. We can understand it as prescribing the inflow flux $\mathbf{f}_B$ or fixing the boundary value u_B (a Dirichlet boundary condition). In the inflow case, discontinuous shock may enter the domain, requiring a swift response inside the domain to align with the specified Dirichlet value. To accommodate such potential discontinuities, we can incorporate an extra constant stencil polynomial to WENO reconstruction near the inflow boundary.

- If $\mathbf{f}(u_B) \cdot \boldsymbol{\nu} = 0$, it is usually referred to as a *noflow* boundary. No additional treatment on the boundary is required, as the value is fixed at zero.

In the following subsections, we present two examples. The first example illustrates how additional higher order stencils affect reconstruction order. The second address an inflow Dirichlet boundary condition that generates an incoming shock wave.

3.11.1 Example on additional stencils reinforced at the boundary

We use logically rectangular meshes, with vertices randomly perturbed in x and y directions within a range of $[-0.15, 0.15]$ times the unperturbed mesh spacing h. A full example mesh is depicted in Figure 3.2.

Let $u(x, y)$ be an arbitrary function defined within the domain $\Omega = [0, 1]^2$. We take

$$u(x,y) = \begin{cases} \sin(x)\cos(y) & x \leq 0.5, \\ \sin\left(\dfrac{(x+0.3)\pi}{2}\right)^6 \cos\left(\dfrac{(y-0.6)\pi}{2}\right)^6 + 0.5 & x > 0.5. \end{cases} \tag{3.212}$$

We consider the standard (5,3) ML-WENO reconstruction for a cell $E_{i,j}$ located far away from boundaries as shown in Figure 3.20. This reconstruction consists of two different stencils with different approximation accuracy: a 5×5 large stencil providing fifth-order accuracy and four 3×3 small stencils providing third-order accuracy. However, when the cell $E_{i,j}$ located within the two-cell distance from any boundary, the large stencil becomes unavailable, leading to a degeneracy in the order of accuracy near the boundary. In this case, ML-WENO grants flexibility of associating an additional large stencil to the cells within the two-cell distance from boundary.

Figure 3.21 illustrates a case where a fifth-order stencil centered at cell $E_{i,j}$ can be used in the ML-WENO reconstruction of different target cells: interior cell (red), cells located one cell away from the boundary (green) and cells on the boundary (yellow).

We define the effective polynomial degree of the reconstructions for each cell as

$$p_{\text{eff}} = \sum_l \tilde{\omega}_l r_l - 1. \tag{3.213}$$

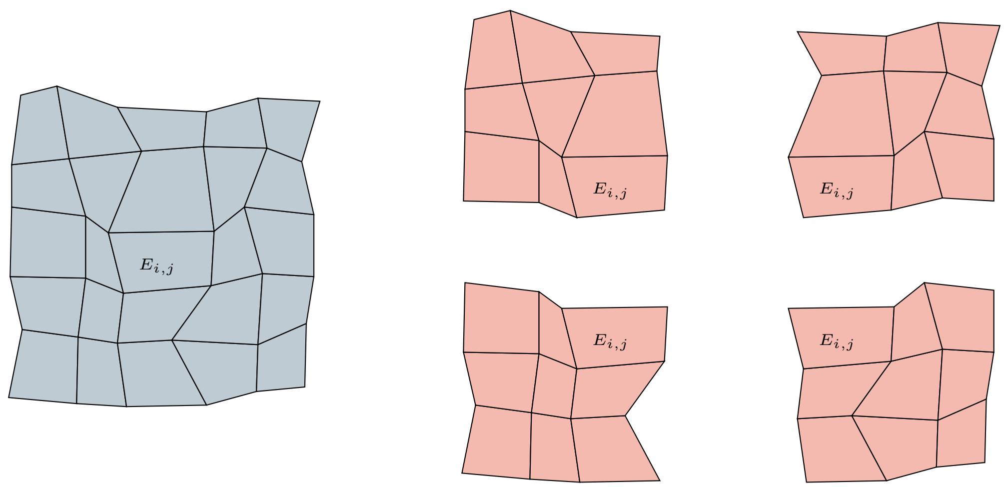


Figure 3.20: An example of a standard (5,3) stencils combination on a logically rectangular quadrilateral mesh.

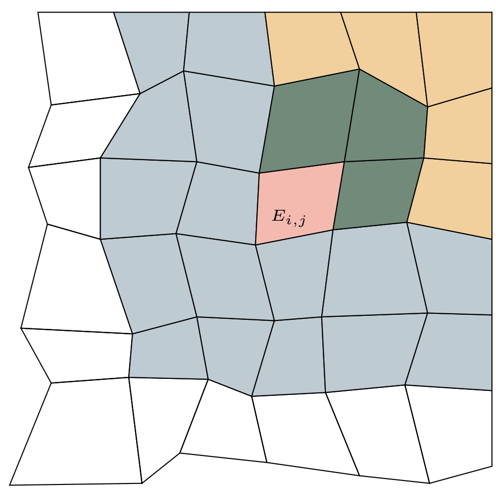


Figure 3.21: An illustration of different cells using one fifth-order stencil to avoid accuracy degeneracy.

In Figure 3.23, we see good approximation (polynomial degree 4) over the middle of the domain except two layers of cells around the shock. However, the reconstruction drops to polynomial degree 3, within the distance of two cells from the boundary. With the help of adding additional large stencils as shown in Figure 3.20, we can achieve good approximation (polynomial degree 4) at the top right corner of the boundary. Figure 3.22 illustrates the reconstructed solution on the 20×20 randomly distorted logically rectangular mesh.

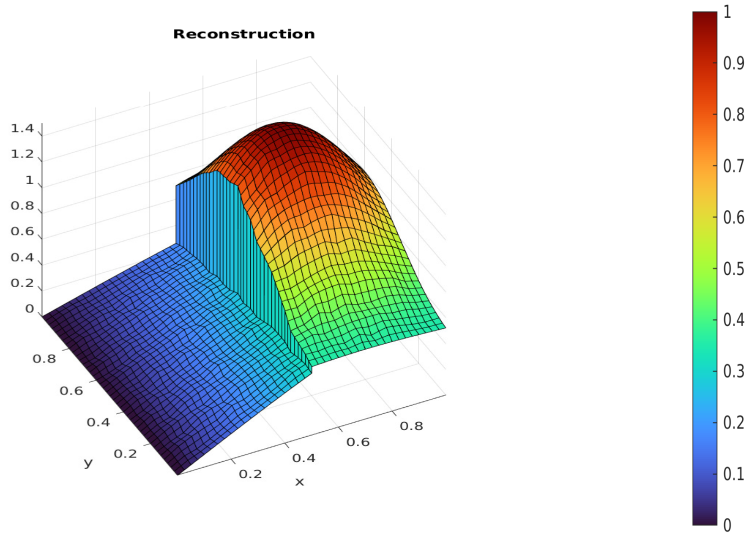


Figure 3.22: The reconstructed solution on the 20×20 randomly distorted logically rectangular mesh.

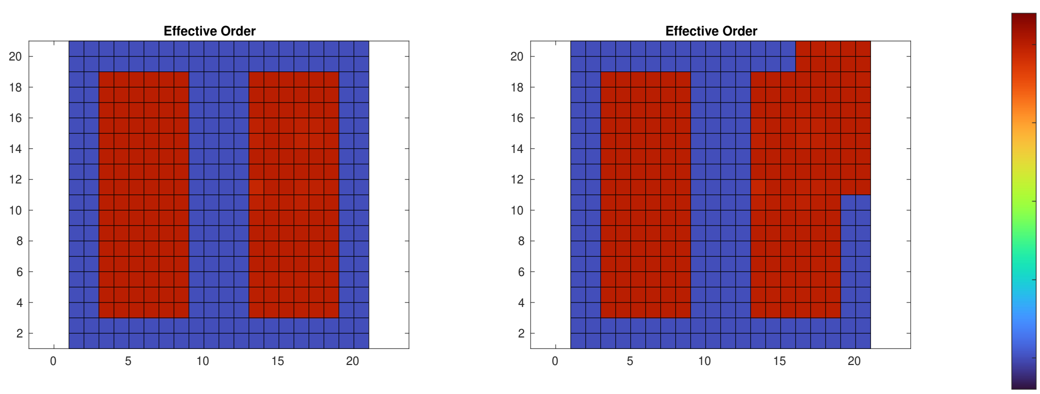


Figure 3.23: Effective polynomial degree with and without some additional fifth-order stencils added near the top right boundary.

3.11.2 Stabilization of a inflow boundary

We present another example demonstrating that adding a "constant" level on the boundary stabilize problems with complex inflow boundary condition. For this case, we consider a Burgers-like equation $u_t + u^3 u_x = 0$ with a fourth order flux

$\mathbf{f}(u) = (u^4/4, 0)$ in the domain $[0,1]^2$. We impose Dirichlet boundary condition on the left boundary as

$$u_D(0,y,t) = \begin{cases} \sin^2(2\pi y), & y \leq 0.25, y \geq 0.75, \\ 1, & 0.25 < y < 0.75. \end{cases} \tag{3.214}$$

Free outflow boundary conditions are assigned to the remaining boundaries. We consider a standard $(3,2)$ reconstruction modified on the inflow boundary $x = 0$ we consider a $(3,2,1)$ multi-level stencils combination as shown in Figure 3.18. We use a rectangular mesh with 80×80 cells and second-order Runge-Kutta time integrator (3.210). Figure 3.24 illustrates the shock wave traveling into the domain with a sharp profile. A slight overshoot can be observed at time 0.125, which is quickly stabilized. Without the additional "constant" level, the inflow rapidly blows up shortly after entering the computational domain. Higher-order simulation results can be found in Arbogast et al. (2024).

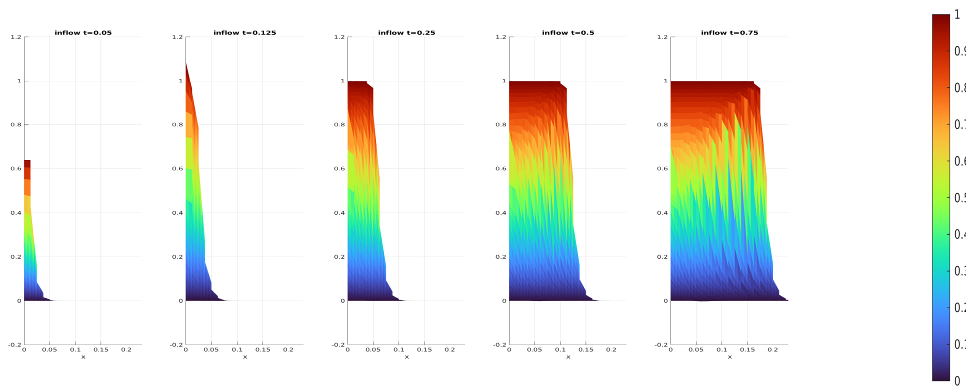


Figure 3.24: A test case demonstrating inflow stabilization using a constant stencil at the inflow boundary on a 80×80 rectangular domain.

3.12 Numerical Test

We test our ML-WENO algorithm with a Riemann shock and rarefaction. Let $\mathbf{f}(u) = (u^2/2, 0)$ be the Burgers flux function. In this example, we take the initial condition $u_0(x,y) = 1$ in the strip $0.5 < x < 1.5$ and 0 elsewhere within the computational domain $\Omega = [0,3] \times [0,1]$. We assign zero flux boundary condition on the two sides $y = 0$ and $y = 1$.

The true solution is

$$u(x,y,t) = \begin{cases} 0, & x < 0.5, x \geq 0.5t + 1.5, \\ (x - 0.5)/t, & 0.5 \leq x < t + 0.5, \\ 1, & t + 0.5 \leq x < 0.5t + 1.5. \end{cases} \tag{3.215}$$

We forward march to the time = 2.0, when the trailing rarefaction reaches the leading shock. For simplification, we test the problem on a 180×60 rectangular mesh. We compute the solution to time 2 using the standard SSP3 Runge-Kutta time stepping scheme, with time step $\Delta t = 0.2h$. In Figure 3.25, we see nonoscillatory and sharp solution at time $t = 1$ and $t = 2$.

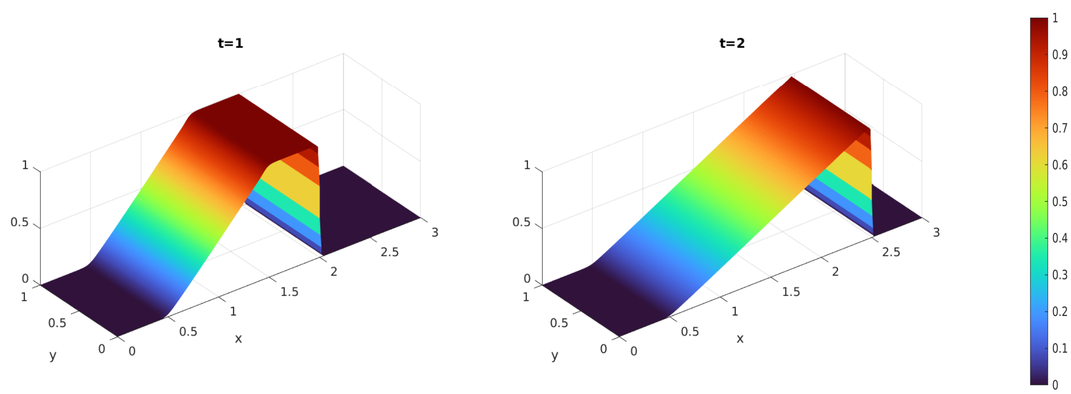


Figure 3.25: The reconstructed solution on the 180×60 rectangular mesh at times 1.0 and 2.0. (The rarefaction and shock meet at time 2.0.)

Chapter 4: Coupled Numerical Experiments

In this chapter, we couple the locally mass-conserving MFEM and FVM methods to perform numerical simulations of a partially molten mantle, with the remaining uncertainty in porosity ϕ addressed through phase package evaluation. We describe the implementation details and coupling strategy that integrate the MFEM, FVM, and the eutectic phase package. Before tackling the fully coupled problem, we first examine an alternative porosity evolution scenario based on a prescribed melting rate. We then present an example that combines the MFEM, FVM, and eutectic phase package into a fully coupled computation. The discussion restricts to pseudo-1D scenarios and subsequently extends to full 2D simulations.

4.1 Coupled Algorithm

Let $\mathcal{U}^n = (C^n, H^n)$ denote the cell-averaged solution to the transport system (2.81) at time t^n. Similarly, let $\mathcal{V}^n = (\check{\mathbf{v}}_r^n, \check{\mathbf{v}}_s^n, \check{q}_l^n, \check{q}_s^n)$ represent the solution to the Darcy-Stokes system at the same time t^n. Finally, let $\mathcal{W}^n = (\phi^n, \phi_1^n, \check{T}^n, c_s^n, c_l^n)$ denote the output of the phase calculation at t^n obtained from the transport result $\mathcal{U}^n$.

For each time step the time t^n, we begin by reconstructing the dimensionless mass composition and enthalpy from the cell-averaged solution $\mathcal{U}^n$ using ML-WENO reconstruction at each data point of interest. The eutectic phase package then provides the corresponding phase parameters $\mathcal{W}^n$ based on these point wise values. With $\mathcal{W}^n$ determinedin particular with ϕ^n, we solve the Darcy-Stokes system at the fixed time t^n to obtain $\mathcal{V}^n$. Finally, the cell-averaged solution $\mathcal{U}^{n+1}$ at the next time t^{n+1} is then obtained by forward marching of the transport system. The complete procedure is summarized as a coupled solver in Algorithm 3.

Algorithm 3 Sequential Coupled Solver

Let $\mathcal{U}^0$ be given as initial data.

1: **for** Time level $n = 0, 1, 2, \ldots, n_{\max} - 1$ **do**

2: Given $\mathcal{U}^n$, compute $\mathcal{W}^n$ by the eutectic phase package.

3: Using ϕ^n in W^n, solve the Darcy-Stokes system (2.77)–(2.80) for $\mathcal{V}^n$.

4: Given $\mathcal{W}^n$ and $\mathcal{V}^n$, compute dimensionless effective velocity (2.38) and phase averaged velocity as

$$\check{\mathbf{v}}_{\text{eff}} = \frac{\phi \check{\mathbf{v}}_l + D_i(1-\phi)\check{\mathbf{v}}_s}{\phi + (1-\phi)D_i}, \quad \check{\bar{\mathbf{v}}} = \phi \check{\mathbf{v}}_r + \check{\mathbf{v}}_s.$$

5: Marching forward the transport system (2.81) to the next time $n+1$ for $\mathcal{U}^{n+1}$.

6: **end for**

4.1.1 Discontinuous Porosity

Due to the presence of no melt regions and distinct eutectic phase regions, the reconstructed values of the dimensionless enthalpy and mass concentration may differ on the two sides of an edge shared by adjacent elements. This, in turn, can lead to two different porosity values being computed for the same edge. However, in a conforming MFEM scheme, it is essential to assign a single consistent and physically appropriate porosity value on each element edge.

Therefore, we evaluate the harmonic average of the two values of porosity ϕ_e^+ and ϕ_e^- at each quadrature points on both sides of an element edge e to obtain

$$\overline{\phi}_e = \frac{2\phi_e^+\phi_e^-}{\phi_e^+ + \phi_e^-}. \tag{4.1}$$

The harmonically averaged porosity $\overline{\phi}_e$ is then used in the continuous compaction equation (3.120), where the face integral is converted to an edge integral via the divergence theorem.

Importantly, when the porosity on one side of the edge vanishes, the geometric average also becomes zero. This property enforces the physical condition that no liquid can pass directly through an edge adjacent to a no-melt region.

4.2 Pseudo One-Dimensional Column Tests

In this section, we test the coupled algorithm on a two-dimensional $M \times N$ grid with a pseudo one-dimensional compaction column problem. We study a compacting column aligned along the y-direction, defined over the domain $y \in [-H, 0]$ and $x \in [-L, L]$. Two scenarios are investigated.

- Prescribed melting case: The MFEM and FVM solvers are coupled without phase computation. Instead, a simple prescribed melting function $\Gamma(\mathbf{x})$ is used.
- Phase calculation case: Phase transition and porosity evolution are computed simultaneously, providing a complete coupling between the MFEM, FVM, and eutectic phase package.

For the two tests presented below, we fix the domain dimension in x-direction $L = 0.1$. We fix mesh resolution in the x-direction as $M = 2$ and pick different resolutions in y-direction. The domain dimension and mesh resolution in y-direction are chosen accordingly. We fix the parameter $\Theta = 0$ as well. The code is implemented in PETSc/C++, with implementation details given in the appendix.

4.2.1 Prescribed Melting Cases

Before coupling the mechanical, transport and phase packages together into one integrated computation, we first illustrate a simpler model that couples MFEM and FVM solvers without involving the phase package. In this reduced setting, the porosity distribution is not determined by phase-splitting calculations. Instead, we prescribe the melting rate or, equivalently, the porosity evolution through a fixed source function distributed in space as

$$\phi(t) = \Gamma(\mathbf{x}). \tag{4.2}$$

We express conservation of liquid mass by introducing the prescribed melting term $\Gamma(\mathbf{x})$ as a source term on the right-hand side of the advection equation as

$$\frac{\partial \phi}{\partial t} + \nabla \cdot (\phi \mathbf{v}_l) = \Gamma, \tag{4.3}$$

where the diffusion term is being dropped for simplicity in this case.

We prescribe the following boundary conditions for the MFEM and the FVM solvers respectively to replicate the behavior of the one-dimensional problem within the two-dimensional grid.

Consider first the Darcy-Stokes MFEM solver. On the bottom edge $y = -H$, impose that:

- Normal Stokes velocity and supplemental edge averaged flux are prescribed with an essential boundary conditions with a constant upwelling velocity v, so

$$\check{\mathbf{v}}_s \cdot \boldsymbol{\nu} = v,\ |e|^{-1} \int_e \mathbf{b} \cdot \boldsymbol{\nu}\, ds = |e|^{-1} \int_e v\, ds;$$

- Tangential Stokes velocity is prescribed with an essential boundary condition as

$$\check{\mathbf{v}}_s \cdot \boldsymbol{\tau} = 0;$$

- Normal Darcy flux is prescribed with an essential boundary condition as

$$|e|^{-1} \int_e \phi \check{\mathbf{v}}_r \cdot \boldsymbol{\nu}\, ds = |e|^{-1} \int_e u\, ds = 0.$$

On the top edge $y = 0$, impose that:

- Normal Stokes velocity and supplemental edge averaged flux are prescribed with natural boundary conditions, i.e., traction free,

$$t_0 = 0;$$

- Tangential Stokes velocity is prescribed with a zero essential boundary condition as

$$\check{\mathbf{v}}_s \cdot \boldsymbol{\tau} = 0;$$

- Normal Darcy flux is prescribed with a natural boundary condition as

$$|e|^{-1} \int_e \phi \check{\mathbf{v}}_r \cdot \boldsymbol{\nu} \, ds = |e|^{-1} \int_e u \, ds = 0.$$

On the left and right Edges $x = \pm L$, impose that:

- Normal Stokes velocity and supplemental edge averaged flux are prescribed with essential boundary conditions as

$$\check{\mathbf{v}}_s \cdot \boldsymbol{\nu} = v = 0, \; |e|^{-1} \int_e \mathbf{b} \cdot \boldsymbol{\nu} \, ds = 0;$$

- Tangential Stokes velocity is prescribed with the natural boundary condition of free traction,

$$t_0 = 0;$$

- Normal Darcy flux is prescribed with a non-permeable essential boundary condition as

$$|e|^{-1} \int_e \phi \check{\mathbf{v}}_r \cdot \boldsymbol{\nu} \, ds = |e|^{-1} \int_e u \, ds = 0.$$

Finally, for the transport FVM solver, on the bottom edge $y = -H$, impose that:

- A corresponding Dirichlet boundary condition is prescribed on the inflow boundary aligning with the essential boundary condition specified in the MFEM solver.

On the top edge $y = 0$, impose that:

- A free outflow boundary condition is imposed, consistent with the free traction boundary condition applied in the MFEM solver.

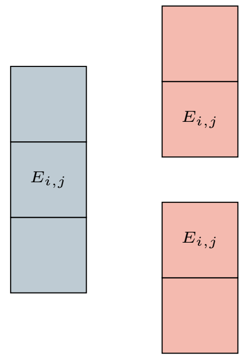


Figure 4.1: An exhibition of a set of 1D ML-WENO (3,2) stencils on a rectagular mesh.

On the left and right edges $x = \pm L$, impose that:

- A no-flow boundary condition is prescribed consistent with the zero-flow Dirichlet boundary used in the MFEM solver.

For simplicity and efficiency in implementation and computation, we employ one-dimensional ML-WENO(3,2) stencils as illustrated in Figure 4.1, since higher-order accuracy in the x-direction is unnecessary for this problem. We use second order SSP Runge-Kutta time integration scheme defined in Equation (3.210).

We consider two different scenarios, each characterized by distinct prescribed melting functions and initial conditions:

Case 1: A piece-wise constant source function.

$$\Gamma_1(z) = \begin{cases} 2.5e-3, & -1.5 < z \leq 0, \\ 0, & -2.0 \leq z \leq -1.5. \end{cases} \tag{4.4}$$

We assign an initial condition of no melting everywhere as

$$\phi_1(z, 0) = 0.0, \quad z \in [-2.0, 0.0]. \tag{4.5}$$

Figure 4.2 illustrates the temporal evolution of porosity, Stokes and Darcy velocities, Stokes and Darcy pressures, as well as the effective pressure, defined as $p_{\text{eff}} = p_s - p_l$.

The third panel, corresponding to dimensionless time $\check{t} = 1325$, demonstrates that the system has reached a steady state after sufficient evolution.

Case 2: We next consider a slightly more complicated scenario in which a melting source is trapped beneath the bottom of an existing porosity channel. The trapped melting source produces a step in porosity, thereby induces an instantaneous perturbation in the compaction rate. The source function is

$$\Gamma_2(z) = \begin{cases} 0, & -1.5 < z \leq 0, \\ 1.5e-3, & -1.8 < z \leq -1.5, \\ 0, & -2.0 \leq z \leq -1.8, \end{cases} \tag{4.6}$$

where we assign an initial condition with an existing porosity channel as

$$\phi_2(z,0) = \begin{cases} 5e-2, & -1.5 < z \leq 0, \\ 0, & -2.0 \leq z \leq -1.5. \end{cases} \tag{4.7}$$

Figure 4.3, 4.4 and 4.5 illustrate the temporal evolution of porosity, Stokes and Darcy velocities and Stokes and Darcy pressures. A propagating discontinuity in porosity forms when the porosity profile decreased upward (See Katz (2022)). Eventually, the system reaches a steady state, characterized by a narrower porosity channel. The simulation demonstrates robust and stable evolution, despite the presence of a highly complicated porosity profile. The solution preserves its overall structure under mesh refinement.

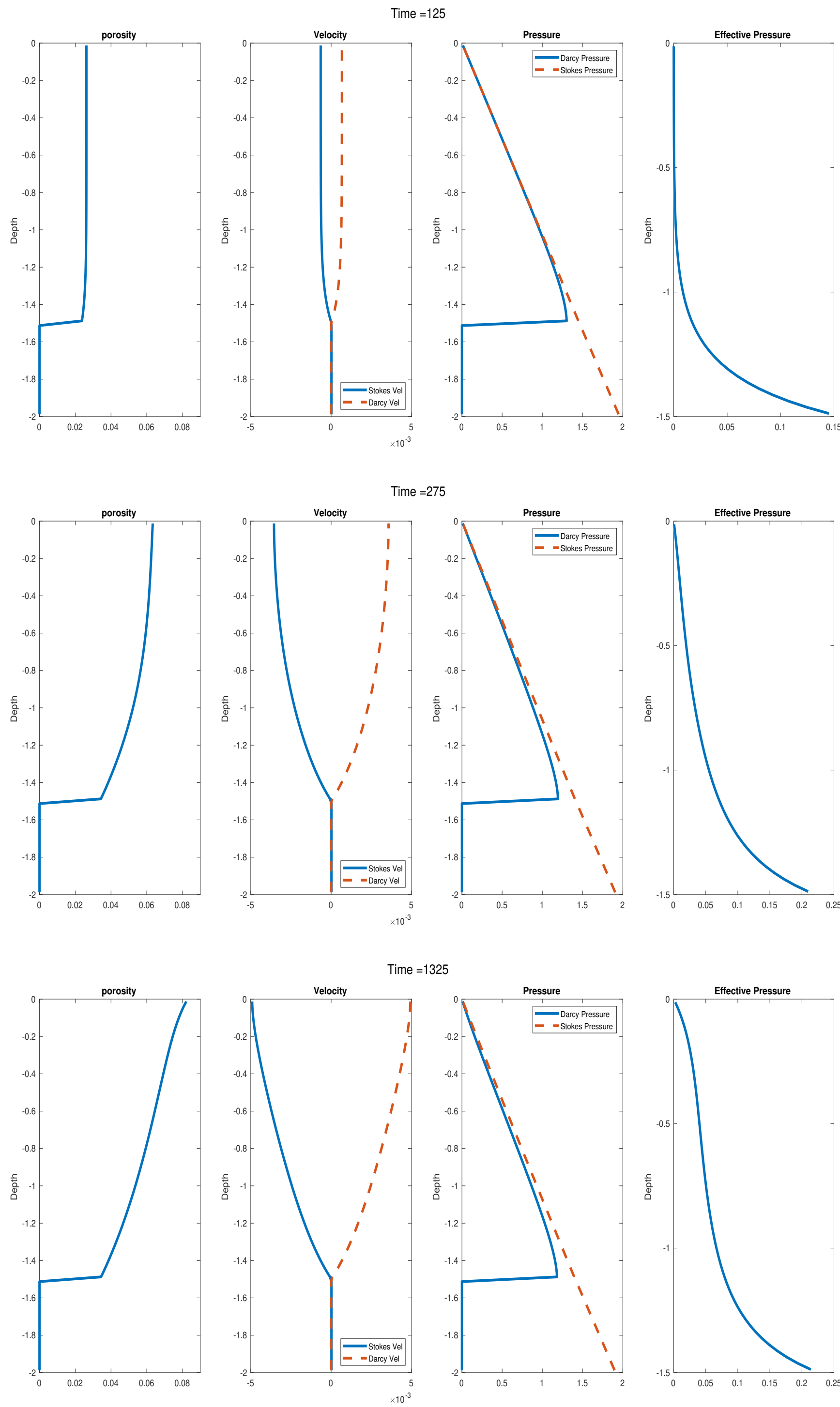


Figure 4.2: Prescribed melting case with constant melting source term 4.4. Shown at time = 125, 275 and 1325, where the system reaches steady state at the time = 1325.

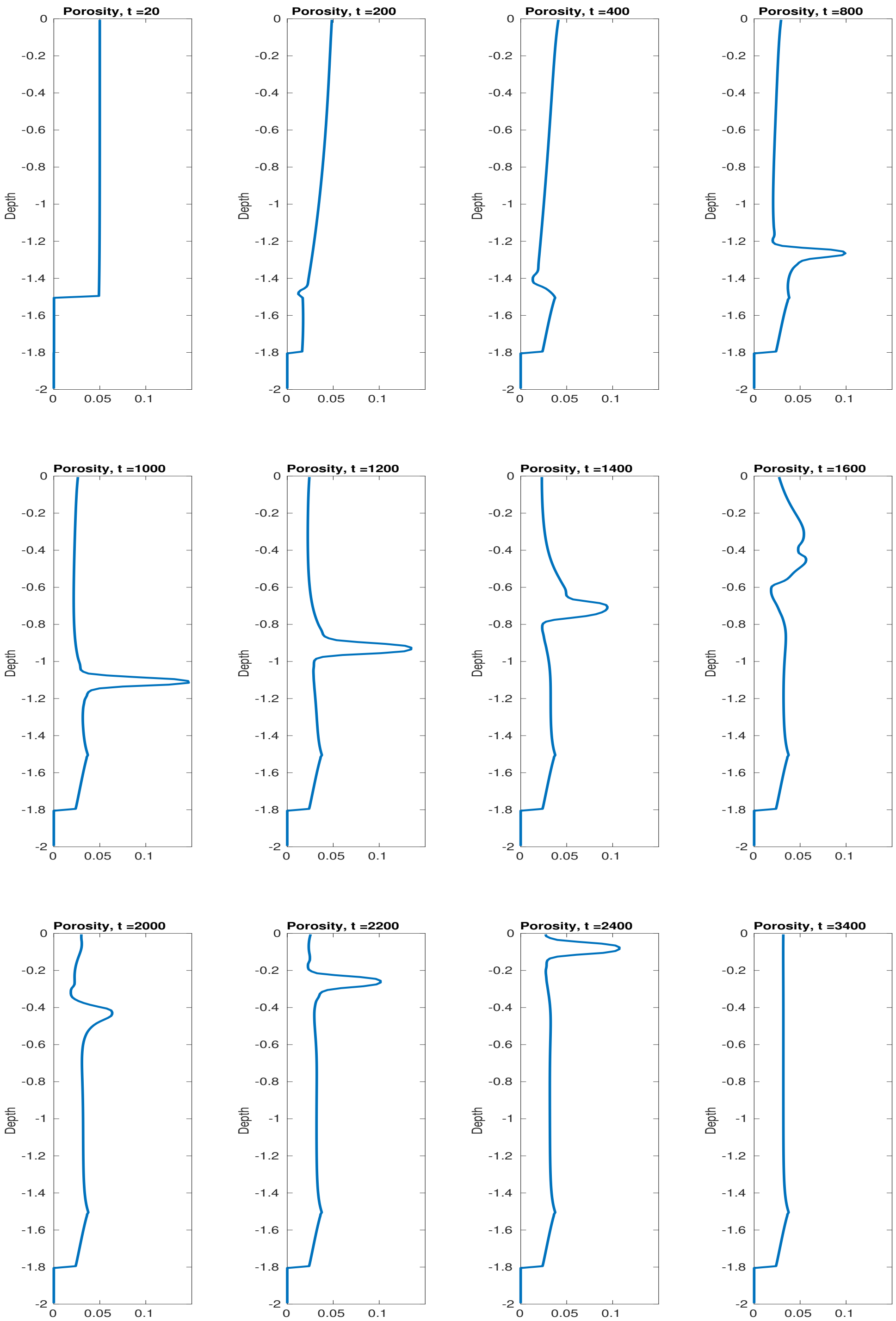


Figure 4.3: Prescribed melting case with trapped melting source term 4.6. Temporal evolution of porosity shown at different time steps.

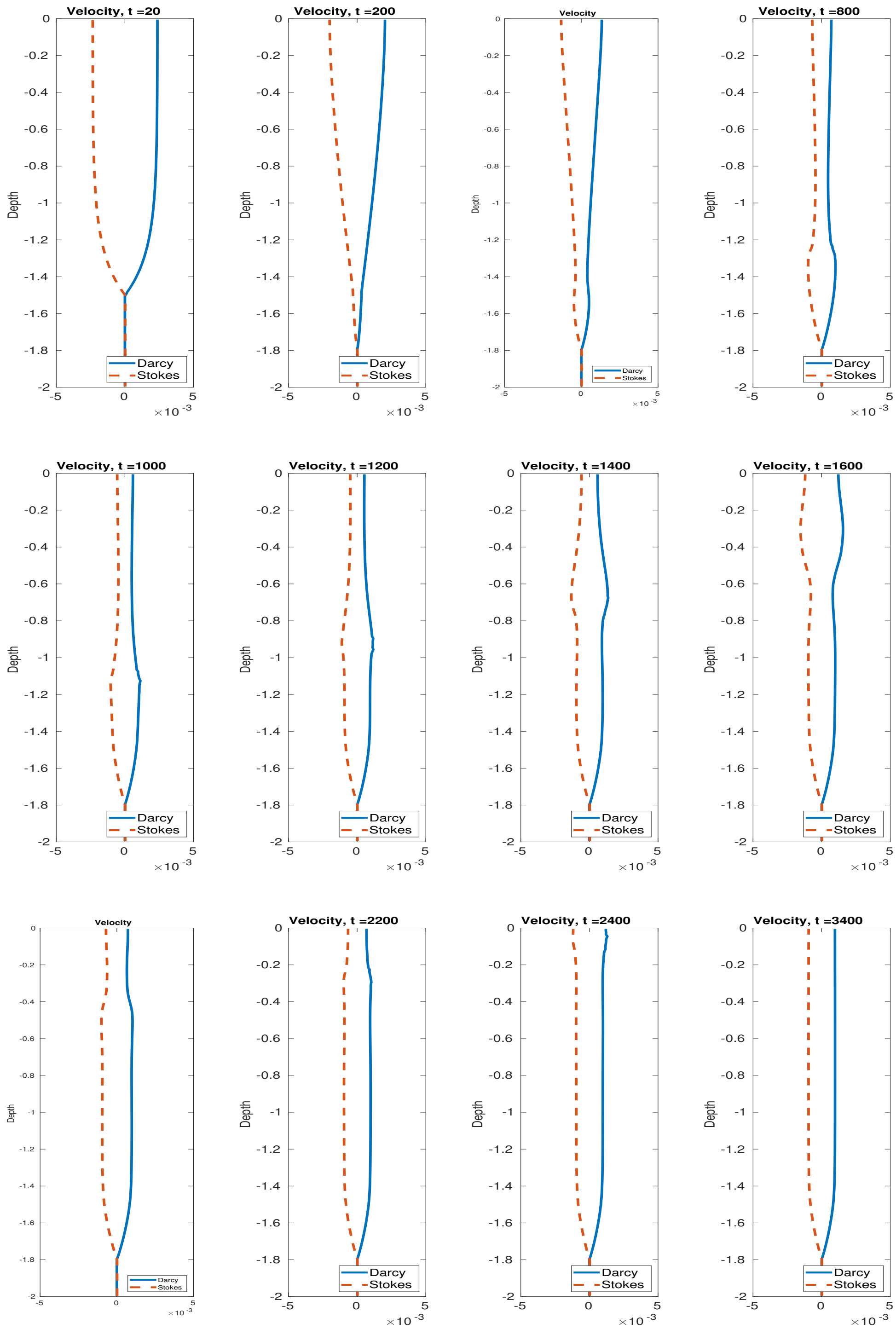


Figure 4.4: Prescribed melting case with trapped melting source term 4.6. Temporal evolution of velocities shown at different time steps.

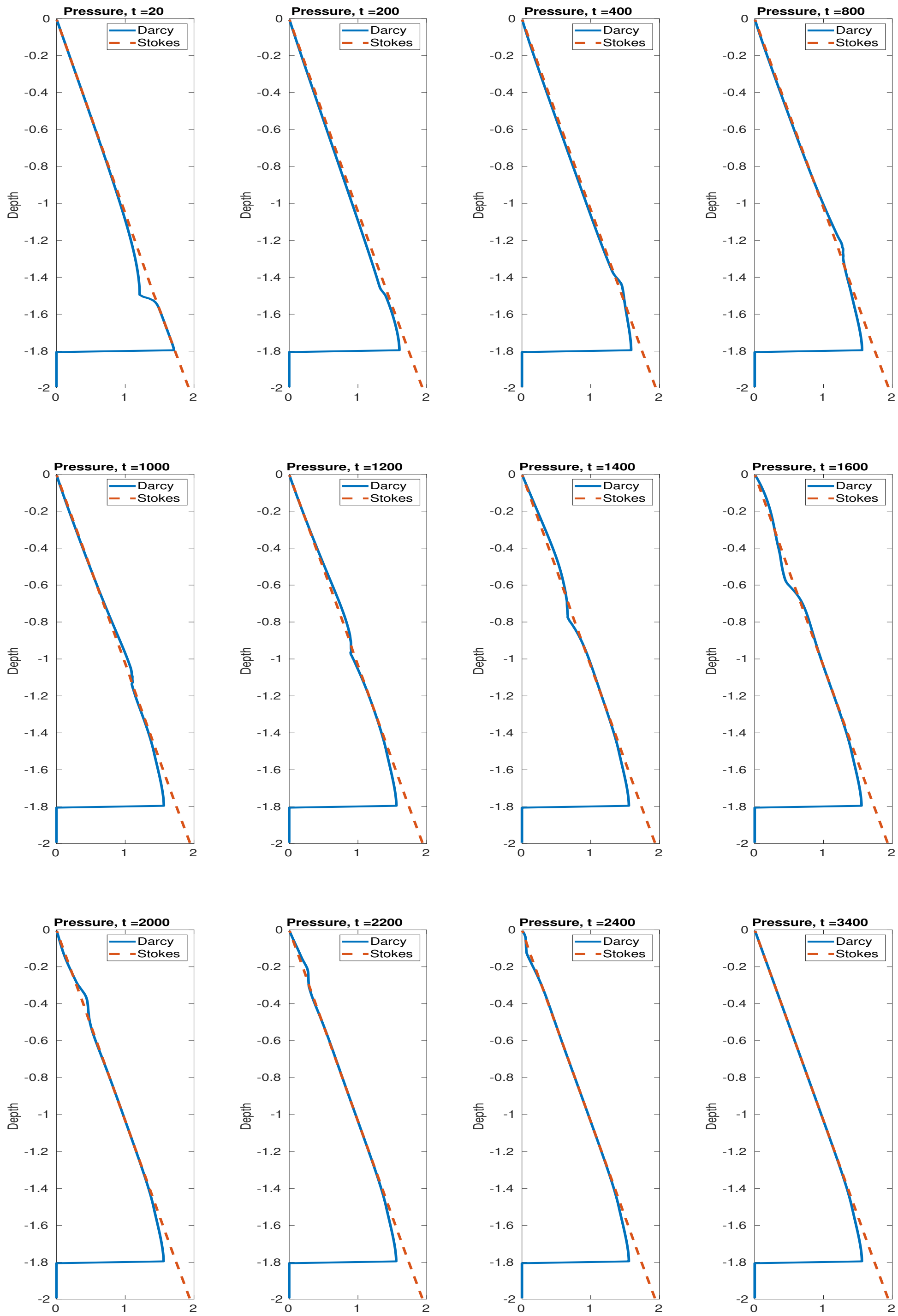


Figure 4.5: Prescribed melting case with trapped melting source term 4.6. Temporal evolution of pressures shown at different time steps.

4.2.2 Phase Coupled Calculation

In this section, we test the MFEM and FVM solvers combined with the eutectic phase package, which gives the fully integrated simulation framework. In this setting, porosity is no longer prescribed through an externally imposed melting function but is instead computed consistently from the transported nondimensional bulk composition and enthalpy.

The computational domain represents an upwelling mantle column, with corner flow neglected near the surface under the simplifying assumption that the column continues its vertical ascent through the lithosphere. This configuration constitutes a minimal yet dynamically representative test case for assessing the stability and accuracy of the coupled algorithm in the presence of phase transitions and melt–solid interactions.

We propose some simplifications for our simulations. First of all, we reconsider the transport equations (2.81) and the simplified version is

$$\begin{aligned} \frac{\partial C}{\partial \check{t}} + \nabla \cdot \check{\mathbf{v}}_{\text{eff}} C = 0, \\ \frac{\partial H}{\partial \check{t}} + \nabla \cdot (\check{\mathbf{v}}\check{T}) - \check{L}\nabla \cdot ((1-\phi)\check{\mathbf{v}}_s) - \frac{\alpha_\rho l_0}{c_p}\check{T}\check{\mathbf{v}} \cdot \mathbf{g} = 0. \end{aligned} \tag{4.8}$$

In this formulation, the diffusive contributions in both the mass composition and enthalpy equations are neglected. Specifically, the diffusion term $\frac{\phi \mathcal{D}_l}{u_0 l_0}\nabla c_l$ in the liquid phase is omitted due to the fact that the porosity ϕ remains small ($\phi \ll 1$) and the associated diffusivity coefficient $\mathcal{D}_l$ is comparatively negligible compared to effective advective velocities and thus can be neglected and thus can be neglected (See Spiegelman (1996)).

Likewise, while thermal conduction is an important mechanism in mantle convection over geological timescales, its influence on the present simulations is negligible. Two considerations motivate the simplification of dropping the thermal conduction from the enthalpy equation (2.81). First, the temperature profile maintains either a nearly constant or linear variation in space, such that the Laplacian of temperature vanishes or remains small throughout the evolution. Second, we do not impose a non-permeable solidification boundary at the surface, which would otherwise establish sharp thermal gradients and necessitate conductive heat transport across a boundary layer, in which case the diffusivity would no longer be negligible (see Hewitt and Fowler (2008)).

Additionally, given that the primary focus of this study is the phenomenon of partial melting under small-porosity conditions, we adopt the lithostatic pressure as the reference pressure in determining the local melting point (see Katz (2008)). This approximation is justified because the dynamic pressure perturbations associated with porous flow are small compared to the background lithostatic gradient. As the mantle column ascends from depth, the concomitant reduction in lithostatic pressure leads to a decrease in the solidus temperature, thereby inducing some degrees of melting in proportion to the degree of decompression.

4.2.2.1 Boundary conditions

In this set of simulations, we do not attempt to prescribe fully physically consistent boundary conditions, owing to the inherent complexity of capturing both the mechanical and thermochemical constraints at the mantle–lithosphere interface. For example, Hewitt and Fowler (2008) formulated boundary conditions that simultaneously satisfy conservation laws and thermodynamic equilibrium at the surface. These constraints introduce additional complexity to the model and the numerical scheme that is not central to the phenomena under investigation here.

By prescribing essential boundary conditions on the bottom and the lateral edges and a natural (free-traction) boundary condition at the top for the Darcy–Stokes MFEM solver, the simulations allow for mechanically reasonable mantle upwelling without artificial confinement. Similarly, the Dirichlet inflow and free outflow conditions for the transport solver ensure a constant supply of chemical species and energy while permitting realistic advective transport, without imposing restrictive thermodynamic constraints. This approach maintains the internal consistency of the coupled MFEM–FVM solver and captures the key dynamics of melt migration, porosity evolution and phase transitions, while reducing computational complexity.

4.2.2.2 Compaction Length

The compaction length is typically defined as

$$\delta_c = \sqrt{\frac{\kappa_\phi(\zeta + 4\eta/3)}{\mu_l}}, \tag{4.9}$$

where ζ and η are bulk and shear viscosities of the solid mantle, respectively. The compaction length an important intrinsic length scale that characterize the interaction between the pressure difference and variations in fluid flux (see Mckenzie (1984) and Spiegelman (1993a)).

In this work, we pick a simple constitutive relationship as shown in equation (2.15), which is $\zeta = \mu_s/\phi$ in the notation of this section and a power-law permeability as $\kappa_\phi = \kappa_0\phi^2$. Assuming that the shear viscosity η is much smaller than the bulk viscosity ζ, the appropriately scaled compaction length can be written in terms of the nondimensionalization length scale as

$$\delta_c = \sqrt{\frac{\kappa_0\mu_s}{\mu_l}\phi} = l_0\sqrt{\phi}, \tag{4.10}$$

where l_0 represents the characteristic length scale used in the nondimensionalization. Thus $\delta_c = \delta(\phi)$ depends on the (evolving) porosity. In geological settings, a constant reference porosity is typically prescribed to define the compaction length. However, this assumption is not adopted in the present study, hence we use dimensionless length in the subsequent plots.

4.2.2.3 Numerical Flux

We modify the semi-discretized formulation of a general conservation law, as presented in equation (4.11), to incorporate a "source" term S. The resulting expression can be written as

$$\frac{\partial \overline{u}_E}{\partial t} + \sum_{i=0}^{4} \frac{|e_i|}{|E|} \sum_{g=0}^{G} \omega_{i,g} F(\mathcal{R}^+_{i,g}, \mathcal{R}^-_{i,g}) + \sum_{h=0}^{H} \xi_h S(\mathcal{R}_h) = 0, \tag{4.11}$$

where $\omega_{i,g}$ represents the weight associated to the g-th quadrature point on edge e_i and ξ_h represents the weight associated to the h-th quadrature point on the cell. In practice, we employ the midpoint rule for the numerical integration over cells. This allows us to approximate the face contributions using the cell-averaged transport variables while retaining second-order accuracy.

The adiabatic cooling term $\frac{\alpha_\rho l_0}{c_p}\check{T}\check{\bar{\mathbf{v}}}\cdot\mathbf{g}$ in equation (4.8) is treated as a "source" term, which is straightforward to implement within the semi-discretized formulation. The corresponding Lax-Friedrichs numerical fluxes of the mass composition and the enthalpy across element edges are expressed in terms of the ML-WENO reconstructed values $\mathcal{R}_C$ and $\mathcal{R}_H$ as

$$\begin{aligned} F_C(\mathcal{R}^+_C, \mathcal{R}^-_C) &= \frac{1}{2}\check{\mathbf{v}}_{\text{eff}} \cdot \boldsymbol{\nu} \Big[\mathcal{R}^+_C + \mathcal{R}^-_C - \alpha_C\big(\mathcal{R}^+_C - \mathcal{R}^-_C\big)\Big], \\ F_H(\mathcal{R}^+_H, \mathcal{R}^-_H) &= \frac{1}{2}\check{\bar{\mathbf{v}}} \cdot \boldsymbol{\nu} \Big[\check{T}^+ + \check{T}^- - \alpha_T\big(\check{T}^+ - \check{T}^-\big)\Big] - \check{L}(1-\phi)\check{\mathbf{v}}_s \cdot \boldsymbol{\nu}, \end{aligned} \tag{4.12}$$

where $\boldsymbol{\nu}$ is the ourward unit normal of the edge, and $\check{\mathbf{v}}_{\text{eff}}$ and $\check{\bar{\mathbf{v}}}$ denote the effective velocity and phase averaged velocity continuous across the edges. The terms $\check{T}^+ = \check{T}(\mathcal{R}^+_H, \mathcal{R}^+_C)$ and $\check{T}^- = \check{T}(\mathcal{R}^+_H, \mathcal{R}^+_C)$ are temperatures evaluated on both sides of the edge using the eutectic phase package, based on the reconstructed pointwise mass composition and enthalpy . The stabilizers α_C and α_T use the local Lax-Friedrich scheme as shown in equation (3.9), ensuring stability of the flux discretization.

The calculation of the numerical flux (4.12) of the mass composition is not entirely straightforward. The primary transport variables—mass composition and enthalpy—may undergo distinct reconstructions across an edge due to discontinuities and the presence of multiple phase regions. As a consequence, the porosity ϕ can take different values, denoted $\phi^+ = \phi(\mathcal{R}^+_H, \mathcal{R}^-_C)$ and $\phi^- = \phi(\mathcal{R}^-_H, \mathcal{R}^-_C)$. To ensure compatibility in the MFEM discretization, we adopt the harmonic average of the two values, as discussed previously in Section 4.1.1, and defined the compatible porosity on the edge as

$$\overline{\phi}_e = \frac{2\phi_e^+\phi_e^-}{\phi_e^+ + \phi_e^-}.$$

In contrast, the bulk mass composition represents a relatively smooth advected quantity. A harmonic average would add artificial damping and suppress mixing, while an arithmetic average is more suitable without over-penalizing discontinuities. Therefore, we adopt

$$c^e = (c^+ + c^-)/2. \tag{4.13}$$

We now revisit the definition of effective velocity defined in equation (2.38) with these interface definitions as

$$\mathbf{v}_{\text{eff}} = \frac{c_l^e\overline{\phi}\mathbf{v}_l + c_s^e(1-\overline{\phi})\mathbf{v}_s}{c_l^e\overline{\phi} + c_s^e(1-\overline{\phi})}, \tag{4.14}$$

where $\overline{\phi}$ is the harmonic-averaged porosity and c_s^e, c_l^e are the arithmetic averaged mass compositions.

Similarly, the phase averaged velocity required in the enthalpy flux is expressed without ambiguity as

$$\check{\mathbf{v}} = \overline{\phi}\mathbf{v}_l + (1-\overline{\phi})\mathbf{v}_s, \tag{4.15}$$

where the same harmonic averaged porosity is employed. This guarantees that in the limiting case $\phi^+ = 0$ or $\phi^- = 0$, the phase-averaged velocity reduces to the solid velocity, thereby preserving the correct physical behavior and preventing artificial transport of enthalpy associated with liquid in no-melt regions.

4.2.2.4 Test results

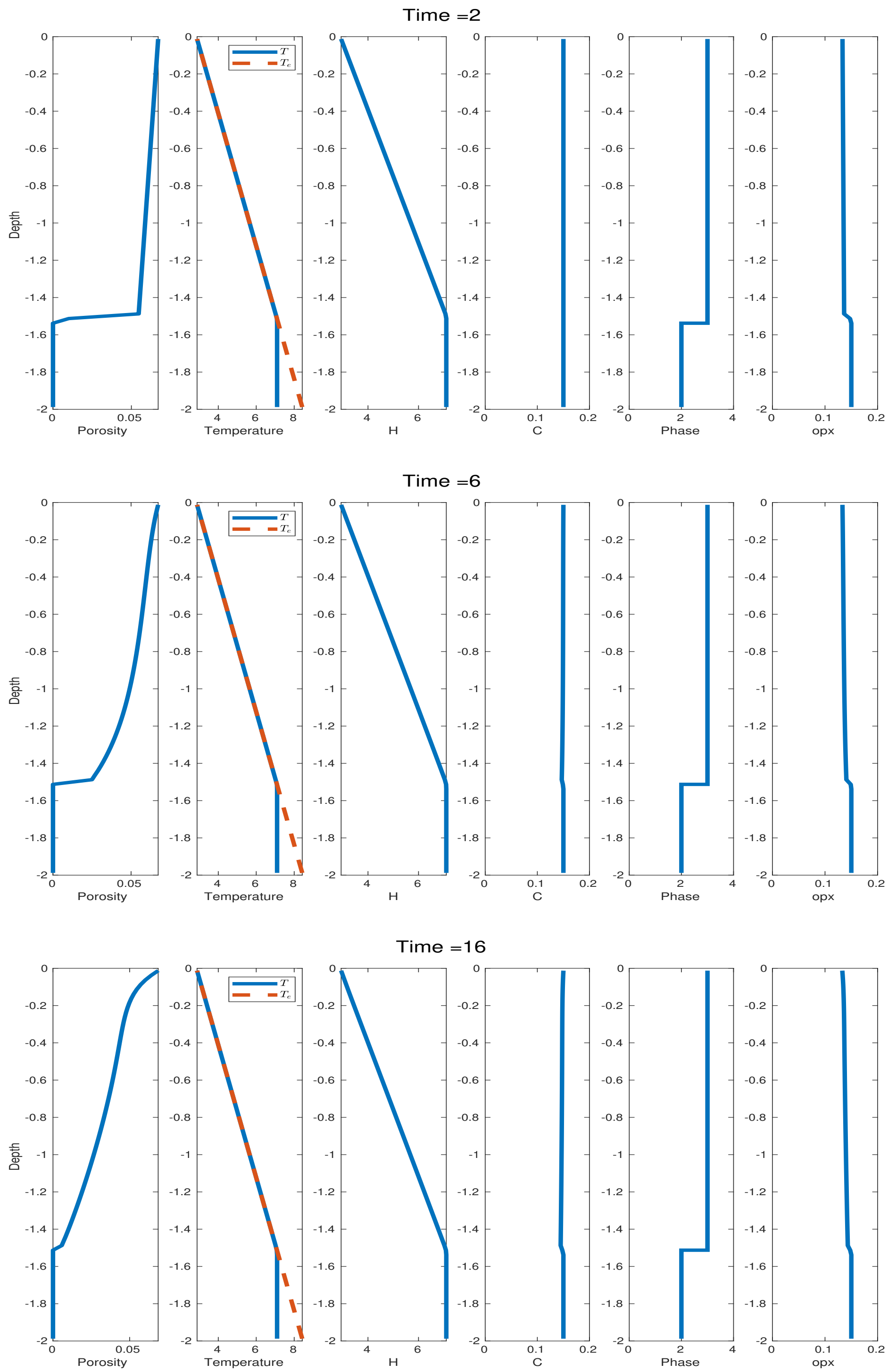
Again, for simplicity and efficiency in implementation and computation, we employ one-dimensional ML-WENO(3,2) stencils as illustrated in Figure 4.1 for both the mass composition and the enthalpy. We use second order SSP Runge-Kutta time integration scheme defined in Equation (3.210) as well.

Full coupled column testing case $H = 2$: The domain in the y-direction spans two dimensionless length $H = 2$ with mesh resolution $N = 80$. We assign an initial condition of the bulk dimensionless mass composition and the bulk dimensionless enthalpy as

$$H = \begin{cases} 2.9 - 2.8y, & -1.5 \leq y < 0, \\ 2.9 + 2.8 \times 1.5, & -2 \leq y < -1.5. \end{cases} \tag{4.16}$$
$$C = 0.15.$$

Figure 4.6 illustrates the temporal evolution of the primary transport variables, bulk dimensionless mass composition and bulk dimensionless enthalpy, along with the corresponding phase calculation determined from the phase package. From time $t = 2$ to time $t = 6$ we observe a transition from phase region 3 (eutectic melting) to phase region 2 (solid, no melting phase) near the cut $y = -1.5$. The porosity field appears to be highly sensitive to these changes in bulk enthalpy and mass composition, leading to the development of a porosity wave near the upper boundary. This wave-shaped pattern arises from the accumulation of porosity near the top boundary, where the local depletion of mass composition occurs more rapidly than it can be replenished from below. This in turn reduces porosity locally and blocks the draining of liquid further down. This wave-shaped pattern near the top boundary subsequently amplifies and ultimately results in an unstable outcome in the numerical solution around time $t = 130$. The wave-shaped pattern persists under mesh refinement, indicating that it reflects a physical feature of the model, which uses a nonphysical BC, rather than a numerical artifact.

Figure 4.7 shows three characteristic velocities defined on the element edges: the effective velocity transporting chemical species, the phase averaged velocity governing the advection of enthalpy, and the solid velocity carrying latent heat. The result confirms that the local mass conservation is well-preserved as evidenced by the phase averaged velocity remaining constant under the Boussinesq approximation.


Time =2
Time =6
Time =16
Depth
Porosity
Temperature
H
C
Phase
opx
T
T_e

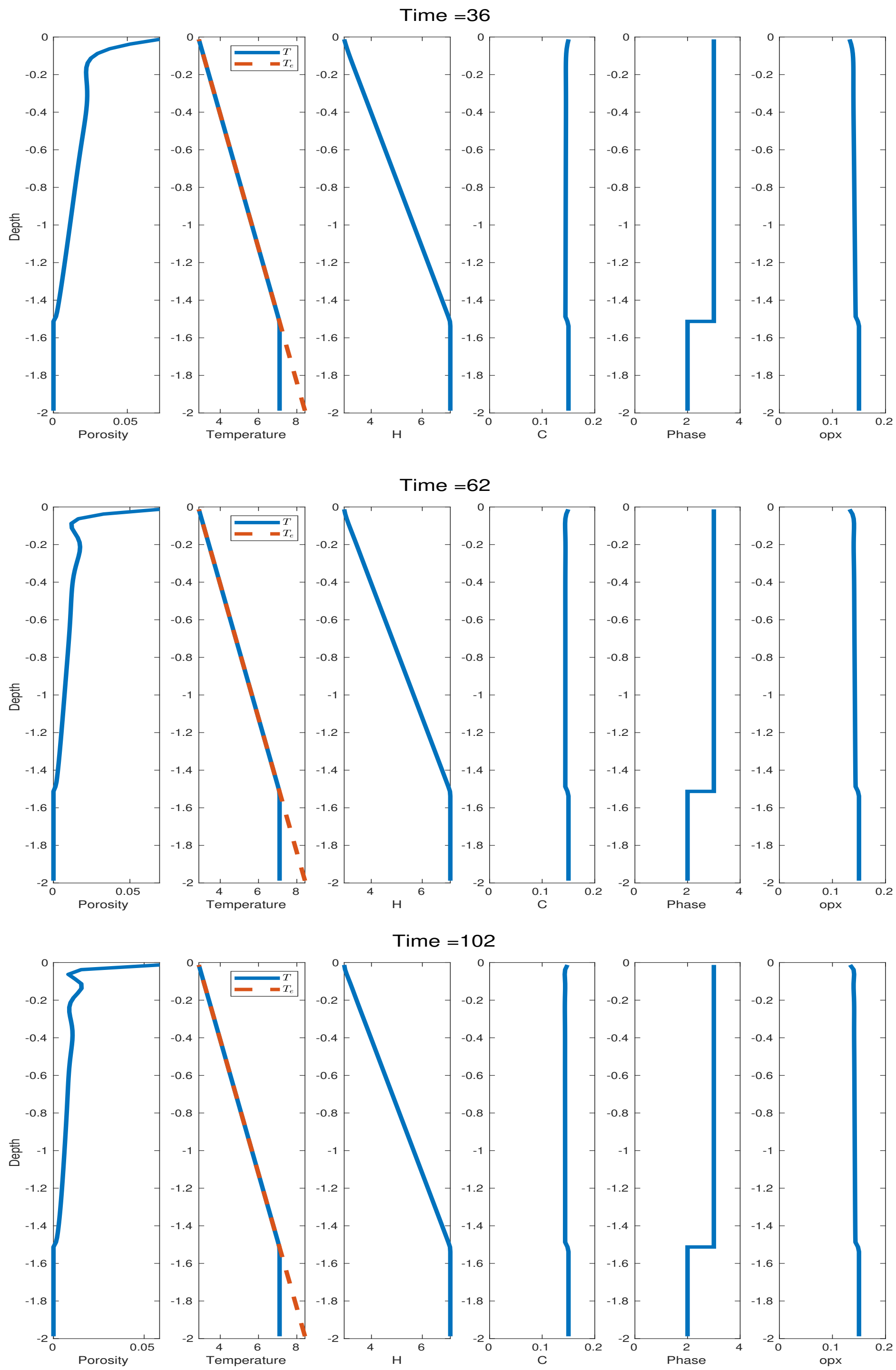
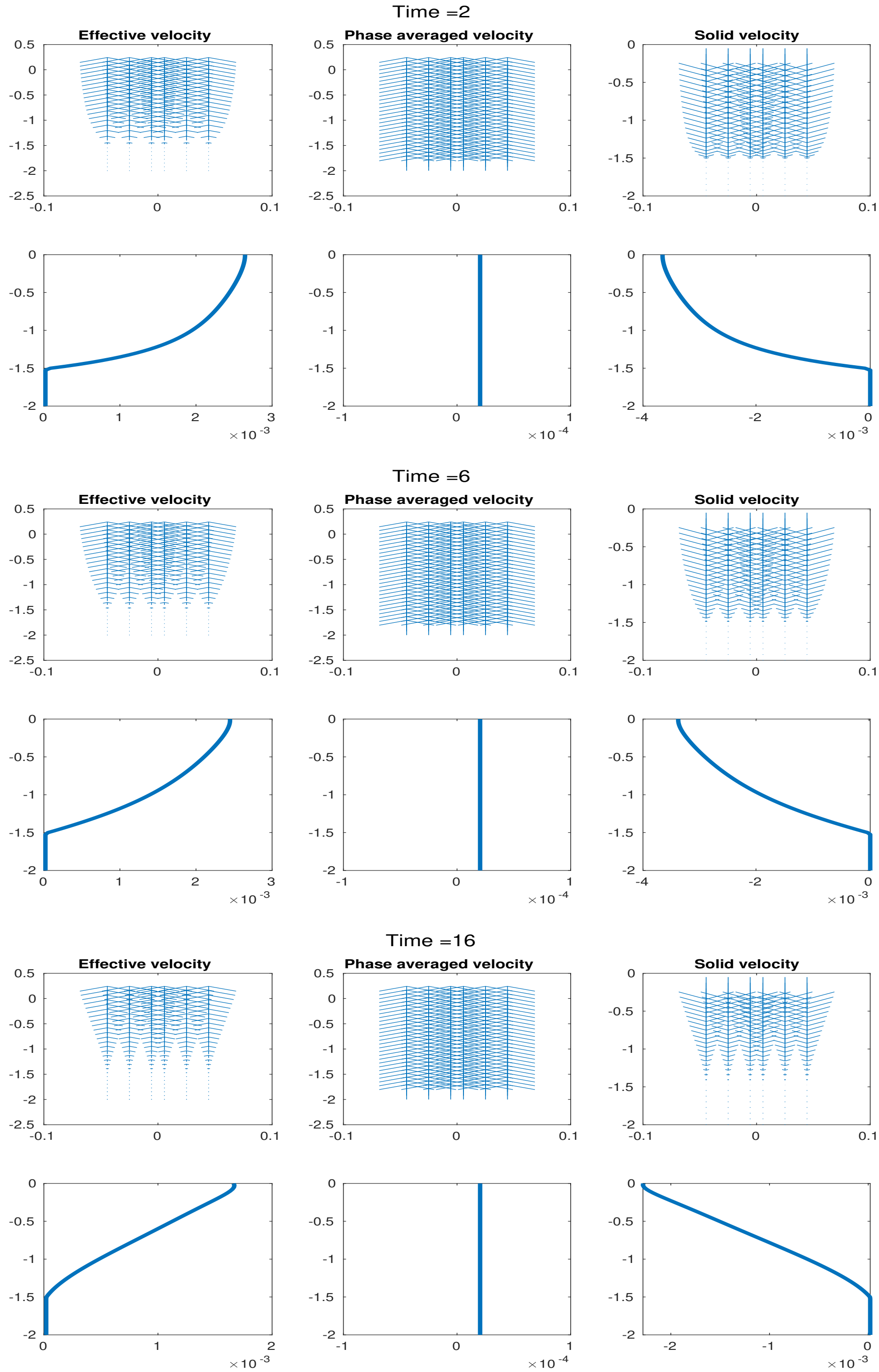


Figure 4.6: Full coupled column testing case $H = 2$. Shown at time = 2, 6, 16, 36, 62 and 102. We present porosity, temperature, dimensionless enthalpy, dimensionless mass composition, phase regions and Opx percentage from left to right.


Time =2
Effective velocity
Phase averaged velocity
Solid velocity
×10^-3
×10^-4
×10^-3
Time =6
Effective velocity
Phase averaged velocity
Solid velocity
×10^-3
×10^-4
×10^-3
Time =16
Effective velocity
Phase averaged velocity
Solid velocity
×10^-3
×10^-4
×10^-3

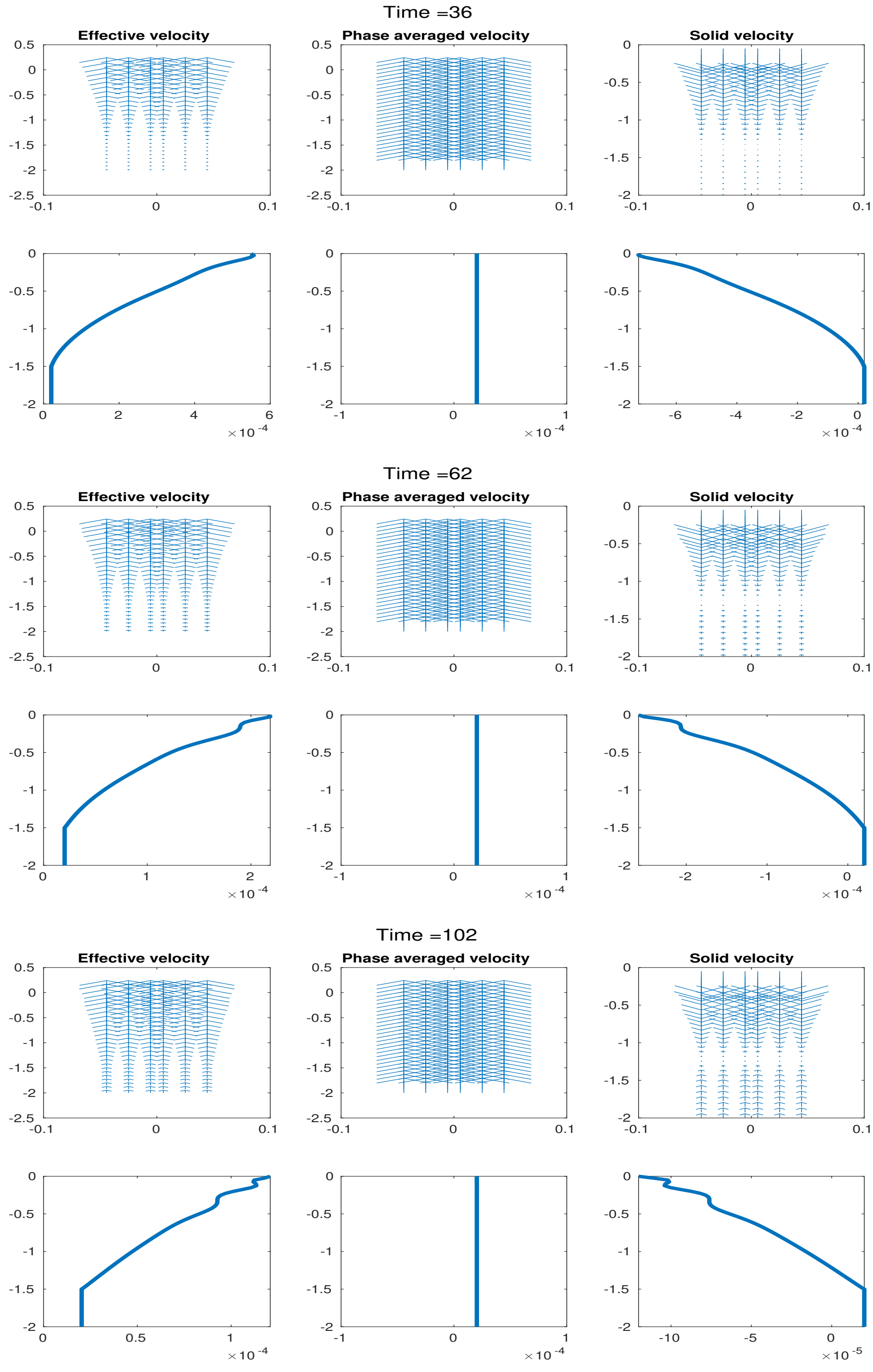


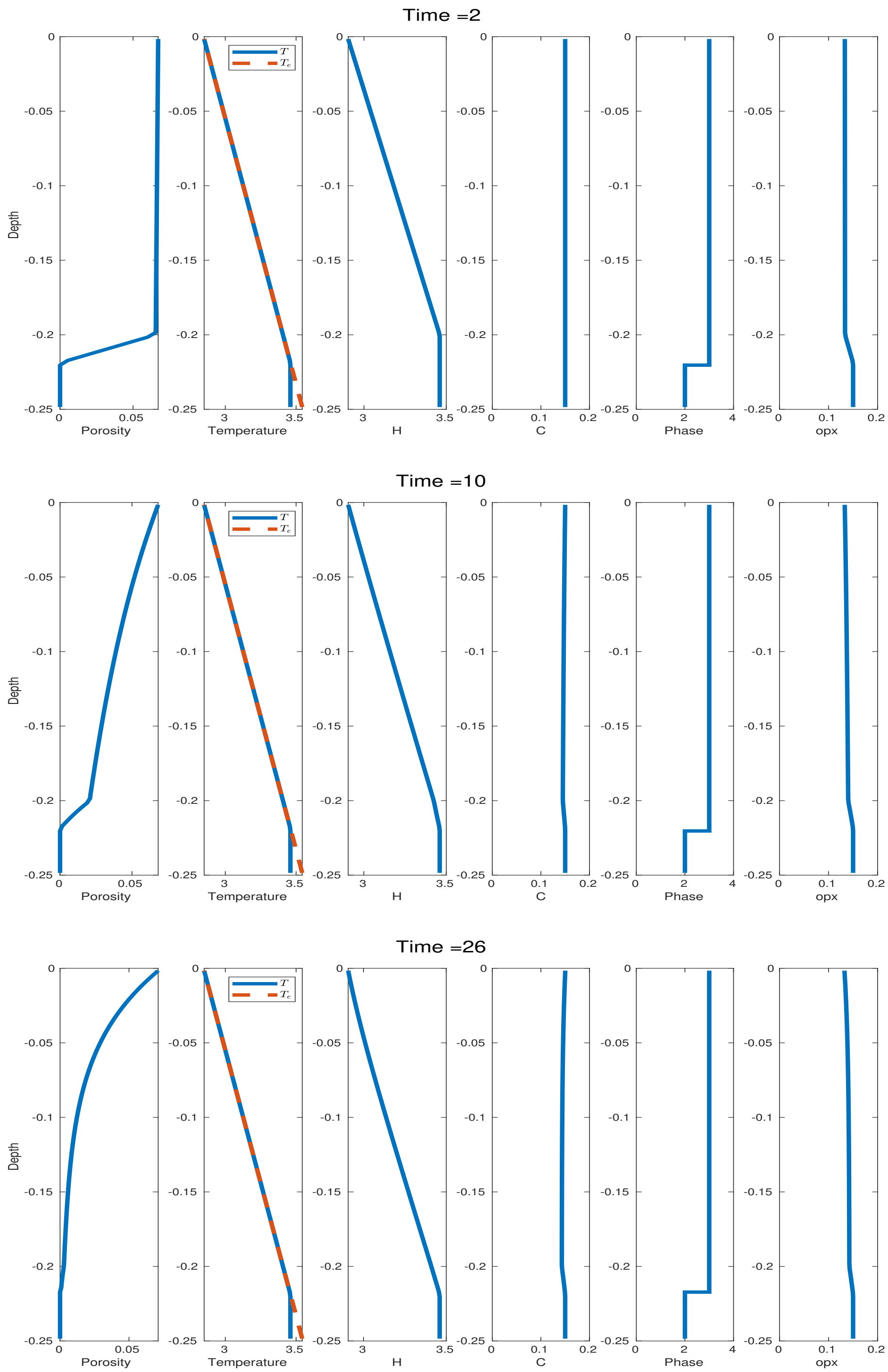
Figure 4.7: Full coupled column testing case $H = 2$. Shown at time = 2, 6, 16, 36, 62 and 102. We present effective velocity, phase averaged velocity and solid velocity from left to right.

Full coupled column testing case $H = 0.25$: The domain in the y-direction spans 0.25 dimensionless length $H = 0.25$ with mesh resolution $N = 80$. We assign the same initial condition of the bulk dimensionless mass composition and a similar initial condition of the bulk dimensionless enthalpy with different cutoff as

$$H = \begin{cases} 2.9 - 2.8y, & -0.2 \leq y < 0, \\ 2.9 + 2.8 \times 0.2, & -0.25 \leq y < -0.2. \end{cases}$$
$$C = 0.15. \tag{4.17}$$

Figure 4.8 illustrates the temporal evolution of the primary transport variables, bulk dimensionless mass composition and bulk dimensionless enthalpy, along with the corresponding phase calculation determined from the phase package. Between time $t = 10$, 26 and 40, the system undergoes successive transitions: initially from Phase Region 3 to Phase Region 2, and subsequently returning to Phase Region 3. These transitions occur while maintaining numerical stability throughout the simulation, illustrating the ability of our framework to handle transition between melting and no melting zones. The system reaches steady state at time $t = 240$, while a notable accumulation of porosity remains concentrated near the top boundary. The instability encountered in the previous larger domain does not arise in this case, as the overall porosity is smaller.

Figure 4.9 again presents the three characteristic velocities defined on the element edges: the effective velocity transporting chemical species, the phase averaged velocity governing the advection of enthalpy and the solid velocity carrying latent heat. As the simulation progresses, the effective velocity exhibits a marked increase near the top boundary, consistent with the porosity accumulation observed in Figure 4.8.



Time =2
Time =10
Time =26
Depth
Porosity
Temperature
H
C
Phase
opx
T
T_e
0
-0.05
-0.1
-0.15
-0.2
-0.25
0.05
3
3.5
0.1
0.2
2
4

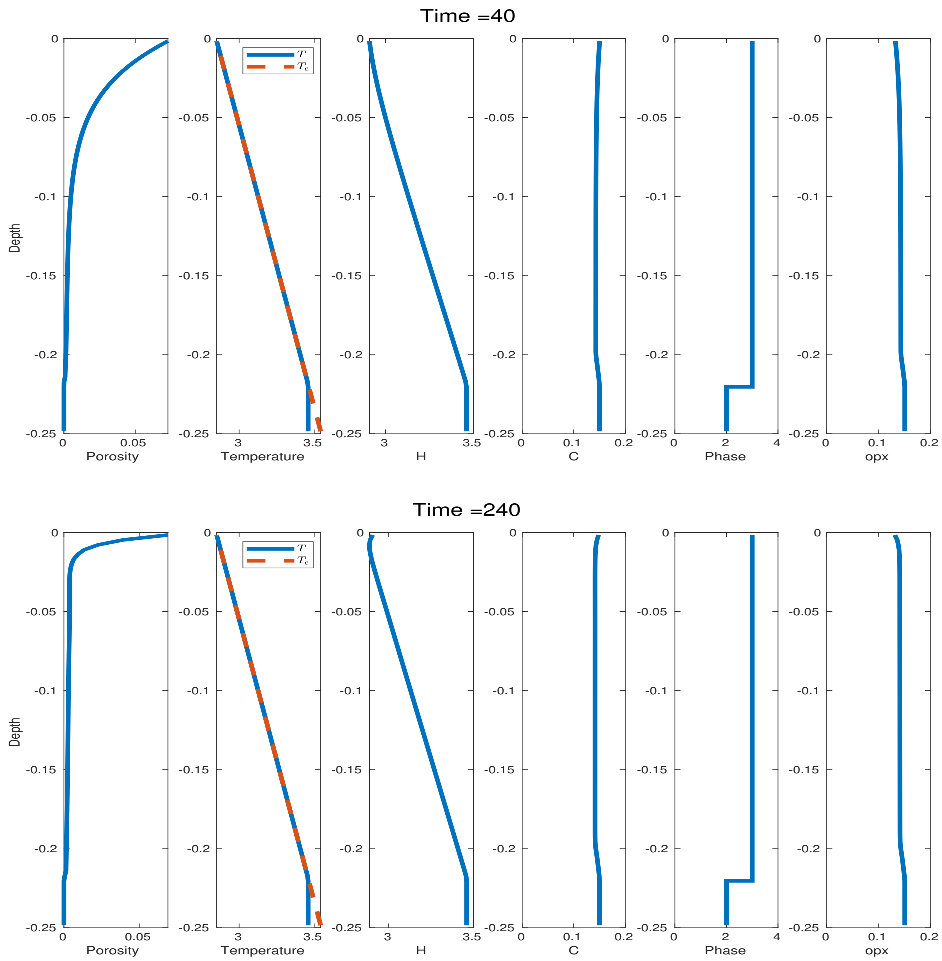
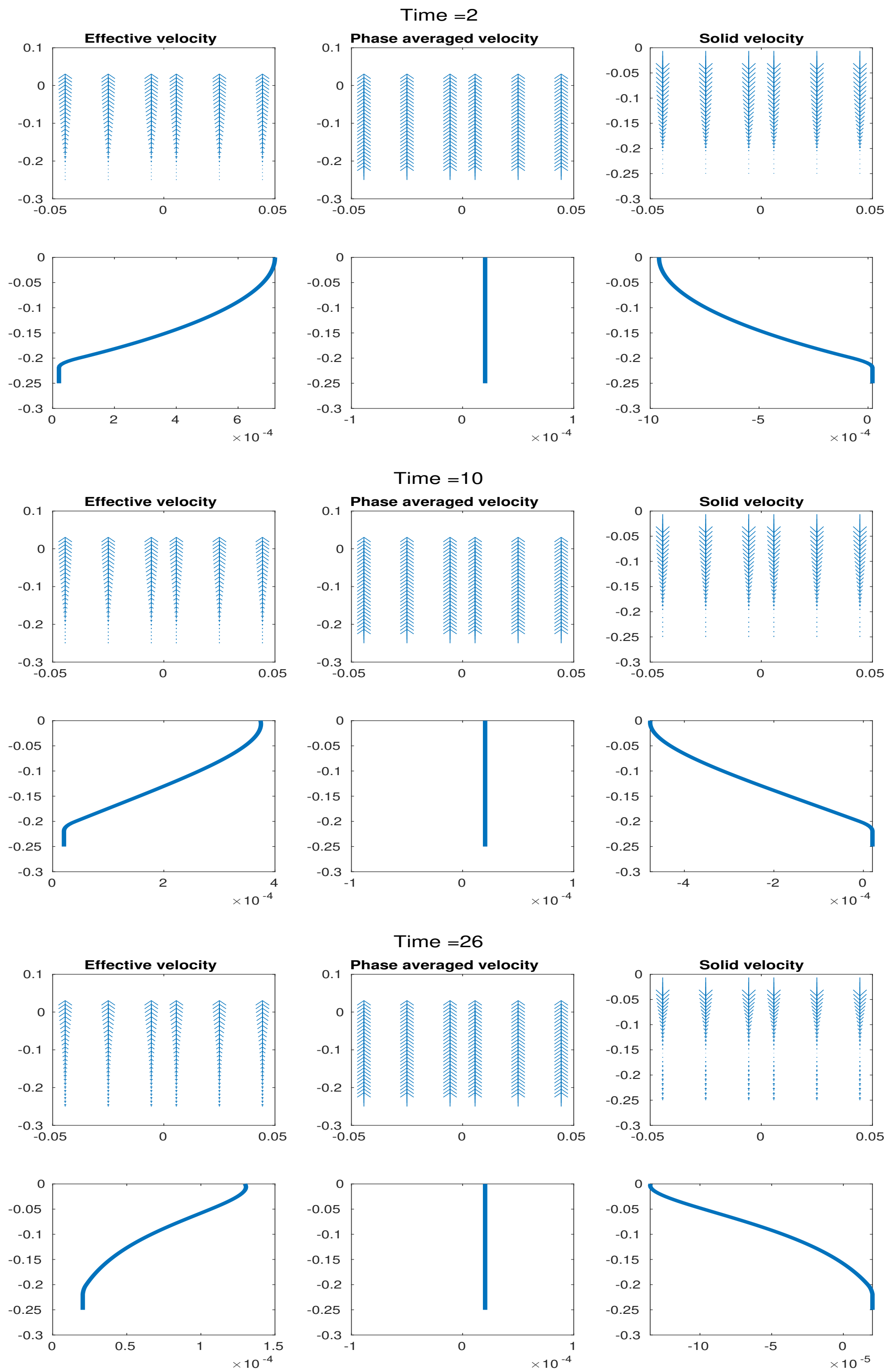


Figure 4.8: Full coupled column testing case $H = 0.25$. Shown at time = 2, 10, 26, 40, and 240. We present porosity, temperature, dimensionless enthalpy, dimensionless mass composition, phase regions and Opx percentage from left to right.


Time =2
Effective velocity
Phase averaged velocity
Solid velocity
×10^-4
×10^-4
×10^-4
Time =10
Effective velocity
Phase averaged velocity
Solid velocity
×10^-4
×10^-4
×10^-4
Time =26
Effective velocity
Phase averaged velocity
Solid velocity
×10^-4
×10^-4
×10^-5

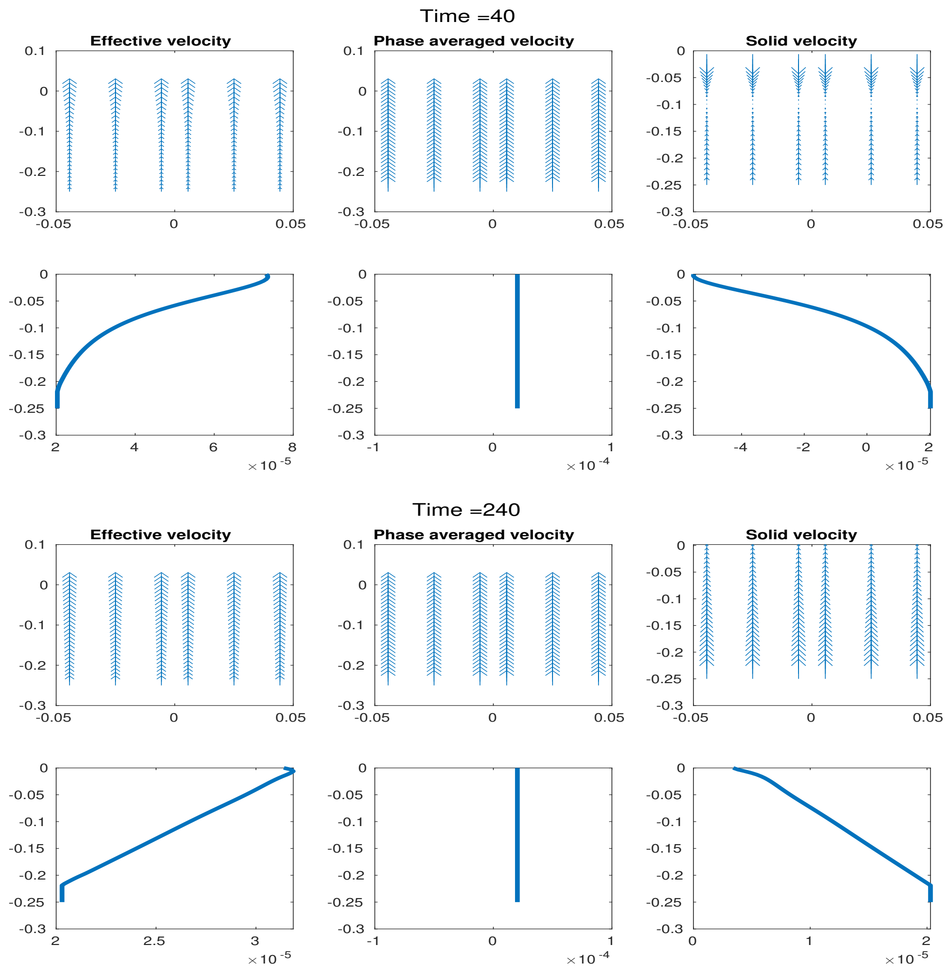


Figure 4.9: Full coupled column testing case $H = 0.25$. Shown at time = 2, 10, 26, 40, and 240. We present effective velocity, phase averaged velocity and solid velocity from left to right.

4.3 Two-Dimensional Test

In this section, we test the fully coupled algorithm on a two-dimensional $M \times N$ grid. The computational domain is defined with dimensions $L = 2$ in the x-direction and $H = 2$ in the y-direction. We reuse the boundary conditions from the pseudo–one-dimensional case described in Section 4.2.2: outflow is permitted only through the top boundary, a constant upwelling velocity is imposed at the bottom boundary, and no-flow conditions are applied on the lateral boundaries. The two-dimensional ML-WENO(3,2) stencils, illustrated in Figure 3.18, are employed without including the constant stencil. We assign a similar initial condition of the bulk dimensionless mass composition and a similar initial condition of the bulk dimensionless enthalpy as

$$\begin{aligned} H &= \begin{cases} 2.9 - 2.8y, & -1.5 \leq y < 0, \\ 2.9 + 2.8 \times 1.5, & -2 \leq y < -1.5, \end{cases} \\ C &= 0.1. \end{aligned} \tag{4.18}$$

This numerical experiment is designed to demonstrate the capability of our solver to handle a fully two-dimensional configuration, although the setup itself is not physically meaningful.

Figure 4.10 illustrates a simple two-dimensional simulation incorporating phase calculations. As porosity is drained, the system undergoes stable phase transitions from phase 2 (eutectic melting) to phase 1 (no melting) and from phase 2 (eutectic melting) to phase 3 (super-eutectic melting), without exhibiting numerical instability.

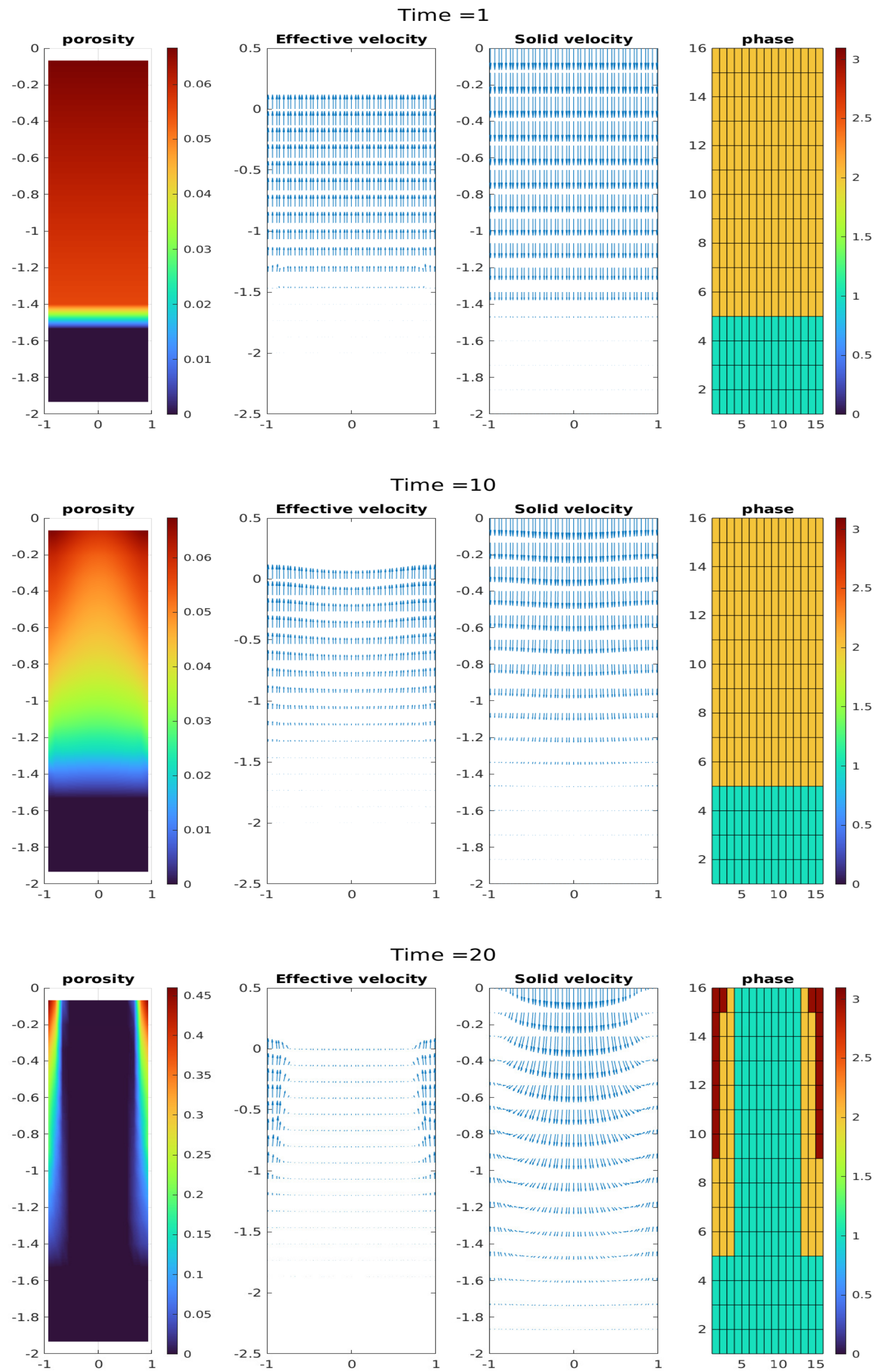


Figure 4.10: Full two-dimensional test case. Shown at time = 1, 10 and 20. We present porosity, effective velocity, solid velocity and phase split.

Chapter 5: Concluding Remarks

5.1 Summary

In this work, we first presented a physics model capturing three key aspects of a two-phase, partially molten mantle system: (1) a degenerate Darcy–Stokes system describing the static mechanical behavior, (2) a nonlinear system governing the transport of chemical species and thermal energy, and (3) a eutectic phase relation characterizing phase transitions. Building upon these governing principles, we developed a locally mass conservative numerical framework that is capable of resolving the coupled multiphysics interactions, with particular emphasis on accurately modeling melt extraction and properly handling degeneracies in regions devoid of melt.

For the degenerate Darcy-Stokes system describing the static mechanics system, we introduced $H(\mathrm{div})$-and H^1- conforming mixed finite element spaces for the Darcy and the Stokes subproblems, respectively, based on the mathematical theory developed in Arbogast et al. (2017). Specifically, we employed the first- and second-order direct serendipity spaces designed for the quadrilateral mesh. The $H(\mathrm{div})$-conforming space relates to the direct serendipity space through the de Rham complex. For the H^1-conforming component, we proposed a modified Bernardi–Raugel element. We proved the stability and approximation properties of this new element. The resulting saddle point system, arising from the coupling of the degenerate Darcy–Stokes equations, was solved using a modified Uzawa iterative method.

To address the nonlinear transport of chemical species and energy in the presence of a porous media, we adopted and modified the weighted essentially non-oscillatory (WENO) method. In particular, we proposed two alternatives to the classic Jiang-Shu smoothness indicator in order to reduce computational cost (see Huang et al. (2023) and Arbogast et al. (2025). Furthermore, we introduce a novel multi-level WENO reconstruction scheme (see Arbogast et al. (2024)) that is more versatile and flexible, allowing the combination of arbitrary stencils of different orders in the reconstruction. This allowed us to be more deal with the complicated multiphysics problems encountered.

Finally, we connected and advanced the multiphysics system by incorporating a eutectic phase module that couples bulk mass composition and thermal energy information with porosity to dynamically capture both the evolution of porosity and phase transitions. This module closes the feedback loop between mechanical, transport, and phase processes, enabling fully coupled simulations of partially molten mantle dynamics.

The computational framework was implemented in C/C++ with PETSc, where, apart from basic linear solvers and standard data management utilities, all components of the numerical algorithms and multiphysics coupling strategies were designed and coded by the author.

5.2 Conclusions

In this dissertation, we developed and analyzed a new Bernardi–Raugel element that achieves the correct rate of convergence on a non-rectangular quadrilateral mesh. We then demonstrated that the $H(\mathrm{div})$- and H^1-conforming mixed finite element two-phase Darcy–Stokes solver can seamlessly and robustly handle the degeneracy induced by no-melt regions while maintaining both stability and computational efficiency. Numerical experiments using manufactured pseudo–one-dimensional problems confirmed second-order accuracy in velocity and first-order accuracy in pressure. Furthermore, we showed that the newly developed multilevel weighted essentially non-oscillatory (ML-WENO) reconstruction scheme effectively captures complex transport phenomena and handles diverse boundary conditions with stability and accuracy.

The proposed coupling strategy between the MFEM and FVM solvers proved to be accurate, stable, and computationally efficient, both with and without the involvement of the phase package. The new ML-WENO reconstruction scheme was shown to couple stably and efficiently with the Darcy–Stokes MFEM solver. When the phase package was incorporated into the porosity computation, porosity exhibited strong sensitivity to variations in mass composition and enthalpy, requiring smaller time steps for stability. Nonetheless, the coupled framework accurately resolved complex porosity structures and demonstrated robust performance across all tested scenarios. Boundary conditions were found to play a critical role in ensuring both the physical realism of the simulations and the numerical stability of the solver.

5.3 Future Work

We propose to enhance the stability, versatility, and usability of the framework. This may include incorporating additional stabilization techniques, improving the coupling strategies between multiphysics components, and extending the implementation to general unstructured meshes. Such improvements will enable the framework to address problems involving complex geological geometries. Furthermore, integration and modification with high-performance parallel computing architectures will allow large-scale simulations.

Moreover, the framework opens the possibility of addressing geologically significant problems that remain poorly understood. For example, while melt extraction at mid-ocean ridges has been extensively studied, the style and spatial variability of melt extraction in subduction zones are less constrained. Applying the present framework to such settings may provide new insights into problems like the relationship between melt generation and devolatilization of the slab and melt migration pathways, thereby contributing to a more complete understanding of volatile cycling and magma genesis in subduction settings.

Finally, the generality of the framework lends itself to applications beyond mantle dynamics. With appropriate modifications to the phase relations and constitutive laws, it can be adapted to study a wide range of two-phase partially molten systems, including problems such as glacial melting, extraterrestrial magmatism, and industrial processes such as eutectic alloy solidification. Such extensions would further demonstrate the flexibility and broad applicability of the numerical methodology developed in this work.

Appendix A: Code Implementation

In this project, all numerical experiments and simulations are implemented in C++/C, with basic iterative solvers and simple parallel data management facilitated by PETSc Balay et al. (2025a). In this section, we describe the implementation of a coupled FEM-FVM code, designed to preserve generality and support future optimization and extensibility.

We first present a general implementation for an unstructured mesh, followed by a detailed discussion of a specfic application involving a logically rectangular mesh. All applications are considered in a two-dimensional setting.

A.1 Geometry

A fundamental step in developing a FEM or FVM code is the indexing of geometric entities. The mesh index data structure implemented in the code is inspired by the classical viewpoint of Boundary Representation (BREP), originally introduced by Baumgart (1975). In the two-dimensional case, each element is bounded by a set of edges, and each edge is defined by a set of vertices. We begin by introducing the following notations for global geometric objects:

$$\begin{aligned} \text{Vertex} \quad & V_i, \quad i \in ID_V, \\ \text{Edge} \quad & E_i, \quad i \in ID_E, \\ \text{Element (i.e. Face)} \quad & F_i, \quad i \in ID_F, \end{aligned} \tag{A.1}$$

where ID_V, ID_E and ID_F denote countable sets containing all unique global indices for vertices, edges, and elements (or cells, i.e., the faces), respectively. Superscripts are used to indicate subsets of these global index sets. For example, $ID_V^{E_i} \subset ID_V$ represents the set of global vertex indices associated with edge E_i, which is a subset of the global vertex index set.

We again present the following notations for local geometric entities:

$$\begin{aligned} \text{Vertex} \quad & v_i, \quad i \in id_v, \\ \text{Edge} \quad & e_i, \quad i \in id_e, \\ \text{Element (i.e. Face)} \quad & f_i, \quad i \in id_f, \end{aligned} \tag{A.2}$$

where id_v, id_e and id_c denote countable sets containing local index for vertices, edges and elements corresponding to a given geometrical object. Superscripts are used to indicate the target geometrical object. For example, $id_v^{E_i} = \{0, 1\}$ represents the local index of the two vertices bounding the edge E_i.

We establish a one-to-one mapping between local and global indices through a local-to-global map, defined as

$$f : id \to ID. \tag{A.3}$$

In practice, it is often unnecessary to explicitly define a local-to-global mapping. On the other hand, for each local array containing global indices, the array indices naturally represent the local index set. This array-based representation enables the implicit construction of a global-to-local mapping within each geometric entity.

A.1.1 General Unstructured Mesh

In both FEM and FVM solvers, an element object F_i (mesh cell or "faces") serves as the fundamental geometry entity. These elements will be assumed to be strictly convex polygons, tessellating space , and sharing complete edges. we start with a brief analysis of the information each face should access from its local perspective. Each element F_i should have the access to the following geometric objects:

- The adjacent elements connected by edges;
- The adjacent elements connected with vertices, but not sharing an edge;
- The vertices of the polygonal element;

- The edges that bound this element.

At the same time, each edge E_j should have access to:

- Two adjacent elements that share this edge;
- Two vertices that bound this edge.

For simplicity, we restrict our discussion to quadrilaterals in the following discussions. In FEM and FVM computations, we assume an orientable mesh, where "clockwise" and "counterclockwise" orientations can be consistently defined. This allows the use of a half-edge data structure to systematically manage edge directions.

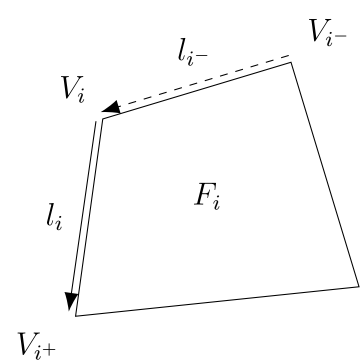


Figure A.1: An illustration of the half-edge data structure for a quadrilateral.

Figure A.1 illustrates a simple half-edge data structure applied to a quadrilateral mesh. For each element F_i, the associated vertex V_i references the outgoing link l_i that points to the next vertex V_{i^+}, while the incoming link l_{i^-}, which points to V_i is not directly referenced. Consequently, a normal "direction" can be determined from the perspective of each element object F_i in accordance to the direction of links.

A.2 Logically Rectangular Mesh

Mesh generation is not the primary focus of this project; therefore, we restrict to a logically rectangular mesh and adopt a simple approach to generating a quadrilateral mesh. The quadrilateral mesh is generated by randomly distorting each interior vertex coordinates of a uniform rectangular mesh. However, the indexing of geometrical entities remain unchanged despite the vertex perturbation.

In our implementation, all indices are represented using Cartesian coordinates. This allows the local-to-global mapping of geometric entities to be established easily through linear functions. All geometric entities can be indexed using Cartesian coordinates straightforwardly. For a mesh with $M \times N$ cells, the total number of vertices is $(M+1) \times (N+1)$, while the total number of edges is $M \times (N+1) + (M+1) \times N$. Horizontal edges are indexed first, and vertical edges are indexed sequentially starting from the offset $M \times (N+1)$.

Before we delve into the indexing conventions of this type of mesh, we introduce a reshape function that facilitates communication between N-dimensional data and its one-dimensional representation.

A.2.1 Reshape of N-dimensional Data

The natural representation of a 2D Cartesian index is a pair (i, j). In practice, it is often reshaped into a 1D array for more efficient computational handling. We introduce a general reshape function that projects N-dimensional data to a one-dimensional data. This reshape function can also be used for tensor data, e.g., The Jiang-Shu smoothness indicators (3.186).

Consider a bijective map between N-dimensional data and its one-dimensional data representation as

$$i \leftrightarrow (i_0, i_1, \ldots, i_{N-1}). \tag{A.4}$$

Algorithm 4 illustrates a simple implementation of the reshape function. The inverse operation, reshaping one-dimensional data back into N-dimensional form, can be achieved using modulo and integer division operations.

Throughout this work, we use these two index representations interchangeably.

Algorithm 4 Reshape N-dimensional data

Consider a N-dimensional array of size $\{n_0 \times n_i \times \dots n_{N-1}\}$.

Let $I = 0$ be the variable storing the output, and $a = 1$ be the intermediate multiplier.

Consider an input N-dimensional index $\{i_0, i_1, \dots, i_{N-1}\}$.

1: **for** rank $r = 0, 1, \dots, N-1$ **do**
2: $\quad I = I + i_r \times a$.
3: $\quad a = a \times n_r$.
4: **end for**

To emphasize the relation between N-dimensional and its one-dimensional representation, we introduce the following notation as

$$i(i_0, i_1, \dots, i_{N_1})_{n_0, n_1, \dots, n_{N-1}}, \tag{A.5}$$

where i denotes the mapping from the N-dimensional data to the one-dimensional data, while $(i_0, i_1, \dots, i_{N-1})$ is the corresponding N-dimensional index. The N-dimensional quantities $n_0, n_1, \dots, n_{N-1}$ represent the respective sizes along each dimension of the N-dimensional array.

A.2.2 Local Vertex Representation

We now represent the local vertex index with the new notation (A.5). For any vertex, we can represent its local index as

$$a(a_0, a_1)_{2,2}, \tag{A.6}$$

where $a_0, a_1 \in \{0, 1\}$. We can link to the next point simply by adding 1 to the one-dimensional index under the understanding of modulo as

$$b \equiv a + 1 \pmod 4, \tag{A.7}$$

which is a simple half-edge data representation. The direction is naturally defined by this addition operation.

A.2.3 Local to Global Index Convention

Recall that local vertices and edges are indexed starting from bottom left corner, as illustrated in Figure 3.1. We now define the local to global index mapping with the help of the new notation (A.5). Consider any local vertex v_i associated to a quadrilateral element $F_{I(I_0,I_1)_{M,N}}$ with the local index

$$id_v \ni i(i_0, i_1)_{2,2}. \tag{A.8}$$

We find the corresponding global index with ease as

$$ID_V \ni I(I_0 + i_0, I_1 + i_1)_{M+1,N+1}. \tag{A.9}$$

We continue our discussion the local-to-global mapping for vertical and horizontal edges. Consider edge e_i defined by the vertex $id_v \ni i(i_0, i_1)$ and its half-edge link l_i:

- Global index of the horizontal edge is

$$ID_E \ni I(I_0, I_1 + i_1)_{M,N+1}; \tag{A.10}$$

- Global index of the vertical edge is

$$ID_E \ni I(I_0 + i_0, I_1)_{M+1,N} + M \times (N + 1). \tag{A.11}$$

A.3 Dofs and MFEM assembly

In FEM and FVM computations, it is necessary to associate geometrical entities with degrees of freedom (dofs). For FVM schemes, the dofs associated with edge fluxes are straightforward. However, in MFEM, the treatment of dofs requires more carefully handling, particularly for those located on the boundary. Following the approach of Hughes (1987), we employ three data processing arrays: ID, IEN and LM arrays. We begin by the introduction of these three arrays as:

- IEN array,

$$\mathrm{IEN}(a, e) = A, \tag{A.12}$$

where a is the local dof index, e is the global element number and A is the global dof index;

- ID array,

$$\mathrm{ID}(A) = \begin{cases} P, & A \notin \mathrm{Essen}, \\ 0, & A \in \mathrm{Essen}, \end{cases} \tag{A.13}$$

where P is the position used in global matrix assembly, and Essen denotes the boundary assigned with essential boundary condition;

- LM array,

$$\mathrm{LM} = \mathrm{ID}(\mathrm{IEN}(a, e)). \tag{A.14}$$

In actual code implementation, we create LM array by screening all dofs prior to computation, while IEN and ID array are not stored explicitly.

A.3.1 MFEM Indexing Example

For a first order $H(\mathrm{div})$ conforming element, the velocity dofs are primarily associated with edges, similar to FVM, making them relatively intuitive to understand and implement. Therefore, we provide a detailed description of how the velocity dofs of a modified Bernardi-Raugel element is implemented using ID, IEN, and LM arrays, where both vertex dofs and edge dofs are present.

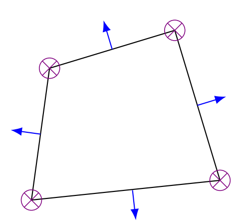


Figure A.2: An exhibition of velocity dofs associated with a modified Bernardi-Raugel element.

Figure A.2 illustrates the dof layout in a Bernardi-Raugel element: two dofs (the x-and y-components) are associated with each vertex node while one normal flux dof is associated with each edge node. The global dofs are ordered as follows: x-and y-component of the vertex dofs, followed by flux dofs on horizontal edges and finally flux dofs on vertical edges.

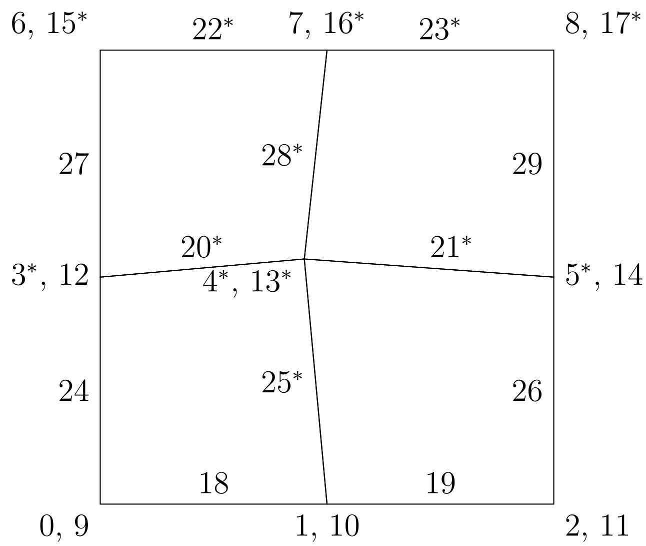


Figure A.3: An exhibition of velocity dofs associated with Bernardi-Raugel on a 4×4 mesh.

Figure A.3 shows a simple 4×4 mesh with corresponding index associated with each velocity dof marked. The total dof counts equal to $2 \times (3 \times 3) + 2 \times 3 + 3 \times 2 = 30$. For example, we assume the following boundary conditions are prescribed.

- Bottom Edge:
  - x-component is fixed with zero;
  - y-component is fixed with a constant value;
  - normal flux is fixed with a constant value.
- Left and Right side Edges:

- x-component is fixed with zero;
- y-component is free, given natural boundary condition (zero stress);
- normal flux is fixed with a zero flux.

• Top Edge:

  - x-component fixed with zero;
  - y-component is free, given natural boundary condition (zero stress);
  - normal flux is free.

In Figure A.3, we mark all the dofs that are not prescribed, i.e., will be determined by solving the linear system, with a superscript $*$. These *free* dofs consist of interior dofs as well as boundary dofs subject to natural boundary conditions. We can construct an ID array accordingly in Table A.1. As a result, the reduced linear

P	0	1	2	3	4	5	6	7	8	9	10	11
A	3	4	4	13	14	16	20	21	22	23	25	28

Table A.1: ID array generated to the dofs in Figure A.3.

system excluding prescribed velocity dofs is 12×12, which is the size of the ID array in Table A.1.

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